

INFOLINK UNIVERSITY COLLEGE

HAWASSA CAMPUS

AUTOMATIC LICENSE PLATE RECOGNITION BASED SMART TRAFFIC MANAGEMENT SYSTEM HAWASSA CITY

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Documentation (Proposal) of Course Software Engineering

Submitted to the Department of Computer Science, Faculty of Engineering, Infolink
University College, in Partial Fulfillment of the Requirement for the Degree of Bachelor of
Science in Computer Science

Hawassa, Ethiopia

January 21, 2026

Table of Content

List of Tables	VII
List of Figures	VIII
Definitions, Acronyms, and Abbreviations	IX
ABSTRACT	X
Problem – Solution – Impact	XII
CHAPTER ONE	- 1 -
INTRODUCTION	- 1 -
1.1 Overview	- 1 -
1.2 Background of the Organization	- 1 -
1.3 Background of the Project	- 2 -
1.4 Team Composition	- 2 -
1.5 Statement of the Problem	- 3 -
1.6 Objectives of the Project	- 4 -
1.6.1 General Objective	- 4 -
1.6.2 Specific Objectives	- 4 -
1.7 Feasibility Analysis	- 5 -
1.7.1 Operational Feasibility	- 5 -
1.7.2 Technical Feasibility	- 5 -
1.7.3 Economic Feasibility	- 6 -
1.7.4 Behavioral / Political Feasibility	- 6 -
1.7.5 Schedule Feasibility	- 6 -
1.8 Cost–Benefit Analysis	- 8 -
1.8.1 Cost of the Project	- 8 -
1.8.2 Consolidated Total Cost	- 13 -
1.8.3 Estimated Cost Reductions	- 13 -
1.9 Scope and Significance of the Project	- 13 -
1.9.1 Scope of the Project	- 13 -
1.9.2 Significance of the Project	- 13 -
1.10 Target Beneficiaries of the System	- 14 -
1.11 Methodology for the Project	- 14 -
1.11.1 Data Source	- 14 -
1.11.2 Fact-Finding Techniques	- 15 -
1.11.3 System Analysis and Design Approach	- 15 -
1.11.4 Overview of Project Development Phases	- 16 -
1.11.5 Testing Procedures	- 18 -
1.11.6 Real World Implementation Strategy	- 19 -
1.11.7 Required Resources and Cost Analysis	- 19 -
1.11.8 Risk Analysis, Assumptions, and Constraints	- 20 -
1.11.9 Limitation of the Project	- 20 -
1.11.10 Ethical, Legal, and Privacy Considerations	- 20 -
CHAPTER TWO	- 22 -
DESCRIPTION OF THE EXISTING SYSTEM	- 22 -
2.1 Introduction to the Existing System	- 22 -
2.2 Stakeholders and Players in the Existing System	- 22 -
2.3 Major Functions and Activities of the Existing System	- 24 -
2.4 Business Rules Governing the Existing System	- 25 -
2.5 Reports Generated in the Existing System	- 26 -

2.5.1 Types of Reports	- 26 -
2.5.2 Limitations of Existing Reports	- 26 -
2.6 Documentation and Forms Used in the Existing System	- 26 -
2.6.1 Common Forms Used	- 26 -
2.6.2 Issues with Manual Documentation	- 27 -
2.7 Bottlenecks of the Existing System (PIECES Framework)	- 27 -
2.7.1 Performance	- 27 -
2.7.2 Information (Input and Output Issues)	- 27 -
2.7.3 Security and Control	- 27 -
2.7.4 Efficiency	- 28 -
2.7.5 Service	- 28 -
2.8 Practices to Be Preserved	- 28 -
2.8.1 Legal and Administrative Compliance	- 28 -
2.8.2 Subcity-Based Administrative Structure	- 28 -
2.8.3 Manual Verification Authority	- 28 -
2.8.4 Existing Communication Culture	- 28 -
2.9 Proposed Solution for the New System	- 28 -
2.9.1 Overview of the Proposed System	- 29 -
2.9.2 Key Features of the Proposed System	- 29 -
2.9.3 Alignment with Existing Practices	- 30 -
2.10 Functional Requirements	- 30 -
2.10.1 Performance Requirements	- 30 -
2.10.2 Process Requirements	- 31 -
2.10.3 Input-Related Requirements	- 32 -
2.10.4 Output-Related Requirements	- 32 -
2.10.5 Storage-Related Requirements	- 32 -
2.11 Non-Functional Requirements	- 33 -
2.11.1 Performance Requirements	- 33 -
2.11.2 User Interface Requirements	- 33 -
2.11.3 Security and Access Permissions	- 33 -
2.11.4 Backup and Recovery Requirements	- 33 -
2.11.5 Scalability and Maintainability	- 33 -
2.11.6 Comparative Analysis of Existing and Proposed Systems	- 34 -
CHAPTER THREE	- 35 -
SYSTEM ANALYSIS	- 35 -
3.1 Introduction	- 35 -
3.2 System Requirements Specification (SRS)	- 35 -
3.3 System Overview	- 36 -
3.3.1 Key Characteristics of the System	- 36 -
3.3.2 Core Functional Capabilities	- 36 -
3.4 Stakeholders and System Actors	- 37 -
3.4.1 Primary Stakeholders	- 37 -
3.4.2 System Actors (Authorized Users and External Systems)	- 37 -
3.4.3 Explicit Exclusion of Driver as a System Actor	- 39 -
3.5 System Boundary Definition	- 40 -
3.5.1 In-Scope System Responsibilities	- 40 -
3.5.2 Out-of-Scope System Responsibilities	- 40 -
3.6 System Constraints and Assumptions	- 40 -
3.7 Operating Environment	- 42 -
3.7.1 Deployment Environment	- 42 -

3.7.2 Client Environment	- 42 -
3.7.3 Integration Environment	- 42 -
3.8 Future Expansion Considerations	- 43 -
3.8.1 Functional Expansion	- 43 -
3.8.2 Intelligence and Analytics Expansion	- 43 -
3.8.3 Integration Expansion	- 43 -
3.9 Use Case Diagram	- 43 -
3.9.1 Identified Main Use Cases	- 43 -
3.10 System Business Rules & Logic (Supplementary Detail)	- 44 -
3.11 Use Case Documentation	- 45 -
3.12 Analysis-Level Class Diagram (Comprehensive Description and Rationale)	- 49 -
3.12.1 Purpose of the Analysis-Level Class Diagram	- 49 -
3.12.2 High-Level System Grouping and Structural Domains	- 49 -
3.12.3 Design Principles Applied	- 50 -
3.12.4 Security & User Management Domain	- 51 -
3.12.5 Vehicle & Registration Domain	- 52 -
3.12.6 Violation & Enforcement Domain (Core)	- 53 -
3.12.7 Surveillance & Evidence Domain	- 54 -
3.12.8 Intelligence & Coordination Domain	- 55 -
3.12.9 Complaint & Oversight Domain	- 56 -
3.12.10 Payment Domain (External Interaction Only)	- 57 -
3.12.11 Shared Value Objects	- 57 -
3.12.12 Conclusion	- 57 -
3.13 SEQUENCE DIAGRAMS – DETAILED OPERATIONAL INTERACTION MODEL	- 58 -
3.13.1 Purpose and Government Context	- 58 -
3.13.2 Detailed Sequence Flow Descriptions	- 59 -
3.14 Activity Diagram – Detailed Functional Documentation	- 70 -
3.14.1 Introduction	- 70 -
3.14.2 Scope and Coverage	- 70 -
3.14.3 Authentication and Role Assignment Phase	- 71 -
3.14.4 Violation Detection Phase (Parallel Activities)	- 71 -
3.14.5 Violation Registration and Evidence Capture	- 72 -
3.14.6 Alert Broadcasting and Coordination Phase	- 72 -
3.14.7 Vehicle Interception and Driver Interaction	- 73 -
3.14.8 Immediate Admission Path (No Complaint)	- 73 -
3.14.9 Complaint Handling Path (Office-Based Only)	- 73 -
3.14.10 Payment Processing Phase (OTP-Based)	- 74 -
3.14.11 Case Closure and Archival	- 75 -
3.14.12 Summary of Activity Diagram Strengths	- 75 -
3.15 User Interface Prototyping	- 76 -
3.15.1 Introduction	- 76 -
3.15.2 User Interface Design Principles	- 77 -
3.15.3 UI Prototypes by User Role	- 77 -
3.15.4 UI Navigation Structure	- 82 -
3.15.5 Non-Functional UI Requirements	- 82 -
3.15.6 User Experience (UX) Justification for Examiners	- 82 -
3.16 Supplementary Specifications	- 86 -
3.16.1 Scope and Purpose	- 86 -
3.16.2 Security Requirements	- 86 -

3.16.3 Legal and Regulatory Constraints	- 87 -
3.16.4 Performance Requirements	- 87 -
3.16.5 Reliability and Availability	- 87 -
3.16.6 Data Integrity and Consistency	- 87 -
3.16.7 External Interface and Dependency Constraints	- 88 -
3.16.8 Usability and Human Factors	- 88 -
3.16.9 Maintainability and Scalability	- 88 -
3.16.10 Assumptions and Dependencies	- 88 -
3.17 Change Management Process	- 88 -
CHAPTER FOUR	- 90 -
SYSTEM DESIGN	- 90 -
4.1 Introduction	- 90 -
4.1.1 Design Objectives	- 90 -
4.1.2 Design Scope	- 91 -
4.1.3 Architectural Philosophy	- 91 -
4.1.4 Compliance and Standards Alignment	- 92 -
4.1.5 Design Readiness Statement	- 92 -
4.1.6 Chapter Organization	- 92 -
4.1.7 System Architecture Design	- 93 -
4.2 Class Type Architecture	- 93 -
4.2.1 Overview	- 93 -
4.2.2 Presentation Layer (User Interaction Layer)	- 94 -
4.2.3 Controller / Process Layer	- 95 -
4.2.4 Domain & Intelligence (Business/Domain) Layer	- 96 -
4.2.5 Data Management (Persistence) Layer	- 96 -
4.2.6 Infrastructure & Integration (System) Layer	- 97 -
4.2.7 Architectural Security Principles	- 98 -
4.2.8 Architectural Quality Attributes	- 99 -
4.2.9 Architecture Validation Statement	- 99 -
4.3 Class Modeling	- 99 -
4.3.1 Purpose of Class Modeling	- 99 -
4.3.2 Class Modeling Scope	- 99 -
4.3.3 User Class Hierarchy	- 100 -
4.3.4 Core Domain Classes	- 100 -
4.3.5 Relationship Design Principles	- 102 -
4.3.6 Compliance and Design Quality Considerations	- 102 -
4.4 State Chart Modeling	- 102 -
4.4.1 Introduction	- 102 -
4.4.2 Scope of State Modeling	- 102 -
4.4.3 Violation State Chart – Conceptual Overview	- 103 -
4.4.4 Complaint State Behavior (Embedded)	- 105 -
4.4.5 Design & Compliance Considerations	- 106 -
4.5 Collaboration Modeling	- 106 -
4.5.1 Introduction	- 106 -
4.5.2 Collaboration Modeling Scope	- 106 -
4.5.3 Key Collaboration Participants	- 107 -
4.5.4 Collaboration Scenario 1: Violation Detection & Reporting	- 107 -
4.5.5 Collaboration Scenario 2: Alert Broadcasting & Inter-Subcity Coordination	- 108 -
4.5.6 Collaboration Scenario 3: CCTV-Assisted Evidence Validation	- 109 -

4.5.7 Collaboration Scenario 4: Complaint Handling & Legal Decision	- 110 -
4.5.8 Collaboration Scenario 5: Penalty Payment & Case Closure	- 111 -
4.5.9 Collaboration Integrity & Governance Controls	- 112 -
4.6 Component Modeling	- 112 -
4.6.1 Introduction	- 112 -
4.6.2 Component Modeling Objectives	- 113 -
4.6.3 High-Level Component Architecture	- 113 -
4.6.4 Presentation Components	- 113 -
4.6.5 Control & Orchestration Components	- 114 -
4.6.6 Core Domain Components	- 114 -
4.6.7 Surveillance & Intelligence Components	- 115 -
4.6.8 Financial & External Integration Components	- 115 -
4.6.9 Governance & Audit Components	- 116 -
4.6.10 Component Dependency Rules	- 117 -
4.6.11 Government-Grade Design Assurance	- 117 -
4.6.12 Transition to Next Section	- 117 -
4.7 Deployment Modeling	- 117 -
4.7.1 Introduction	- 117 -
4.7.2 Deployment Design Principles	- 118 -
4.7.3 Deployment Environments	- 118 -
4.7.4 Physical Deployment Nodes	- 119 -
4.7.5 Scalability & Future Deployment Considerations	- 122 -
4.7.6 Government Compliance Assurance	- 122 -
4.8 Persistence Modeling	- 122 -
4.8.1 Introduction	- 122 -
4.8.2 Persistence Design Objectives	- 122 -
4.8.3 Data Categories and Classification	- 123 -
4.8.4 Database Architecture	- 124 -
4.8.5 Core Persistent Entities	- 125 -
4.8.6 Data Lifecycle Management	- 127 -
4.8.7 Data Security Controls	- 127 -
4.8.8 Backup and Disaster Recovery	- 127 -
4.8.9 Compliance and Governance Assurance	- 128 -
4.9 User Interface design	- 128 -
4.9.1 Introduction	- 128 -
4.9.2 UI Design Objectives	- 128 -
4.9.3 UI Architectural Pattern	- 129 -
4.9.4 Common UI Components	- 129 -
4.9.5 Role-Specific UI Designs	- 130 -
4.9.6 UI Validation and Error Handling	- 132 -
4.9.7 Accessibility and Government Standards Compliance	- 132 -
4.9.8 UI Security Considerations	- 132 -
4.9.9 UI Evolution and Future Readiness	- 133 -
4.9.10 UI Design Summary	- 133 -
4.10 System Transition and Deployment Strategy	- 133 -
4.10.1 Introduction	- 133 -
4.10.2 Transition Objectives	- 133 -
4.10.3 Transition Scope	- 134 -
4.10.4 Phased Transition Strategy	- 134 -
4.10.6 Training and Capacity Building	- 136 -

4.10.7 Data Migration and Initialization	- 136 -
4.10.8 Compliance, Governance, and Audit Readiness	- 137 -
4.10.9 Transition Completion Criteria	- 137 -
CONCLUSION	A
RECOMMENDATIONS	C
APPENDIX	E
Appendix A: Actors Summary	E
Appendix B: Legal Basis	E
Appendix C: Assumptions and Limitations	E
Appendix D: Detailed Project Task and Schedule	E
REFERENCES	I

List of Tables

<u>Table</u>	<u>Page No</u>
Table 1.1 Team Composition and Role	3
Table 1.2 Traffic Officer End-User Devices	9
Table 1.3 Data Center Server Infrastructure	10
Table 1.4 Supporting Infrastructure	10
Table 1.5 Connectivity & Electricity (Annual Operational Cost)	10
Table 1.6 Training Cost (All Phases)	10
Table 1.7 Phase one cost table	11
Table 1.8 Phase two cost table	11
Table 1.9 Phase three cost table	12
Table 1.10 Phase four cost table	12
Table 1.11 Total cost breakdown table	13
Table 1.12 Cost reduction	12
Table 1.13 Tools and Technologies Used	18
Table 2.1 Proposed System Addresses Existing Problems	30
Table 2.2 Comparison of Existing and Proposed Systems	34
Table 3.1 Stakeholders and System Interaction	39
Table 3.2 Alignment with Use Cases	58
Table 3.3 Legend and Modeling Conventions	59
Table 3.4 Role and UX Design Rationale	83
Table 4.1 Relationship Between Analysis Artifacts and Design Artifacts	92
Table 4.2 Actor to Layer Mapping	98
Table 4.3 Architectural Support for Phased Deployment	99
Table 4.4 Architectural Benefits Summary	99
Table 4.5 Specialized User Classes	100
Table 4.6 Network & Security Boundaries	121
Table 4.7 Retention and Archival Policy	128
Table 4.8 User Roles and UI Access Scope	129
Table 4.9 Risk Management and Mitigation	137
Table A.1 Summary of Project Tasks, Duration, and Focus	145

List of Figures

<u>Figure</u>	<u>Page No</u>
Fig 3.1 Use case UML Diagram of ALPR STMS, Hawassa _____	45
Fig 3.2 Class UML Diagram ALPR STMS, Hawassa _____	58
Fig 3.3 Security Login Sequence UML Diagram ALPR STMS, Hawassa _____	61
Fig 3.4 Manual Violation Reporting UML Diagram ALPR STMS, Hawassa _____	62
Fig 3.5 CCTV-Assisted Violation Sequence UML Diagram ALPR STMS, Hawassa _____	65
Fig 3.6 Inter-Subcity Coordination Sequence UML Diagram ALPR STMS, Hawassa _____	66
Fig 3.7 Complain Resolution Sequence UML Diagram ALPR STMS, Hawassa _____	67
Fig 3.8 OTP based Payment Sequence UML Diagram ALPR STMS, Hawassa _____	68
Fig 3.9 System Closure Sequence UML Diagram ALPR STMS, Hawassa _____	69
Fig 3.10 Full-System Sequence UML Diagram ALPR STMS, Hawassa _____	70
Fig 3.11 Full-System Activity UML Diagram ALPR STMS, Hawassa _____	76
Fig 4.1: Class-type layered architectural Diagram UML Diagram ALPR STMS, Hawassa _	89
Fig 4.2: Violation State Chart Diagram _____	103
Fig 4.3: Collaboration Diagram – Violation Detection and Reporting _____	108
Fig 4.4: Collaboration Diagram – Alert & Coordination _____	109
Fig 4.5: Collaboration Diagram – CCTV Validation _____	110
Fig 4.6: Collaboration Diagram – Complaint Handling _____	111
Fig 4.7: Collaboration Diagram – Payment _____	112
Fig 4.8: Component Interaction Diagram _____	116
Fig 4.9: Deployment Diagram _____	121
Fig 4.10: Persistence Architecture Diagram _____	127
Figure A.1: Detailed Project Tasks and Schedule – ALPR STMS, Hawassa _____	143

Definitions, Acronyms, and Abbreviations

<u>Term / Acronym</u>	<u>Description</u>
ALPR	Automatic License Plate Recognition
SRS	Software Requirements Specification
CCTV	Closed-Circuit Television
STMS	Smart Traffic Management System
HCA TRDD	Hawassa City Administration Transport and Road Development Department
Subcity	Administrative traffic enforcement zone within Hawassa City
Wanted Vehicle	A vehicle flagged for violations, crimes, or investigations
FR	Functional Requirement
NFR-P0	Non-Functional Requirements of Performance
NFR-UI	Non-Functional Requirements of User Interface
NFR-S	Non-Functional Requirements of Security and Access Permissions
NFR-BR	Non-Functional Requirements of Backup and Recovery
NFR-SM	Non-Functional Requirements of Scalability and Maintainability
UC	Use Case
OTP	One-Time Password
UML	Unified Modeling Language
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective

ABSTRACT

Urban traffic management in rapidly growing in Ethiopia in such cities like Hawassa; has become increasingly complex due to rising vehicle volumes, limited enforcement capacity, and heavy reliance on manual traffic control mechanisms. Existing traffic enforcement practices are largely dependent on human observation, walkie-talkie communication, and fragmented paper-based records, resulting in delayed response to violations, weak inter-subcity coordination, limited traceability of enforcement actions, and inadequate use of digital evidence. These constraints hinder effective traffic regulation, transparency, and long-term planning.

This project presents the design and specification of a **Smart Traffic Management System based on Automatic License Plate Detection and Recognition (ALPR)**, developed to modernize traffic law enforcement in Hawassa City in alignment with national transport regulations and smart city initiatives. The proposed system integrates **computer vision, deep learning, location intelligence, and structured software engineering methodologies** to enhance detection, reporting, coordination, and resolution of traffic violations while preserving legal authority and human decision-making.

The system supports both **manual violation reporting by traffic officers** and **automated CCTV-assisted detection**, enabling a hybrid enforcement model that reflects real-world operational conditions. Detected or reported violations are digitally recorded, enriched with time-stamped location data, visual evidence, and vehicle information, and broadcast to relevant officers and subcity units through a centralized coordination mechanism. Advanced services such as **route prediction and location-based alerting** are incorporated to improve interception efficiency and proactive enforcement without replacing institutional oversight.

A key design principle of the system is strict **role separation and legal compliance**. All human operators—including traffic officers, subcity officers, CCTV operators, complaint officers, and system administrators—access the system through secure authentication and role-based authorization, while drivers interact with the system exclusively through **OTP-based payment channels** after official enforcement decisions. Complaint handling is conducted entirely through traffic offices, where assigned officers review evidence, consult CCTV records or reporting officers, and formally confirm, revoke, or uphold violations before any payment is authorized.

The system architecture follows a **layered, service-oriented design**, incorporating violation management, ALPR processing, penalty calculation, complaint resolution, payment integration, audit logging, and reporting services. All critical actions are recorded in immutable audit logs to ensure accountability, transparency, and evidentiary integrity. The solution is designed for **incremental deployment**, allowing Hawassa City to transition gradually from manual digital reporting to semi-automated and predictive enforcement without disrupting existing operations.

The proposed system complies with **ISO/IEC software quality and usability standards**, public-sector IT governance principles, and the **Road Transport Traffic Control COM Regulation 557-2024**, ensuring regulatory alignment, legal admissibility, and long-term sustainability. By improving enforcement accuracy, operational coordination, and institutional

accountability, this project provides a scalable and legally grounded foundation for intelligent traffic management and contributes meaningfully to Hawassa City's smart city transformation.

Problem – Solution – Impact

Problem

Urban traffic enforcement in Hawassa City is constrained by operational, technological, and institutional limitations. Current practices rely heavily on **manual observation**, **verbal radio communication**, and **paper-based documentation**, which leads to fragmented enforcement records, delayed response to violations, and weak coordination across subcity boundaries. The absence of an integrated digital system makes it difficult to reliably track violations, preserve admissible evidence, or monitor the movement of suspect vehicles across different locations.

Furthermore, existing traffic control mechanisms lack **real-time intelligence**, **automated detection capabilities**, and **predictive support**, limiting proactive enforcement. Complaint handling is time-consuming and inconsistently documented, while payment processes are disconnected from enforcement decisions, raising concerns related to transparency, accountability, and legal defensibility. These challenges collectively undermine effective implementation of traffic regulations, including those mandated under **Road Transport Traffic Control COM Regulation 557-2024**, and restrict the city’s ability to scale toward smart mobility governance.

Solution

To address these challenges, this project proposes an **ALPR-Based Smart Traffic Management System** designed according to modern software engineering principles, public-sector IT governance standards, and applicable transport regulations. The system introduces a **hybrid enforcement model** that combines manual reporting by traffic officers with **automated CCTV-assisted detection** using computer vision and deep learning techniques.

The solution provides a centralized platform for **violation management**, **digital evidence handling**, **location-aware alert broadcasting**, and **route prediction**, enabling coordinated enforcement across subcities. Role-based authentication ensures that all authorized personnel—traffic officers, subcity officers, CCTV operators, complaint officers, and administrators—interact with the system according to their legal mandates, while drivers engage solely through **OTP-based payment mechanisms** after formal validation.

Complaint resolution is fully office-based and evidence-driven, preserving due process and institutional authority. Integrated audit logging, reporting, and data retention mechanisms ensure traceability, compliance, and long-term analytical value. The system is designed for **incremental deployment**, allowing seamless integration with existing CCTV infrastructure without disrupting current security operations.

Impact

The proposed system significantly enhances **enforcement accuracy**, **operational efficiency**, and **institutional accountability** in Hawassa City’s traffic management framework. By

digitizing violation detection, documentation, and coordination, the system reduces response time, minimizes human error, and strengthens evidence-based decision-making.

From a governance perspective, the solution supports **regulatory compliance**, **legal admissibility of digital evidence**, and transparent enforcement aligned with national traffic regulations. Predictive analytics and real-time alerts improve interception success rates, while structured reporting enables informed policy formulation and urban mobility planning.

In the long term, the system contributes to Hawassa City’s smart city objectives, providing a scalable, secure, and sustainable foundation for intelligent traffic enforcement. It also serves as a replicable model for other Ethiopian cities seeking to modernize traffic control systems while maintaining legal rigor, human oversight, and public trust.

CHAPTER ONE

INTRODUCTION

1.1 Overview

Rapid urbanization, population growth, and increasing motorization have significantly transformed transportation systems in Ethiopian cities. Hawassa City, the capital of the Sidama National Regional State, has experienced sustained growth in vehicle ownership, traffic density, and urban mobility demands. While this growth supports economic and social development, it also introduces complex challenges related to traffic congestion, law enforcement, road safety, and inter-agency coordination.

Traditional traffic management approaches—largely dependent on human observation, manual reporting, and analog communication—are increasingly inadequate to address the scale, speed, and complexity of modern urban traffic. These limitations highlight the need for intelligent, technology-driven solutions capable of improving traffic law enforcement, operational efficiency, and public safety.

This document presents the Automatic License Plate Detection and Recognition (ALPR)–Based Smart Traffic Management System proposed for Hawassa City Administration Transport and Road Development Department. The system is designed to modernize traffic enforcement operations by introducing automated vehicle identification, digital evidence management, real-time inter-sub-city coordination, and data-driven decision support.

This chapter provides background information on the responsible organization, the project context, team composition, the problem statement, and the objectives of the proposed system.

1.2 Background of the Organization

The Hawassa City Administration Transport and Road Development Department, operating under the Sidama National Regional State, is the primary government authority responsible for planning, regulating, and managing transportation and road infrastructure within Hawassa City.

The department's key mandates include:

- ✓ Traffic regulation and enforcement
- ✓ Road safety management and accident prevention
- ✓ Vehicle monitoring and control
- ✓ Coordination with traffic police, city security units, and sub-city administrations
- ✓ Development and maintenance of urban transport systems

Within the department, the Traffic and Vehicle Management Sector is responsible for day-to-day traffic operations. This sector oversees traffic police deployment across sub-cities, major intersections, and entry/exit checkpoints. Officers monitor traffic flow, enforce traffic laws, respond to violations, and coordinate incident handling.

Currently, operational coordination relies heavily on:

- ✓ Manual observation by traffic police officers
- ✓ Verbal and radio-based communication using walkie-talkie
- ✓ Paper-based or fragmented digital record-keeping

Although closed-circuit television (CCTV) cameras exist in some locations for general security purposes, they are not integrated into a centralized, intelligent traffic management or vehicle identification system. Consequently, the department lacks a unified digital platform for automated vehicle tracking, violation management, and historical data analysis.

1.3 Background of the Project

Traffic control and law enforcement in Hawassa City remain predominantly human-centered and reactive. When a traffic violation, accident, or vehicle-related offense occurs, the process typically involves:

- ✓ Manual identification of the vehicle by a traffic police officer
- ✓ Verbal reporting of the incident to the nearest sub-city station
- ✓ Broadcasting of vehicle details to other sub-cities or exit checkpoints using walkie-talkie
- ✓ Reliance on officer memory and experience to predict vehicle movement

This approach presents several operational limitations:

- ✓ Lack of digital records and traceability
- ✓ No centralized database of vehicles and violations
- ✓ Difficulty in real-time tracking of wanted or suspect vehicles
- ✓ Absence of automated evidence (images or video) to support enforcement decisions
- ✓ Inefficient coordination across sub-cities and checkpoints

The proposed ALPR-Based Smart Traffic Management System addresses these limitations by introducing an intelligent, integrated, and scalable digital platform. Using computer vision and deep learning technologies, the system automatically detects and recognizes vehicle license plates, records violations with supporting evidence, and distributes real-time alerts to relevant traffic officers and subcity stations based on geographic location, severity, and operational priority.

The system is designed to align with Hawassa City's current operational structure while laying the foundation for future smart city initiatives, including fully automated CCTV-based enforcement and predictive traffic analytics.

1.4 Team Composition

The project is developed by a team of Computer Science students from Infolink College, Hawassa Campus, as part of the Software Engineering course requirement.

The team collaboratively undertakes:

- ✓ System analysis and requirement elicitation
- ✓ Software Requirements Specification (SRS) development

- ✓ System design and architecture modeling

Automatic license plate detection & recognition using deep learning	Automatic License Plate Recognition Based Smart Traffic Management System (ALPR STMS) for Hawassa City				
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Table 1.1 Team Composition and Role

The project follows established software engineering methodologies and best practices to ensure the development of a reliable, maintainable, and well-documented system suitable for real-world deployment and future enhancement.

1.5 Statement of the Problem

Despite ongoing efforts by the Hawassa City Administration Transport and Road Development Department, the existing traffic management and enforcement framework faces significant operational and technical challenges, including:

- ✓ Traffic violations are recorded manually, increasing the likelihood of errors, inconsistencies, and data loss
- ✓ Communication between traffic officers and subcity stations relies on walkie-talkies, which provide no digital logging, audit trails, or prioritization
- ✓ There is no automated mechanism for vehicle identification using license plate recognition
- ✓ Tracking wanted or suspect vehicles across subcities in real time is inefficient and unreliable
- ✓ Existing CCTV infrastructure is not integrated into traffic enforcement or vehicle monitoring workflows

- ✓ Lack of digital evidence (images or video) complicates dispute resolution and complaint handling
- ✓ Penalty calculation, violation history management, and reporting are time-consuming and fragmented

These challenges reduce enforcement effectiveness, delay response times, limit accountability, and hinder proactive traffic management, ultimately affecting road safety and public trust.

1.6 Objectives of the Project

1.6.1 General Objective

The general objective of this project is to design and develop an Automatic License Plate Detection and Recognition (ALPR)–Based Smart Traffic Management System that enhances the operational capacity of the Hawassa City Administration Transport and Road Development Department by:

- ✓ Automating vehicle identification
- ✓ Improving inter-subcity coordination and communication
- ✓ Supporting evidence-based traffic law enforcement
- ✓ Enabling real-time monitoring and intelligent decision-making

The system aims to transform the current manual and radio-based traffic enforcement process into a digitally integrated, intelligent, and scalable platform, capable of supporting present operational needs while enabling future expansion toward comprehensive smart city traffic management.

1.6.2 Specific Objectives

The specific objectives of the project are to:

1. Design and implement a system capable of automatically detecting and recognizing vehicle license plates using deep learning and computer vision techniques
2. Digitally store recognized license plate numbers along with comprehensive vehicle profiles and violation histories
3. Provide traffic police officers with a user-friendly application for reporting and managing traffic violations
4. Generate and broadcast real-time alerts to relevant subcity stations and nearby traffic officers based on geographic location, urgency, and operational priority
5. Support map-based visualization and manual path prediction of suspect or wanted vehicles by reporting officers
6. Assist traffic police in tracking wanted vehicles across multiple subcities in near real time
7. Capture and store digital evidence, including images and video frames, to support traffic violation enforcement and dispute resolution
8. Establish a scalable foundation for future system phases, including full CCTV-based automated detection, predictive vehicle tracking, and advanced traffic analytics.

1.7 Feasibility Analysis

Feasibility analysis evaluates whether the proposed **ALPR-Based Smart Traffic Management System** can be successfully developed, implemented, and sustained within the operational, technical, economic, political, and time constraints of Hawassa City Administration. This analysis ensures that the project is practical, realistic, and aligned with organizational capacity and strategic goals.

1.7.1 Operational Feasibility

Operational feasibility assesses how well the proposed system fits within existing organizational structures, workflows, and operational practices of the Hawassa City Administration Transport and Road Development Department.

The proposed system is **operationally feasible** for the following reasons:

- ✓ The system is designed to **support and enhance existing traffic police workflows**, not replace them.
- ✓ Traffic officers will continue performing enforcement duties, but with **digital tools** for reporting, evidence capture, and communication.
- ✓ Subcity stations will benefit from **centralized dashboards and alert systems**, improving coordination without altering command hierarchies.
- ✓ The system aligns with current enforcement responsibilities such as violation tracking, vehicle monitoring, and inter-subcity communication.
- ✓ Minimal organizational restructuring is required, as the system integrates with existing traffic police roles.

Proper training and phased deployment will further ensure smooth adoption and operational acceptance by traffic officers and administrators.

1.7.2 Technical Feasibility

Technical feasibility evaluates whether the required technology, infrastructure, and expertise are available or attainable to develop and operate the system.

The system is **technically feasible** due to the following factors:

- ✓ ALPR technology using **computer vision and deep learning** is mature and widely used globally.
- ✓ Open-source frameworks (e.g., OpenCV, TensorFlow, PyTorch) support license plate detection and recognition.
- ✓ The system can be deployed using **standard server infrastructure**, with optional cloud or hybrid deployment models.
- ✓ Existing CCTV infrastructure can be integrated incrementally in future phases.
- ✓ Mobile and web-based applications can be developed using well-established technologies.
- ✓ The development team possesses foundational skills in software engineering, databases, and system design, with supervision and learning aligned to academic standards.

The system architecture is modular and scalable, allowing gradual enhancement as technical capacity grows.

1.7.3 Economic Feasibility

Economic feasibility examines whether the benefits of the system justify the costs associated with development, deployment, and operation.

The project is **economically feasible** because:

- ✓ It leverages **existing infrastructure** such as traffic police personnel and partial CCTV installations.
- ✓ Open-source technologies reduce licensing costs.
- ✓ Improved enforcement efficiency can increase traffic fine collection accuracy.
- ✓ Reduced manual work lowers administrative overhead.
- ✓ Long-term benefits outweigh initial investment through improved road safety, reduced violations, and better data-driven planning.

Cost–Benefit Analysis

Benefits:

- ✓ Reduced operational inefficiencies
- ✓ Improved violation detection and enforcement
- ✓ Enhanced public trust through transparency and evidence-based decisions
- ✓ Better traffic planning through historical data analysis
- ✓ Reduced accident rates and enforcement costs over time

Costs:

- ✓ Initial system development
- ✓ Hardware and infrastructure setup
- ✓ Training and maintenance

Overall, the tangible and intangible benefits significantly outweigh the projected costs.

1.7.4 Behavioral / Political Feasibility

Behavioral and political feasibility assesses user acceptance, institutional support, and policy alignment.

The system is **behaviorally and politically feasible** because:

- ✓ It strengthens the authority and effectiveness of traffic police rather than diminishing it.
- ✓ Digital evidence reduces conflict between officers and drivers.
- ✓ Transparency supports fair enforcement, improving public perception.
- ✓ The project aligns with **government digital transformation and smart city initiatives**.
- ✓ No legal or policy barriers exist to prevent system implementation.

Resistance to change may occur initially; however, training, awareness programs, and leadership support will mitigate this risk.

1.7.5 Schedule Feasibility

The proposed project schedule has been carefully evaluated to ensure **realism, achievability, and alignment with academic and institutional constraints**.

✓ The feasibility of the schedule is justified based on the following factors:

a) Phased Development Strategy

The system is structured into **four incremental phases**, with the current development cycle focusing exclusively on **Phase One (Manual Digital Traffic Reporting System)**.

This controlled scope significantly reduces complexity and ensures deliverability within the allocated time frame.

A detailed task-level schedule, including weekly breakdowns, milestones, and phase alignment, is provided in **Appendix D**.

b) Academic Calendar Alignment

The schedule explicitly accounts for:

- ✓ University holiday periods
- ✓ Final examination breaks
- ✓ Reduced productivity windows
- ✓ No critical development tasks are scheduled during these periods, ensuring realistic workload distribution.

c) Institutional Authorization Considerations

Adequate time is allocated for:

- ✓ Document collection
- ✓ Legal and administrative approvals
- ✓ Stakeholder coordination
- ✓ These activities are known to require extended durations in government environments and have been appropriately planned.

d) Progressive Deliverables

Key milestones, particularly the **Software Requirements Specification (SRS) submission**, are positioned early to:

- ✓ Enable formal validation
- ✓ Reduce rework risk
- ✓ Support subsequent design and implementation stages

e) Reusability Across Future Phases

Artifacts produced during this schedule (requirements, UML models, architecture) are designed for reuse in Phases Two through Four, improving long-term efficiency without extending the current timeline.

1.8 Cost–Benefit Analysis

Although the project requires initial development and deployment investment, the long-term benefits significantly outweigh the costs.

Tangible Benefits

- ✓ Reduced time to intercept wanted vehicles
- ✓ Improved violation detection accuracy
- ✓ Reduced manpower requirements
- ✓ Faster penalty processing
- ✓ Reliable Data store

Intangible Benefits

- ✓ Improved public safety
- ✓ Increased transparency and accountability
- ✓ Enhanced public trust in traffic enforcement
- ✓ Foundation for future smart city initiatives

Overall, the system represents a **high return on investment** for the city administration.

1.8.1 Cost of the Project

1.8.1.1 Introduction

The ALPR-Based Smart Traffic Management System (STMS) is a multi-phase government initiative aimed at digitizing traffic violation reporting, integrating CCTV monitoring, predictive path tracking, and advanced automation for smart enforcement. The following cost analysis presents a detailed breakdown of all resources, infrastructure, devices, connectivity, electricity, development, testing, training, and operational expenses for **all four phases**.

This analysis ensures transparency, audit readiness, and budget planning for the Finance Department and other regulatory bodies.

The total cost of the project is minimized by using existing infrastructure and open-source technologies. Costs are categorized as follows:

One-Time Costs

- ✓ System development and customization
- ✓ Initial training programs
- ✓ System integration and configuration
- ✓ Data center build-up

Recurrent Costs

- ✓ System maintenance
- ✓ Model retraining and updates
- ✓ Server maintenance

- ✓ User support and administration
- ✓ Network access

1.8.1.2 Assumptions

- ✓ **Prices** reflect market averages including taxes, installation, and logistics.
- ✓ Imported products are calculated in ETB; local price equivalents used where possible.
- ✓ Quantities are **planning estimates**: e.g., 500 traffic officers, 100 CCTV cameras (distributed over phases).
- ✓ System implementation is incremental:
 - Phase 1: Digital traffic reporting (manual reporting, GPS, OTP payments)
 - Phase 2: CCTV integration (video capture and storage)
 - Phase 3: Route prediction with partial CCTV expansion
 - Phase 4: Smart CCTV automation (full city-scale video monitoring & rule-based detection)
- ✓ **Training costs** apply across all phases.
- ✓ Tablets provided for officers; personal devices are not used.
- ✓ Annual operational costs (connectivity & electricity) are recurring per phase allocation.

1.8.1.3 Core Infrastructure

Item	Description	Quantity	Unit Cost (ETB)	Total Cost (ETB)
Android Tablets	Enterprise, GPS + LTE, 6–8GB RAM	500	24,750	12,375,000
Rugged Covers & Mounts	Shock-resistant	500	2,750	1,375,000
Subtotal – Officer Devices				13,750,000

These devices are mandatory from Phase One and remain valid for subsequent phases.

Table 1.2 Traffic Officer End-User Devices

Server Role	Model	CPU	RAM	Storage (Initial → Phase 4)	Primary Function	Avg. Cost (ETB)
Application Server	Dell PowerEdge R750	Dual Intel Xeon	128GB	2TB NVMe → 4TB	APIs, services, dashboards	900,000
Database Server	HPE DL380 Gen10	Dual Xeon	256GB	4TB SSD → 8TB RAID	Violation, payment, logs	890,000
Analytics Server	Lenovo SR665 V3	Dual AMD EPYC	256GB	4TB SSD → 8TB	Prediction & analytics	1,150,000
GPU	GPU	Multi-	256GB	6TB NVMe	ALPR, video	3,000,000

Server Role	Model	CPU	RAM	Storage (Initial → Phase 4)	Primary Function	Avg. Cost (ETB)
Processing Server	Optimized Server	core + GPU		→ 12TB	processing	
Storage Server	Synology RS4021xs+	Multi-core	64GB	80TB → 200TB	Video & evidence storage	1,700,000
Subtotal - Servers						7,640,000

Table 1.3 Data Center Server Infrastructure

Item	Description	Cost (ETB)
Network Switches (Core & Access)	Managed, enterprise-grade	450,000
Structured Cabling & Fiber	Server room + CCTV backbone	1,100,000
Server Racks	Lockable, ventilated	280,000
UPS & Power Conditioning	Redundant protection	420,000
Cooling & Ventilation	Standard server room	350,000
Subtotal – Supporting Infrastructure		2,600,000

Table 1.4 Supporting Infrastructure

Item	Description	Annual Cost (ETB)
Mobile Internet (Traffic Officers)	500 × 14,000 ETB yearly (Unlimited)	7,000,000
Fixed Broadband (Data Center)	Fixed BB 50Mbps for 5255 ETB monthly	63,060
Electricity (Servers & Cooling)	Based on EEU tariff	180,000
Subtotal – Connectivity & Electricity		7,243,060

Table 1.5 Connectivity & Electricity (Annual Operational Cost)

Training Type	Description	Cost (ETB)
Traffic Officer Training	System usage, reporting, OTP payment	200,000
Administrative Training	System administration, monitoring	200,000
Manuals & Materials	Guides, cheat sheets	100,000
Refresher & Updates	Phase updates & optimization	150,000
Total Training Cost		650,000

Table 1.6 Training Cost (All Phases)

1.8.1.4 Phase-Wise Cost Breakdown

Phase One – Digital Traffic Reporting System

Scope:

Manual traffic violation reporting, GPS capture, subcity alerts, OTP-based payments. No CCTV integration yet.

Cost Component	Description	Cost (ETB)
Officer Devices	Tablets & covers (500 units)	13,750,000
Application Server	Core backend services	900,000
Database Server	Violation & payment records	890,000
Supporting Infrastructure	Network switches, racks, UPS	2,600,000
Connectivity & Electricity (Annual)	Mobile internet, fixed broadband, power	7,243,060
System Design & Architecture	Requirements & models	700,000
System Development	Core system build	1,800,000
Testing & Quality Assurance	Functional & security	600,000
Training	Included from section 1.8.2.E	650,000
Phase One Total		28,133,060

Table 1.7 Phase one cost table

Phase Two – CCTV Integration

Scope:

Introduction of CCTV feeds and storage server. Storage server now used. Cameras initially installed (10–20 units). System integration software, design, testing, and load validation included.

Cost Component	Description	Cost (ETB)
GPU Server	Video ingestion	3,000,000
Storage Server	NAS for CCTV data	1,700,000
CCTV Integration Software	Video streaming ingestion	1,800,000
Development & Testing	Integration & QA	1,600,000
Supporting Infrastructure	Additional cabling & racks	600,000
Connectivity & Electricity	Annual allocation	7,243,060
Training	Included from section 1.8.2.E	650,000
Phase Two Total		16,593,060

Table 1.8 Phase two cost table

Phase Three – Route Prediction & Partial CCTV Expansion

Scope:

- ✓ 40 additional CCTV cameras installed
- ✓ Predictive path algorithms designed, trained, and validated
- ✓ Storage expansion for new video feeds

Cost Component	Description	Cost (ETB)
Additional CCTV Cameras	40 units, enterprise-grade	1,680,000
Predictive Model	Algorithm design & training	2,400,000
Storage Expansion	Additional NAS capacity	1,400,000
Development & Testing	Model validation & integration	800,000
Connectivity & Electricity	Annual allocation	7,243,060
Training	Included from section 1.8.2.E	650,000
Phase Three Total		14,173,060

Table1.9 Phase three cost table

Phase Four – Smart CCTV Automation

Scope:

- ✓ Additional 60 CCTV cameras
- ✓ Full GPU server utilization for ALPR and real-time processing
- ✓ Rule-based detection system fully operational
- ✓ Stress testing, load balancing, optimization, and city-scale monitoring

Cost Component	Description	Cost (ETB)
CCTV Cameras	60 units, high-end	2,700,000
GPU Server Expansion	High-volume real-time processing	2,800,000
Automation Software	Rule-based detection & alerting	2,100,000
Storage Expansion	High-resolution video storage	1,400,000
Development, Testing & Optimization	Load & system validation	1,100,000
Connectivity & Electricity	Annual allocation	7,243,060
Training	Included from section 1.8.2.E	650,000
Phase Four Total		17,993,060

Table1.10 Phase four cost table

1.8.2 Consolidated Total Cost

Phase	Cost (ETB)
Phase One	28,133,060
Phase Two	16,593,060
Phase Three	14,173,060
Phase Four	17,993,060
Grand Total	76,892,240

Table 1.11 Total cost breakdown table

1.8.3 Estimated Cost Reductions

Cost Area	Annual Savings (ETB)
Elimination of Traffic Charge Pads	1,200,000
Reduced Fuel Consumption	2,800,000
Reduced Manual Coordination	1,500,000
Faster Complaint Resolution	1,700,000
Total Estimated Annual Savings	7,200,000

Table 1.12 Cost reduction

1.9 Scope and Significance of the Project

1.9.1 Scope of the Project

The scope of the ALPR-Based Smart Traffic Management System includes:

- ✓ Automated and manual license plate recognition
- ✓ Digital traffic violation reporting and management
- ✓ Centralized vehicle and violation database
- ✓ Real-time alerts and inter-subcity coordination
- ✓ Digital evidence storage (images and video frames)
- ✓ Map-based visualization and tracking support

Out of scope for the initial phase:

- ✓ Fully autonomous enforcement without human intervention
- ✓ National-level vehicle registration integration
- ✓ Advanced AI-based behavior prediction (future phase)

1.9.2 Significance of the Project

The project is significant because it:

- ✓ Modernizes traffic enforcement operations in Hawassa City
- ✓ Improves road safety and compliance with traffic laws
- ✓ Enhances transparency, accountability, and public trust

- ✓ Supports data-driven traffic planning and policy decisions
- ✓ Lays the foundation for smart city traffic management systems

1.10 Target Beneficiaries of the System

The primary and secondary beneficiaries of the system include:

Primary Beneficiaries

- ✓ Hawassa City Administration Transport and Road Development Department
- ✓ Traffic police officers and subcity stations
- ✓ City traffic management authorities

Secondary Beneficiaries

- ✓ Road users and vehicle owners
- ✓ City residents and pedestrians
- ✓ Law enforcement and security agencies
- ✓ Urban planners and policy makers

Long-Term Beneficiaries

- ✓ Hawassa City Administration
- ✓ Sidama National Regional State
- ✓ Future smart city initiatives

1.11 Methodology for the Project

The methodology defines the systematic and disciplined approach followed in the planning, analysis, design, development, testing, and implementation of the **Automatic License Plate Detection and Recognition (ALPR) Based Smart Traffic Management System**. Given the safety-critical nature of traffic systems and the involvement of multiple operational stakeholders, a clearly defined methodology is essential to ensure correctness, reliability, scalability, and sustainability.

The proposed methodology integrates **software engineering best practices**, **incremental development**, and **reuse-oriented design**, allowing the system to evolve gradually from a manual digital support tool into a fully automated intelligent traffic management platform. This approach minimizes operational disruption, reduces risk, and ensures continuous validation with real-world traffic operations in Hawassa City.

1.11.1 Data Source

The project uses both **primary and secondary data sources** to ensure that the developed system accurately reflects real traffic management practices, legal requirements, and technical constraints.

Primary data is collected directly from the **Hawassa City Administration Transport and Road Development Department**, particularly from traffic police officers, subcity coordinators, and supervisory officials. These stakeholders provide practical insights into

daily enforcement activities, reporting procedures, communication flows, and coordination challenges among subcities.

Secondary data sources include Ethiopian road traffic regulations, penalty and violation documents, administrative directives, academic research papers on ALPR and intelligent transportation systems, and software engineering reference materials. These sources ensure that the system design complies with legal frameworks and aligns with established technical standards.

1.11.2 Fact-Finding Techniques

To obtain complete, accurate, and verifiable system requirements, multiple fact-finding techniques are applied.

Interview

Structured and semi-structured interviews are conducted with traffic police officers, subcity station heads, and administrative personnel. The interviews focus on understanding how violations are currently identified, how information is shared, how wanted vehicles are tracked, and what challenges officers face during real enforcement operations. Interview results are used to define functional requirements, user roles, and system constraints.

Practical Observation

Practical observation is carried out at selected intersections, checkpoints, and subcity stations. By observing traffic control activities directly, the development team gains insight into response times, communication delays, coordination gaps, and operational limitations that are not always expressed verbally. This ensures the system addresses actual operational realities.

Document Analysis

Existing documents such as violation forms, penalty records, traffic rules, and reporting templates are analyzed. This helps identify mandatory data fields, penalty structures, evidence requirements, and compliance obligations that the system must support.

1.11.3 System Analysis and Design Approach

The project follows a **structured system analysis approach** combined with **object-oriented analysis and design (OOAD)**. During analysis, system requirements are identified, categorized, validated, and documented to ensure clarity and completeness.

During the design phase, **Unified Modeling Language (UML)** is used to model system behavior and structure. UML diagrams provide a visual representation of user interactions, data flow, and system components. These models serve as a blueprint for development and ensure a shared understanding among stakeholders.

To manage complexity and support future expansion, the system adopts a **hybrid Incremental and Reuse-Oriented Development Model**. Existing libraries, frameworks, and proven algorithms are reused where appropriate to reduce development time and improve reliability.

1.11.4 Overview of Project Development Phases

The system is developed in **four clearly defined and logically progressive phases**. Each phase builds upon the previous one, allowing gradual improvement of intelligence, automation, and system capability.

Phase One: Digital Manual Reporting and Intelligent Alert Broadcasting

Phase One focuses on replacing the existing **walkie-talkie-based communication system** with a **digital, location-aware reporting and alert platform**. In this phase, traffic police officers manually enter traffic violation information into the system using a mobile or desktop interface.

Once a violation is reported, the system automatically records:

- ✓ License plate number
- ✓ Violation type
- ✓ Time and date
- ✓ Geographic location of the reporting officer

The system then broadcasts alerts digitally to:

- ✓ The relevant subcity station
- ✓ Nearby traffic officers
- ✓ All other subcities (with different priority levels)

Officers can also draw an estimated escape path on a digital map. Officers located within the defined radius or along the drawn path receive **high-priority alerts**, while others receive general notifications. This phase establishes the digital foundation of the system while remaining fully compatible with existing enforcement practices.

Phase Two: CCTV-Assisted Vehicle Detection and Location Updates

Phase Two introduces **CCTV-assisted intelligence** without disrupting the existing city security surveillance system. CCTV feeds are routed to a central control environment where the ALPR system analyzes video frames for license plates marked as wanted.

When a wanted vehicle is detected:

- ✓ The system extracts the license plate
- ✓ Confirms it against the database
- ✓ Records the camera location and timestamp
- ✓ Broadcasts a new alert to nearby officers

A **circular alert radius** is generated around the detection point, prioritizing officers most likely to intercept the vehicle. This phase significantly improves tracking efficiency and reduces dependence on human observation alone.

Phase Three: Predictive Route Intelligence and Proactive Enforcement

Phase Three adds **predictive intelligence** to the system. Using historical movement data, road network structure, traffic patterns, and officer input, the system predicts possible routes a wanted vehicle may take.

Based on these predictions:

- ✓ Officers positioned along likely routes receive **priority alerts**
- ✓ Long-range alerts are issued ahead of the vehicle's potential movement
- ✓ Response coordination becomes proactive rather than reactive

This phase enables smarter deployment of enforcement resources and increases the probability of successful interception.

Phase Four: Fully Automated Smart Traffic Enforcement

Phase Four represents the highest level of system intelligence. In this phase, the system automatically detects traffic violations using smart cameras without manual reporting.

Violations such as red-light crossing, wrong turn, or restricted-lane usage are:

- ✓ Detected automatically
- ✓ Documented with image/video evidence
- ✓ Linked to vehicle and owner information
- ✓ Processed for penalty calculation

Officers intervene only for confirmation, enforcement, and dispute handling. This phase significantly reduces manual workload and supports scalable smart city traffic management.

Activity / Function	Tools, Frameworks, and Technologies
User Interface Development	HTML5, CSS3, JavaScript (Responsive Web Design Standards)
Mobile Application Development	Flutter (Cross-Platform: Android & iOS)
Backend Application Development	Python (FastAPI or Flask – RESTful API Architecture)
Computer Vision Processing	OpenCV (Image Processing and Video Stream Handling)
Deep Learning & AI Models	YOLO (Object Detection), Convolutional Neural Networks (CNN)
Development Database Environment	Firebase (Authentication, Realtime Database, Prototyping)
Production / Deployment Database	City Data Center Servers (PostgreSQL or MySQL – Relational DBMS)
Mapping & Location Services	OpenStreetMap / Google Maps API (GPS Tracking and Route Visualization)
Notification & Alert System	Firebase Cloud Messaging (FCM), SMS Gateway (OTP & Alerts)
Operating Platforms	Linux (Primary), Windows, macOS

Activity / Function	Tools, Frameworks, and Technologies
Integrated Development Environments (IDEs)	Visual Studio Code, PyCharm
Documentation & Reporting Tools	Microsoft Word, Microsoft Excel
User Training & Capacity Building Tools	Microsoft PowerPoint, Recorded Video Tutorials

Table 1.13 Tools and Technologies Used

These tools are selected for reliability, scalability, and compatibility with intelligent transportation systems.

1.11.5 Testing Procedures

Testing is a critical activity in the development of the **ALPR-Based Smart Traffic Management System** because the system directly supports law enforcement decisions, public safety, and legal accountability. Any malfunction, incorrect detection, or delayed alert could lead to operational failure or public mistrust. Therefore, a **multi-layered and continuous testing strategy** is adopted throughout all development phases.

Unit Testing

Unit testing is conducted at the component level to verify the correctness of individual modules. Each functional unit—such as license plate detection, plate number recognition, violation record creation, alert broadcasting, route prediction, and penalty calculation—is tested independently using controlled inputs. This ensures that each module produces accurate and expected outputs before integration.

Integration Testing

Integration testing ensures that individual modules interact correctly when combined. For example, the integration between the CCTV feed processing module and the alert broadcasting module is tested to confirm that a detected vehicle automatically triggers location-based notifications. This phase verifies data consistency, communication reliability, and error handling between system components.

System Testing

System testing evaluates the complete system as a unified whole under realistic operational conditions. Simulated traffic scenarios, multiple concurrent violation reports, and high alert volumes are used to test system performance, response time, scalability, and fault tolerance. This ensures the system can handle real-world traffic conditions without degradation.

User Acceptance Testing (UAT)

User Acceptance Testing is conducted with selected traffic police officers, subcity coordinators, and system operators. Their feedback is used to validate usability, clarity of alerts, accuracy of predictions, and ease of reporting. UAT ensures the system aligns with operational workflows and does not introduce unnecessary complexity.

Validation and Evidence Accuracy Testing

Special attention is given to validating image and video evidence used for violations. This includes testing image clarity, timestamp accuracy, location tagging, and plate recognition confidence scores. Manual override and verification options are tested to ensure human judgment can correct system errors.

1.11.6 Real World Implementation Strategy

The implementation of the system follows a **phased, controlled, and risk-aware deployment strategy** designed to minimize disruption to existing traffic operations and ensure smooth adoption by enforcement personnel.

Incremental Deployment Approach

Rather than deploying the entire system at once, the system is implemented incrementally. Each phase is deployed only after successful testing and validation of the previous phase. This allows early detection of issues, gradual training of users, and progressive improvement of system reliability.

Parallel Operation with Existing System

During the initial deployment stages, the system operates in parallel with the existing walkie-talkie communication system. This ensures continuity of enforcement operations and provides a fallback mechanism in case of technical failure. Over time, reliance on manual communication is reduced as system confidence increases.

Pilot Area Implementation

The system is first deployed in selected subcities and high-traffic areas. Performance metrics such as response time, interception success rate, and officer satisfaction are evaluated before city-wide rollout.

Training and Change Management

Structured training sessions are provided to traffic officers, CCTV operators, and administrators. Training includes system usage, interpretation of alerts, handling false positives, and escalation procedures. This ensures effective adoption and reduces resistance to change.

1.11.7 Required Resources and Cost Analysis

The successful implementation of the project requires a combination of **technical, human, and organizational resources**. The project prioritizes cost efficiency by leveraging existing infrastructure wherever possible.

Human Resources

- ✓ Software developers (backend, frontend, UI UX, AI)
- ✓ System analysts and designers

- ✓ Traffic enforcement domain experts
- ✓ CCTV operators and supervisors
- ✓ System administrators

Hardware Resources

- ✓ CCTV cameras (Existing plus new additional)
- ✓ Central servers (city data center)
- ✓ Workstations for control centers
- ✓ Mobile devices for traffic officers
- ✓ Network infrastructure

Software Resources

Most software components are open-source or freely available, reducing licensing costs. Firebase is used for development, while deployment relies on government-owned servers.

1.11.8 Risk Analysis, Assumptions, and Constraints

Risks

- ✓ Poor image quality from some CCTV cameras
- ✓ Network connectivity issues
- ✓ False positives in automated detection
- ✓ Resistance to technology adoption

Assumptions

- ✓ Cooperation from traffic and security departments
- ✓ Availability of legal approvals
- ✓ Access to CCTV feeds

Constraints

- ✓ Limited camera coverage
- ✓ Budgetary limitations
- ✓ Legal and privacy regulations

Risk mitigation strategies include manual verification, phased deployment, and policy compliance.

1.11.9 Limitation of the Project

Despite its advanced capabilities, the system has limitations. Full automation depends on camera quality, coverage, and lighting conditions. Certain violations still require human judgment. However, the system is designed to evolve and improve over time through updates and infrastructure expansion.

1.11.10 Ethical, Legal, and Privacy Considerations

The system strictly follows Ethiopian traffic laws and data protection principles. Access control, role-based permissions, encrypted data storage, and audit logs ensure privacy and prevent misuse. Public awareness and transparency are emphasized to maintain trust.

CHAPTER TWO

DESCRIPTION OF THE EXISTING SYSTEM

2.1 Introduction to the Existing System

The existing traffic management and law enforcement system in **Hawassa City** is predominantly **manual, reactive, and communication-driven**, relying heavily on **human observation, physical presence, and verbal information exchange**. Traffic enforcement activities are carried out by traffic police officers stationed at major roads, intersections, roundabouts, and checkpoints throughout the city.

Traffic police officers are responsible for observing traffic flow, identifying violations, stopping offending vehicles where possible, and reporting incidents to their respective subcity traffic stations. These activities are largely dependent on individual judgment, visibility conditions, and the immediate physical reach of officers.

Although Hawassa City has deployed **closed-circuit television (CCTV) cameras** at selected strategic locations, the primary purpose of these cameras is **general city security and crime monitoring**, not traffic law enforcement. CCTV feeds are managed by the city police security unit and are not integrated into a dedicated traffic management or intelligent transportation system. Consequently, traffic enforcement operations do not benefit from real-time automated surveillance support.

Furthermore, the existing system lacks a **centralized digital platform** for storing and managing traffic violations, vehicle information, enforcement history, or penalty records. Most operational data is communicated verbally and recorded manually, significantly limiting the system's **efficiency, accuracy, consistency, scalability, and auditability**.

2.2 Stakeholders and Players in the Existing System

The existing traffic management system involves multiple stakeholders who interact through **loosely coordinated and predominantly manual processes**.

2.2.1 Traffic Police Officers

Traffic police officers serve as the frontline enforcers of traffic laws in Hawassa City. Their primary responsibilities include:

- ✓ Monitoring traffic behavior
- ✓ Identifying traffic violations such as speeding, illegal parking, reckless driving, and failure to obey traffic signals
- ✓ Taking immediate enforcement action when possible

When a violation occurs, officers attempt to physically stop the offending vehicle. If the vehicle escapes, the officer communicates incident details—such as the license plate number, vehicle description, and location—via **walkie-talkie** to the relevant subcity traffic station.

Officers rely entirely on:

- ✓ Personal judgment
- ✓ Physical visibility
- ✓ Manual memory and verbal descriptions

There is no automated support for license plate identification, route prediction, structured alert dissemination, or evidence capture.

2.2.2 Subcity Traffic Stations

Hawassa City is administratively divided into several subcities, each operating its own traffic control station. Subcity traffic stations function as **local coordination centers** responsible for:

- ✓ Receiving violation reports from traffic officers within their jurisdiction
- ✓ Relaying information to other subcity stations when a vehicle crosses administrative boundaries
- ✓ Managing local enforcement records

Communication between subcity stations is conducted verbally through radio communication. This process:

- ✓ Lacks standardized reporting formats
- ✓ Is dependent on officer availability
- ✓ Does not provide digital traceability or prioritization

Despite these limitations, subcity stations play a critical role in maintaining jurisdictional order and basic coordination.

2.2.3 City Traffic and Transport Administration

At the city level, the **Hawassa City Administration Transport and Road Development Department** provides strategic oversight, policy direction, and administrative supervision of traffic enforcement activities.

The department is responsible for:

- ✓ Implementing traffic regulations
- ✓ Managing penalty structures
- ✓ Supervising subcity traffic operations
- ✓ Ensuring regulatory compliance

However, due to the absence of centralized digital systems, the department has **limited real-time visibility** into field operations, enforcement efficiency, and violation trends.

2.2.4 CCTV Surveillance Unit

The CCTV surveillance infrastructure in Hawassa City is managed by the **city police security department**. Camera operators monitor live feeds primarily for:

- ✓ Crime prevention
- ✓ Public safety monitoring
- ✓ General security incidents

Traffic monitoring is not a primary responsibility of the CCTV unit, and there is **no automated traffic violation detection or license plate recognition** mechanism in place.

Traffic police may request CCTV footage after an incident, but this process is:

- ✓ Manual
- ✓ Time-consuming
- ✓ Reactive rather than proactive

2.2.5 Drivers and Vehicle Owners

Drivers and vehicle owners interact with the existing system only when:

- ✓ Physically stopped by traffic police
- ✓ Summoned to traffic offices for violations
- ✓ Submitting complaints or appeals

They do not receive:

- ✓ Real-time alerts
- ✓ Digital penalty notifications
- ✓ Automated enforcement updates

All interactions related to penalties, payments, complaints, and appeals are conducted **manually at traffic offices**.

2.3 Major Functions and Activities of the Existing System

The existing traffic management system operates through a sequence of **manual inputs, processes, and outputs**.

2.3.1 Inputs

The primary inputs to the system include:

- ✓ Visual observation of traffic violations by officers
- ✓ Verbal communication of license plate numbers and incident details
- ✓ Physical driver documents (driver's license, vehicle registration)
- ✓ Handwritten notes and paper-based reports

These inputs are often:

- ✓ Incomplete
- ✓ Inconsistent
- ✓ Prone to human error
- ✓ Affected by environmental conditions such as weather and visibility

2.3.2 Processes

Key operational processes include:

- ✓ Manual identification of traffic violations
- ✓ Physical attempts to stop offending vehicles
- ✓ Verbal reporting via walkie-talkie
- ✓ Subcity-level coordination
- ✓ Manual determination of penalties
- ✓ Temporary physical confiscation of driver licenses

These processes are **sequential, slow, and highly dependent on human coordination**, making them difficult to scale and standardize.

2.3.3 Output

The primary outputs of the existing system include:

- ✓ Verbal alerts communicated to other officers
- ✓ Paper-based violation records
- ✓ Physical penalty notices
- ✓ Manual payment receipts

There is no centralized digital repository or searchable system for outputs, limiting historical analysis and reporting.

2.4 Business Rules Governing the Existing System

Traffic enforcement in Hawassa City operates under **Ethiopian federal and regional traffic laws**, including regulations issued by the Federal Transport Authority and implemented at the city level.

Key business rules include:

- ✓ Traffic violations are categorized based on type and severity
- ✓ Certain violations result in monetary penalties, while others include penalty points
- ✓ Repeat offenses lead to increased penalties
- ✓ Traffic police officers have the authority to temporarily confiscate driver licenses
- ✓ Drivers have the right to submit complaints and appeals
- ✓ Enforcement jurisdiction is divided among subcities, requiring inter-subcity cooperation

Although these rules are legally well-defined, their enforcement under the existing system is:

- ✓ Manual
- ✓ Inconsistent
- ✓ Difficult to audit
- ✓ Largely undocumented digitally

There is no automated rule validation, digital penalty calculation, or real-time monitoring of repeat offenders.

2.5 Reports Generated in the Existing System

Reporting within the existing traffic management system is **manual, fragmented, and decentralized**, with most reports maintained in physical form.

2.5.1 Types of Reports

The main reports generated include

- ✓ **Daily Traffic Violation Reports:** Prepared by individual officers or subcity stations, summarizing daily violations by type and frequency.
- ✓ **Incident and Escape Reports:** Generated when a vehicle commits a serious violation and escapes. These reports contain manually recorded details such as license plate number, vehicle description, location, and time.
- ✓ **Penalty and Payment Reports:** Document penalties imposed and payments collected. Records are maintained using physical receipts and manual logs.
- ✓ **License Confiscation Reports:** Prepared when driver licenses are temporarily confiscated due to severe violations.
- ✓ **Complaint and Appeal Reports:** Created when drivers submit complaints or appeals against penalties. These records are stored and processed manually.

2.5.2 Limitations of Existing Reports

The current reporting mechanism suffers from several limitations:

- ✓ Reports are not available in real time
- ✓ No centralized database exists for historical analysis
- ✓ Repeat offenders are difficult to identify
- ✓ Reports cannot be easily shared across subcities
- ✓ High risk of data loss, duplication, and inconsistency

These limitations significantly reduce enforcement effectiveness and make **data-driven traffic management and planning nearly impossible**.

2.6 Documentation and Forms Used in the Existing System

Documentation in the existing traffic management system of Hawassa City is **entirely manual**, relying on paper-based forms that are physically completed, stored, and retrieved at subcity or city traffic offices.

2.6.1 Common Forms Used

The most frequently used forms in the existing system include:

- ✓ **Traffic Violation Ticket Forms** – used to record details of traffic offenses

- ✓ **Vehicle and Driver Information Forms** – used to capture driver identity and vehicle registration details
- ✓ **License Confiscation Slips** – issued when licenses are temporarily confiscated
- ✓ **Payment Receipts** – issued for penalty payments
- ✓ **Complaint and Appeal Forms** – used by drivers to challenge penalties

All forms are filled manually and archived in **physical filing systems**, with no digital backup or centralized repository.

2.6.2 Issues with Manual Documentation

Manual documentation introduces several critical challenges:

- ✓ Handwritten forms are prone to errors and illegibility
- ✓ Physical storage consumes significant office space
- ✓ Retrieval of historical records is time-consuming and inefficient
- ✓ Documents are vulnerable to damage, loss, or misplacement
- ✓ Lack of standardized formats across subcities leads to inconsistency

The absence of digitized documentation **prevents automation, data integration, and real-time information sharing** across enforcement units.

2.7 Bottlenecks of the Existing System (PIECES Framework)

To systematically analyze weaknesses in the existing system, the **PIECES Framework**—covering **Performance, Information, Economy, Control, Efficiency, and Service**—is applied.

2.7.1 Performance

- ✓ Slow response to traffic violations
- ✓ Delayed communication between subcities
- ✓ Absence of real-time alerting mechanisms
- ✓ Inability to track escaped vehicles promptly

2.7.2 Information (Input and Output Issues)

- ✓ Inaccurate or incomplete license plate recordings
- ✓ Missing or inconsistent violation details
- ✓ Redundant data entry across multiple offices
- ✓ No standardized reporting formats or validation

2.7.3 Security and Control

- ✓ Unauthorized access to paper-based records
- ✓ Absence of role-based access control
- ✓ No audit trails or activity logs
- ✓ High risk of record manipulation or tampering

2.7.4 Efficiency

- ✓ High manpower requirements for routine tasks
- ✓ Excessive time spent searching and verifying records
- ✓ Manual coordination across offices
- ✓ Reduced productivity during peak traffic hours

2.7.5 Service

- ✓ Poor service delivery to drivers
- ✓ Long waiting times for complaints and appeals
- ✓ Limited transparency in penalty handling
- ✓ No digital notifications or confirmations

2.8 Practices to Be Preserved

Despite its limitations, the existing system contains **essential practices** that must be preserved to ensure legal validity, administrative continuity, and user acceptance.

2.8.1 Legal and Administrative Compliance

- ✓ Enforcement based on federal and regional traffic laws
- ✓ Official authority of traffic police officers
- ✓ Established penalty structures and violation classifications

2.8.2 Subcity-Based Administrative Structure

- ✓ Jurisdictional enforcement by subcities
- ✓ Existing communication hierarchy
- ✓ Clear responsibility allocation

2.8.3 Manual Verification Authority

- ✓ Human judgment in violation confirmation
- ✓ Supervisor approval mechanisms
- ✓ Driver rights to complaint and appeal

2.8.4 Existing Communication Culture

- ✓ Alert broadcasting during critical incidents
- ✓ Inter-subcity collaboration and coordination

These practices ensure **fairness, legality, and administrative acceptance** and are integrated into the proposed system rather than replaced.

2.9 Proposed Solution for the New System

To overcome the limitations of the existing system, this project proposes an **Automatic License Plate Detection and Recognition–Based Smart Traffic Management System**

(ALPR-STMS) tailored for the **Hawassa City Administration Transport and Road Development Department**.

The proposed system **digitizes, automates, and enhances** current practices while preserving the existing legal, administrative, and operational framework.

2.9.1 Overview of the Proposed System

The proposed system introduces a **centralized digital platform** integrating:

- ✓ Traffic police officers
- ✓ Subcity traffic offices
- ✓ CCTV surveillance systems
- ✓ Central control center administrators
- ✓ Drivers and payment gateways

The system supports **manual, semi-automated, and fully automated enforcement**, enabling **incremental deployment** based on infrastructure readiness.

2.9.2 Key Features of the Proposed System

1. Digital Violation Reporting

- ✓ Officers submit violations via mobile or web applications
- ✓ Plate numbers, violation types, and locations are digitally recorded
- ✓ Image and CCTV evidence are attached where available

2. Centralized Vehicle and License Plate Database

- ✓ Digital storage of vehicle and owner records
- ✓ License plate numbers serve as unique identifiers
- ✓ Instant retrieval of violation history

3. Smart Alert Broadcasting

- ✓ Automatic alerts to nearby officers and subcities
- ✓ Priority alerts based on radius and predicted routes
- ✓ Border alerts for vehicles exiting city limits

4. CCTV-Assisted Detection (Phases 2 and 3)

- ✓ Automated analysis of CCTV feeds
- ✓ License plate detection and verification
- ✓ Manual validation and override by operators

5. Predictive Path Tracking

- ✓ Officers define estimated escape routes on maps
- ✓ System generates radius-based and route-based alerts
- ✓ Predictive intelligence improves interception success

6. Automated Penalty Processing (Phase 4)

- ✓ Automatic calculation of penalties and points
- ✓ Direct linkage to traffic regulations
- ✓ Digital notification and payment requests

7. Complaint and Appeal Management

- ✓ Digital submission of complaints
- ✓ Evidence-based review process
- ✓ Penalties may be confirmed, modified, or dismissed

Existing Problem	Proposed Solution
Manual reporting	Digital reporting via mobile/web
Delayed communication	Real-time alert broadcasting
No vehicle tracking	ALPR with predictive routing
Data inconsistency	Centralized digital database
Weak evidence	Image and video-based proof
Poor transparency	Digital logs and audit trails

Table 2.1 Proposed System Addresses Existing Problems

2.9.3 Alignment with Existing Practices

The proposed system:

- ✓ Preserves subcity jurisdictional authority
- ✓ Maintains human decision-making and verification
- ✓ Provides manual fallback during technical failures
- ✓ Fully complies with federal and city traffic regulations

2.10 Functional Requirements

Functional requirements define what the system must do to support traffic law enforcement, coordination, and administrative control. These requirements are derived from stakeholder needs, operational workflows, and system objectives of the ALPR-Based Smart Traffic Management System.

2.10.1 Performance Requirements

These requirements define how the system must perform in terms of responsiveness, processing, and operational behavior.

- ✓ **FR-07** ~ The system shall broadcast violation alerts to nearby traffic officers based on geographic proximity.
- ✓ **FR-08** ~ The system shall notify relevant subcity traffic offices when violations require escalation.

- ✓ **FR-09** ~ The system shall support priority-based alerting for high-risk or escaped vehicles.
- ✓ **FR-11** ~ The system shall prioritize alerts for officers located along predicted escape paths.
- ✓ **FR-13** ~ The system shall automatically extract license plate numbers from camera feeds when available.
- ✓ **FR-14** ~ The system shall compare detected plates against active violation and wanted vehicle records.
- ✓ **FR-17** ~ The system shall calculate penalty fees based on violation type and predefined traffic rules.

2.10.2 Process Requirements

These requirements define the **business workflows and operational procedures** supported by the system.

Authentication & Access Control

- ✓ **FR-01** ~ The system shall require all users to authenticate before accessing any functionality.
- ✓ **FR-02** ~ The system shall enforce role-based access control (RBAC) based on user roles such as Traffic Police Officer, Subcity Officer, CCTV Operator, and System Administrator.

Violation Reporting & Coordination

- ✓ **FR-03** ~ The system shall allow traffic police officers to manually report traffic violations using a digital interface.
- ✓ **FR-10** ~ The system shall allow officers to draw estimated escape routes on a map interface.
- ✓ **FR-12** ~ The system shall integrate with CCTV feeds to support license plate detection.
- ✓ **FR-15** ~ The system shall allow CCTV operators to validate, confirm, or dismiss detected events.
- ✓ **FR-19** ~ The system shall prevent penalty finalization until reviewed by authorized officers.

Complaint Handling (Internal Process)

- ✓ **FR-20** ~ The system shall allow subcity officers to retrieve violation records using violation record IDs.
- ✓ **FR-21** ~ The system shall allow authorized officers to review attached evidence, including images and video.
- ✓ **FR-22** ~ The system shall support requesting CCTV footage from surveillance units.
- ✓ **FR-23** ~ The system shall allow officers to revoke, modify, or confirm violations based on review outcomes.

Payment Processing Workflow

- ✓ **FR-24** ~ The system shall initiate OTP-based payment requests only after official penalty confirmation.

- ✓ **FR-25** ~ The system shall integrate with an external payment gateway for transaction processing.
- ✓ **FR-27** ~ The system shall not expose payment interfaces directly to drivers.

Administrative Processes

- ✓ **FR-28** ~ The system shall allow administrators to manage users and assign roles.
- ✓ **FR-29** ~ The system shall allow configuration of traffic rules, penalties, and system parameters.

2.10.3 Input-Related Requirements

These requirements define all data that must be **entered, captured, or received** by the system.

- ✓ **FR-04** ~ The system shall automatically capture the date, time, and geographic location of each reported violation.
- ✓ **FR-05** ~ The system shall associate each violation with a unique violation record ID.
- ✓ **FR-06** ~ The system shall retrieve vehicle details using the provided license plate number.
- ✓ **FR-10** ~ The system shall allow officers to draw estimated escape routes on a map interface.
- ✓ **FR-16** ~ The system shall record detection confidence levels and validation outcomes.

2.10.4 Output-Related Requirements

These requirements define the **information and results produced** by the system.

- ✓ **FR-07** ~ The system shall broadcast violation alerts to nearby traffic officers.
- ✓ **FR-08** ~ The system shall notify relevant subcity traffic offices when violations require escalation.
- ✓ **FR-09** ~ The system shall support priority-based alerting for high-risk or escaped vehicles.
- ✓ **FR-11** ~ The system shall prioritize alerts for officers located along predicted escape paths.
- ✓ **FR-18** ~ The system shall optionally assign penalty points according to regulation.
- ✓ **FR-26** ~ The system shall record payment confirmation status for each violation.

2.10.5 Storage-Related Requirements

These requirements define how data must be **stored, retained, and audited**.

- ✓ **FR-05** ~ The system shall associate each violation with a unique violation record ID.
- ✓ **FR-14** ~ The system shall compare detected plates against active violation and wanted vehicle records.
- ✓ **FR-16** ~ The system shall record detection confidence levels and validation outcomes.
- ✓ **FR-21** ~ The system shall allow authorized officers to review attached evidence, including images and video.
- ✓ **FR-26** ~ The system shall record payment confirmation status for each violation.
- ✓ **FR-30** ~ The system shall maintain detailed audit logs of all system actions.

2.11 Non-Functional Requirements

Non-functional requirements specify **how** the system performs its functions and define the **quality attributes, constraints, and operational standards** under which the ALPR-Based Smart Traffic Management System operates. These requirements ensure that the system is secure, reliable, usable, and suitable for government-level deployment.

2.11.1 Performance Requirements

These requirements define the system's responsiveness, processing speed, and operational efficiency.

- ✓ **NFR-P01** ~ The system shall process violation reports in real time.
- ✓ **NFR-P02** ~ Alerts shall be delivered within seconds of reporting or detection.
- ✓ **NFR-P03** ~ CCTV-based detection shall operate with minimal processing latency.

2.11.2 User Interface Requirements

These requirements define usability, accessibility, and interaction quality for system users.

- ✓ **NFR-UI01** ~ User interfaces shall be simple and intuitive to minimize operational errors.
- ✓ **NFR-UI02** ~ Mobile access shall be supported for field traffic police officers.
- ✓ **NFR-UI03** ~ Minimal training shall be required for basic system operation.

2.11.3 Security and Access Permissions

These requirements define how the system protects data, controls access, and ensures accountability.

- ✓ **NFR-S01** ~ All system access shall be authenticated and authorized.
- ✓ **NFR-S02** ~ Role-based access control shall restrict functionality based on assigned user roles.
- ✓ **NFR-S03** ~ Sensitive data shall be encrypted both in storage and during transmission.
- ✓ **NFR-S04** ~ CCTV data access shall be restricted to authorized roles only.
- ✓ **NFR-S05** ~ All system actions shall be logged for audit purposes.

2.11.4 Backup and Recovery Requirements

These requirements ensure data durability, continuity of operations, and fault tolerance.

- ✓ **NFR-BR01** ~ The system shall be available during official traffic operation hours.
- ✓ **NFR-BR02** ~ Automatic backups shall prevent data loss.
- ✓ **NFR-BR03** ~ System failures shall not affect historical data integrity.

2.11.5 Scalability and Maintainability

Although not explicitly listed in the guideline outline, these requirements are essential for a phased, government-scale system and are therefore included as supporting attributes.

- ✓ **NFR-SM01** ~ The system shall support the addition of new cameras and subcities without major redesign.
- ✓ **NFR-SM02** ~ A modular architecture shall support phased upgrades (Phase 1 to Phase 4).
- ✓ **NFR-SM03** ~ System components shall be maintainable without full system downtime.

2.11.6 Comparative Analysis of Existing and Proposed Systems

To further justify the non-functional improvements introduced by the proposed system, a comparative analysis is provided below.

Feature	Existing System	Proposed System
Report	Manual	Digital
Communication	Walkie-talkie	Real-time alerts
Tracking	None	ALPR-based
Data Storage	Paper	Centralized Database
Evidence	Limited	CCTV-based

Table 2.2 Comparison of Existing and Proposed Systems

CHAPTER THREE

SYSTEM ANALYSIS

3.1 Introduction

System analysis is a critical phase of the Software Development Life Cycle (SDLC) that focuses on understanding organizational processes, identifying stakeholder needs, defining system requirements, and modeling system interactions before system design and implementation. A well-executed system analysis ensures that the proposed solution addresses real operational challenges, aligns with institutional policies, and remains technically and legally viable.

This chapter presents a detailed system analysis of the **Automatic License Plate Detection and Recognition–Based Smart Traffic Management System (ALPR-STMS)** proposed for the **Hawassa City Administration Transport and Road Development Department**. The analysis is grounded in requirement engineering best practices, structured stakeholder engagement, and UML-based system modeling.

Currently, traffic enforcement operations in Hawassa City rely on manual observation, walkie-talkie communication, and fragmented documentation. Vehicle tracking, inter-subcity coordination, violation evidence handling, and penalty enforcement depend heavily on human judgment and verbal reporting. These practices lead to delayed response times, weak traceability, data inconsistency, and limited enforcement efficiency.

The proposed ALPR-STMS introduces a **centralized digital platform** that supports traffic officers, subcity coordination units, and CCTV surveillance teams through digital reporting, real-time alert broadcasting, CCTV-assisted detection, and phased intelligent automation. Importantly, the system is designed to **support and enhance existing legal and administrative workflows**, not replace them.

This chapter focuses on:

- ✓ Identifying system stakeholders and actors
- ✓ Defining the system scope and boundaries
- ✓ Establishing a clear understanding of system responsibilities
- ✓ Preparing the foundation for functional modeling and system design

3.2 System Requirements Specification (SRS)

The **System Requirements Specification (SRS)** formally defines the functional and non-functional requirements of the ALPR-STMS in a precise, verifiable, and implementation-independent manner. It serves as a reference document for developers, system owners, evaluators, and future maintainers.

The SRS for the ALPR-STMS is derived from:

- ✓ Structured interviews with traffic police officers, subcity officers, and administrative personnel
- ✓ Observation of real traffic enforcement and coordination activities
- ✓ Review of Ethiopian federal and regional traffic regulations
- ✓ Analysis of limitations in the existing traffic management system
- ✓ Consideration of future system expansion through phased implementation

The system is intentionally designed as a **phased solution**, starting with digital support for manual enforcement and evolving toward semi-automated and fully automated traffic intelligence. Therefore, the SRS addresses both **current operational realities** and **future smart city objectives**.

3.3 System Overview

The **Automatic License Plate Detection and Recognition–Based Smart Traffic Management System (ALPR-STMS)** is a centralized information system designed to support traffic law enforcement, vehicle tracking, and inter-subcity coordination within Hawassa City.

- ✓ The system provides digital tools and intelligence to:
- ✓ Traffic police officers operating in the field
- ✓ Subcity traffic coordination offices
- ✓ CCTV surveillance operators and systems
- ✓ City-level traffic administration

3.3.1 Key Characteristics of the System

- ✓ The system **does not directly serve drivers** as system users.
- ✓ All enforcement actions, decisions, and communications occur **within authorized traffic and administrative units**.
- ✓ Drivers interact with the enforcement process **physically** through officers and subcity offices.
- ✓ Electronic payment (OTP-based) is supported **only after official confirmation** by authorized personnel.

3.3.2 Core Functional Capabilities

The ALPR-STMS integrates:

- ✓ Manual digital reporting of traffic violations by officers
- ✓ Automatic capture of location and timestamp data
- ✓ Real-time alert broadcasting across subcities
- ✓ CCTV-assisted license plate detection and tracking
- ✓ Digital storage of violations, penalties, and evidence
- ✓ Internal coordination for complaint handling and verification
- ✓ Secure audit logging and administrative control

The system acts as a **decision-support and coordination platform**, enhancing speed, accuracy, accountability, and evidence availability while preserving human authority in enforcement decisions.

3.4 Stakeholders and System Actors

This section identifies **stakeholders** (entities with interest or authority over the system) and **system actors** (entities that directly interact with the system).

In alignment with the revised architecture, **drivers are explicitly excluded as system actors**.

3.4.1 Primary Stakeholders

Hawassa City Administration Transport and Road Development Department

The principal system owner responsible for policy oversight, regulatory compliance, and administrative governance. The department ensures that the system aligns with traffic laws, operational mandates, and city-level objectives.

Traffic Police Department

Responsible for day-to-day traffic law enforcement. Traffic police officers are the primary users of the system for reporting violations, receiving alerts, and coordinating enforcement actions.

Subcity Traffic Coordination Offices

Subcity offices act as operational coordination hubs. They oversee enforcement activities within their jurisdiction, manage escalations across subcities, handle complaints, and authorize penalties or revocations.

CCTV Surveillance Units

Responsible for monitoring camera feeds, validating system detections, and providing video evidence when required. They support both proactive tracking and post-incident verification.

3.4.2 System Actors (Authorized Users and External Systems)

The system involves multiple authorized human actors and external systems, each with clearly defined responsibilities and access boundaries. This separation of roles ensures accountability, security, and compliance with institutional traffic enforcement procedures.

Traffic Police Officer (System Actor)

A frontline law enforcement officer responsible for on-road traffic control and violation reporting. The Traffic Police Officer:

- ✓ Logs into the system using secure credentials
- ✓ Reports traffic violations digitally
- ✓ Automatically captures time and geographic location of violations
- ✓ Receives real-time alerts related to reported or detected violations
- ✓ Draws estimated escape routes using the map interface
- ✓ Acts on system-generated intelligence during vehicle interception

Sub-city Officer (System Actor)

An authorized supervisory officer responsible for administrative coordination and decision-making at the subcity level. The Subcity Officer:

- ✓ Coordinates inter-subcity enforcement activities
- ✓ Handles escalation of violations beyond local jurisdictions
- ✓ Reviews detailed violation records and attached evidence
- ✓ Oversees complaint resolution outcomes
- ✓ Authorizes penalty confirmation or violation revocation

Complaint Officer (System Actor)

An authorized office-based official responsible for handling formal complaints submitted by drivers. The Complaint Officer operates **strictly within office environments** and does not interact with drivers digitally. The Complaint Officer:

- ✓ Logs into the system with restricted administrative credentials
- ✓ Retrieves violation records using a unique violation record ID
- ✓ Reviews digital evidence, including images and CCTV footage
- ✓ Requests additional CCTV video evidence from CCTV operators when required
- ✓ Communicates internally with reporting Traffic Police Officers for clarification
- ✓ Decides whether a violation should be confirmed, modified, or revoked
- ✓ Ensures all complaint decisions are documented and auditable

CCTV Operator (System Actor)

An authorized operator responsible for managing surveillance and video evidence. The CCTV Operator:

- ✓ Monitors live CCTV camera feeds
- ✓ Validates system-detected license plate recognition events
- ✓ Confirms or dismisses detected violations based on visual evidence
- ✓ Provides video or image evidence upon official request

CCTV System (External System Actor)

An automated surveillance subsystem integrated with the traffic management platform. The CCTV System:

- ✓ Supplies live video feeds
- ✓ Performs license plate detection using computer vision techniques
- ✓ Generates detection events for system evaluation and validation

Payment Gateway (External System Actor)

An external financial system responsible solely for payment processing. The Payment Gateway:

- ✓ Processes OTP-based traffic penalty payments

- ✓ Confirms transaction success or failure
- ✓ Does not store or manage violation or enforcement data

System Administrator (System Actor)

A technical authority responsible for system governance and configuration. The System Administrator:

- ✓ Manages system users and assigns roles
- ✓ Configures traffic rules, penalties, and system parameters
- ✓ Maintains audit logs and system integrity
- ✓ Oversees system availability and security controls

Driver (External Human Actor)

A vehicle operator indirectly interacting with the system. The Driver:

- ✓ Receives OTP payment requests after official penalty confirmation
- ✓ Completes penalty payment via the payment gateway
- ✓ Submits complaints in person at subcity offices
- ✓ Does not log into or directly interact with the system

Stakeholder	Primary Interaction
Traffic Police Officer	Report violations & intercept vehicles
Subcity Officer	Coordinate enforcement & authorize decisions
Complaint Officer	Review complaints & decide outcomes
CCTV Operator	Validate detections & provide evidence
CCTV System	Detection and recognition
Payment Gateway	Facilitate penalty payment
System Administrator	Manage system configuration
Driver	Pay penalties & submit complaints

Table 3.1 Stakeholders and System Interaction

3.4.3 Explicit Exclusion of Driver as a System Actor

Drivers and vehicle owners **are not system users** and **do not directly interact with the ALPR-STMS**.

Driver interaction occurs through:

- ✓ Physical engagement with traffic police officers
- ✓ Complaint submission at subcity traffic offices using a violation record ID
- ✓ Manual review and decision by authorized officials
- ✓ OTP-based payment facilitated by officers or subcity offices after confirmation

This design:

- ✓ Preserves legal and administrative authority

- ✓ Prevents unauthorized access to enforcement systems
- ✓ Aligns with existing operational practices
- ✓ Ensures accountability and controlled decision-making

3.5 System Boundary Definition

Defining the system boundary is essential to clearly distinguish what responsibilities belong to the ALPR-STMS and what activities remain outside the scope of the system. This separation prevents ambiguity, scope creep, and unrealistic expectations during development and deployment.

3.5.1 In-Scope System Responsibilities

The ALPR-STMS is responsible for the following within its boundary:

- ✓ Digital reporting of traffic violations by authorized officers
- ✓ Automatic capture of violation metadata (time, location, officer ID)
- ✓ Centralized storage of vehicle, violation, and penalty records
- ✓ CCTV-assisted license plate detection and tracking
- ✓ Real-time alert broadcasting to officers and subcity offices
- ✓ Inter-subcity coordination and escalation management
- ✓ Penalty calculation based on predefined traffic rules
- ✓ Internal complaint review support through evidence retrieval
- ✓ OTP-based payment request initiation after official confirmation
- ✓ Audit logging of all system actions
- ✓ Administrative management of users, roles, and system rules

These responsibilities are executed strictly by **authorized personnel and system components**.

3.5.2 Out-of-Scope System Responsibilities

The following activities are explicitly outside the system boundary:

- ✓ Physical stopping of vehicles and roadside enforcement
- ✓ Direct interaction with drivers through mobile or web interfaces
- ✓ Legal judgment or court-level adjudication
- ✓ National vehicle registration management
- ✓ Banking operations and financial transaction processing
- ✓ Physical document storage and handling

The ALPR-STMS acts as a **support and coordination system**, not as a replacement for legal or physical enforcement processes.

3.6 System Constraints and Assumptions

This section identifies the constraints and assumptions under which the ALPR-Based Smart Traffic Management System (ALPR-STMS) must operate. Clearly defining these factors is

essential for realistic system design, risk mitigation, and expectation management during development and deployment.

3.6.1 System Constraints

System constraints represent **non-negotiable limitations** imposed by technical, legal, organizational, or environmental factors.

3.6.1.1 Infrastructure Constraints

- ✓ The system initially operates with a **limited number of CCTV cameras**, which may not provide full city-wide coverage.
- ✓ Camera placement and quality vary, affecting detection accuracy and coverage consistency.
- ✓ Existing CCTV infrastructure is primarily designed for security surveillance rather than traffic enforcement, requiring adaptation rather than replacement.

3.6.1.2 Legal and Regulatory Constraints

- ✓ Surveillance data usage is governed by **Ethiopian federal and regional laws**, restricting:
- ✓ Who can access video feeds
- ✓ How long data can be stored
- ✓ How evidence can be used in enforcement and complaints
- ✓ Automated enforcement decisions **cannot replace human authority**; final decisions is gonna be validated by authorized officers.

3.6.1.3 Technical Constraints

- ✓ System operation depends on **network connectivity** between:
 - Traffic officers
 - Subcity offices
 - CCTV systems
 - Central servers
- ✓ Temporary connectivity failures may delay alerts or synchronization.
- ✓ Integration with external systems (e.g., payment gateways) depends on third-party availability.

3.6.1.4 Organizational Constraints

- ✓ Enforcement workflows must align with existing **subcity jurisdictional structures**.
- ✓ The system cannot override established command hierarchies or administrative procedures.

3.6.2 System Assumptions

Assumptions define conditions expected to hold true for the system to operate effectively.

- ✓ Traffic police officers and subcity personnel will **adopt digital reporting practices** after training.
- ✓ CCTV camera feeds will be **accessible to the system** through approved integration channels.
- ✓ Vehicle registration and plate information are **available in digital form** or can be digitized.
- ✓ Legal approval exists for the use of ALPR technology under defined operational conditions.
- ✓ Officers will continue to exercise **human judgment**, especially in complaint resolution and penalty confirmation.

3.7 Operating Environment

The operating environment describes the **technical and organizational context** in which the ALPR-STMS will be deployed and executed.

3.7.1 Deployment Environment

- ✓ The system will be deployed on **server infrastructure located at the Hawassa City Administration data center**.
- ✓ Centralized deployment ensures:
 - Controlled access
 - Data security
 - Regulatory compliance
- ✓ Redundant storage and backup mechanisms support reliability and data protection.

3.7.2 Client Environment

Authorized users access the system through:

- ✓ Web-based interfaces (subcity offices, control centers)
- ✓ Mobile-friendly interfaces (traffic officers in the field)

Interfaces are optimized for low-latency operation and limited bandwidth environments.

3.7.3 Integration Environment

The system integrates with:

- ✓ Existing CCTV infrastructure
- ✓ External OTP-based payment gateways

During development and testing:

- ✓ **Firebase** is used for rapid prototyping and validation.

During production:

- ✓ **City-owned servers and databases** (e.g., PostgreSQL/MySQL) are used for full control and compliance.

3.8 Future Expansion Considerations

The ALPR-STMS is intentionally designed as a **scalable and extensible platform**, capable of supporting future smart traffic and smart city initiatives.

3.8.1 Functional Expansion

Future system versions may support:

- ✓ Fully automated traffic violation detection (speeding, red-light violations)
- ✓ AI-based pattern recognition for repeated offenders
- ✓ Automated blacklist and watchlist management

3.8.2 Intelligence and Analytics Expansion

- ✓ Machine learning-based route prediction
- ✓ Historical traffic behavior analysis
- ✓ Predictive congestion and enforcement planning
- ✓ Performance analytics for traffic enforcement units

3.8.3 Integration Expansion

- ✓ Integration with national traffic and transport systems
- ✓ Smart city platforms (emergency response, urban planning)
- ✓ Inter-city enforcement coordination

3.9 Use Case Diagram

The **Use Case Diagram** represents the functional scope of the ALPR-STMS and its interaction with **authorized external actors**.

Important UML Interpretation Rule

Only **primary ovals representing user goals** are considered valid use cases.

All other ovals shown inside the system boundary represent **included behaviors or internal scenarios**, not independent use cases.

3.9.1 Identified Main Use Cases

Based on UML standards and diagram analysis, the system contains **exactly ten (10) main use cases**:

- ✓ Login
- ✓ Signout
- ✓ Violation

- ✓ View
- ✓ Detect Plate Number
- ✓ Manage Camera Feed
- ✓ Complain
- ✓ Payment
- ✓ Manage
- ✓ Coordinate

All other actions (e.g., Draw Escape Path, Broadcast Violation, Penalty Calculation, Predict Escape Path) are **scenarios** occur within these **use cases**.

3.10 System Business Rules & Logic (Supplementary Detail)

Penalty Logic (Point vs. Fee):

- ✓ *Rule:* When Violation is reported (e.g., Red Light pass), the system must calculate both a **Penalty Fee** and **Penalty Points**.
- ✓ *Rule:* If Penalty Points > Threshold (e.g., 60 points), the system flags the license for suspension.

Escape Path Prediction:

- ✓ *Logic:* The **Predict Escape Path** function uses the Current Location and road network topology to highlight the 3 most probable exit routes on the map.

OTP Security:

- ✓ *Constraint:* The **OTP Payment** is valid for certain minutes. If expired, Request Payment must be triggered again.

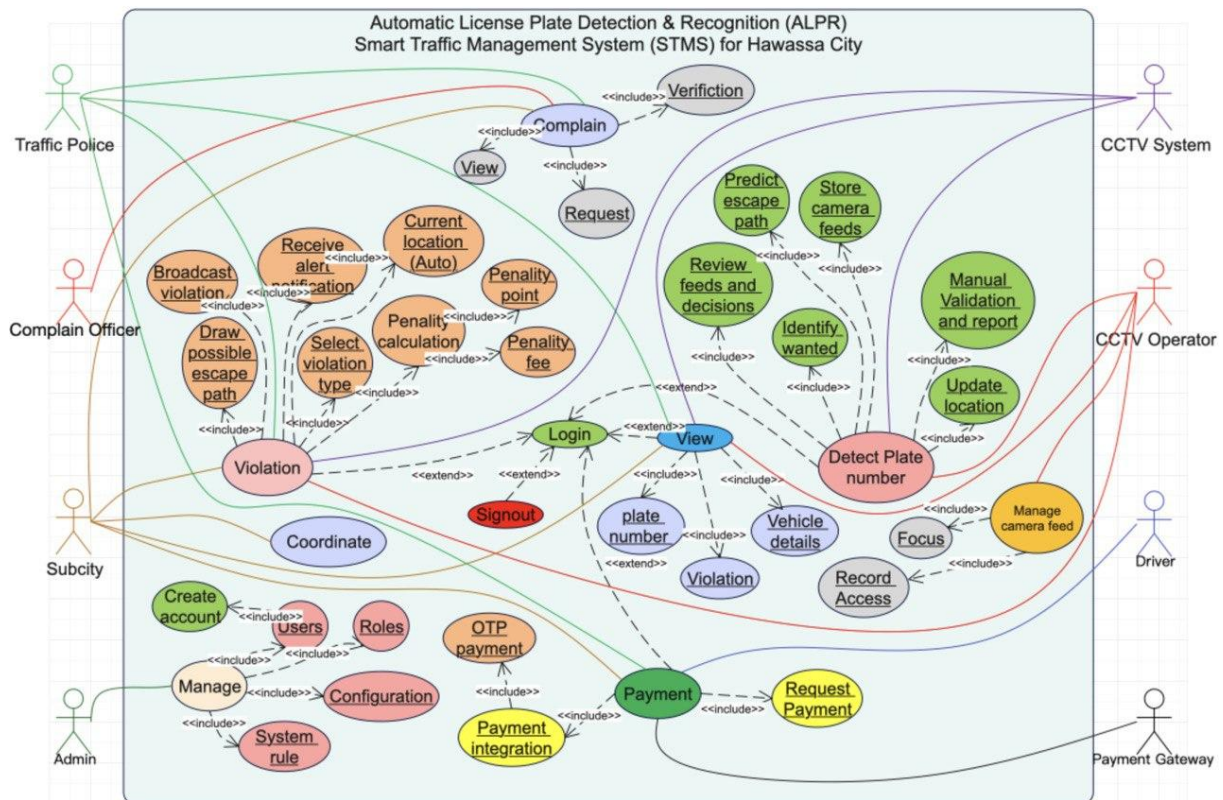


Fig 3.1 Use case UML Diagram of ALPR STMS, Hawassa

3.11 Use Case Documentation

UC-01: Login

Actors:

Traffic Police, Complain Officer, Subcity Officer, Admin, CCTV Operator

Description:

The primary security gateway for the system. It authenticates users based on their credentials and directs them to the appropriate dashboard based on their assigned role (defined in UC-03).

Preconditions:

User account must exist in the system.

Postconditions:

User is successfully authenticated and session is active.

UC-02: Signout

Actors:

All System Users

Description:

Allows any authenticated user to securely terminate their active session to prevent unauthorized access.

Relationships:

Extends: UC-01 Login

Postconditions:

Session is terminated; user is returned to the Login screen.

UC-03: Manage System (Administration)

Actors:

Admin

Description:

This is the administrative hub where the system environment is maintained. The Admin manages the human resources (users) and the logical rules that the system enforces.

Internal Scenarios (Included Behaviors):

- ✓ **Users** (Profile management); Includes **Create Account**: Registering new personnel into the database
- ✓ **Roles** (Permission assignment)
- ✓ **Configuration**: adjusting technical system settings
- ✓ **System Rule** (Defining penalty logic, speed limits, and enforcement thresholds)

Postconditions:

System configurations or User access rights are immediately updated.

UC-04: Detect Plate Number (Core Function)

Actors:

CCTV System (Primary/Auto), CCTV Operator (Secondary/Verification)

Description:

The core automated loop of the STMS. The system continuously processes video feeds to extract license plate data, cross-reference it with criminal databases, and track vehicle movement.

Internal Scenarios (Included Behaviors):

- ✓ **Store Camera Feeds**: Continuous archiving of video data for evidence.
- ✓ **Predict Escape Path**: Algorithmically determining potential routes for fleeing/wanted vehicles.
- ✓ **Review Feeds and Decisions**: Operator oversight of the automated detections.
- ✓ **Identify Wanted**: Real-time database check for stolen or flagged vehicles.
- ✓ **Update Location**: Logging the real-time GPS/Zone coordinates of the detected plate.
- ✓ **Manual Validation and Report**: Operator confirms the detection and finalizes the event report.

Preconditions:

Cameras are active and feeding data to the server.

Postconditions:

Vehicle presence is logged, location is updated, and alerts are ready for the violation module.

UC-05: Violation

Actors:

Traffic Police, Subcity Officer

Description:

This use case handles the enforcement workflow. Once a violation is detected, this module processes the specific offense, calculates the appropriate penalty, and alerts relevant units.

Internal Scenarios (Included Behaviors):

- ✓ **Receive Alert Notification:** Officer receives real-time data of an incident.
- ✓ **Current Location (Auto):** System automatically pulls location data from the detection module.
- ✓ **Select Violation Type:** Categorizing the offense (e.g., Speeding, Illegal Parking).
- ✓ **Penalty Calculation:** The logic engine that computes the fine; *Includes:* **Penalty Point** (Demerit points added to license), **Penalty Fee** (Financial amount calculated).
- ✓ **Draw Possible Escape Path:** Visualizing the predicted route on the officer's map interface.
- ✓ **Broadcast Violation:** Sending alerts to nearby patrol units or sub-cities.

Relationships:

Extends: UC-01 Login (and interacts with View).

Postconditions:

Violation is officially recorded and penalty is assigned.

UC-06: View Data

Actors:

Traffic Police, Subcity Officer, CCTV Operator

Description:

A generalized use case allowing authorized roles to access read-only data for verification and monitoring.

Internal Scenarios (Included Behaviors):

- ✓ **View Plate Number:** Displaying the alphanumeric recognition result.
- ✓ **View Vehicle Details:** displaying ownership, model, and registration status.
- ✓ **View Violation:** Accessing the history of penalties associated with a plate.

Relationships:

Extends: UC-01 Login

Postconditions:

Requested data is displayed on the UI.

UC-07: Manage Camera Feed

Actors:

CCTV Operator

Description:

Allows the operator to manage the technical quality and retrieval of video data.

Internal Scenarios (Included Behaviors):

- ✓ **Focus:** Remote adjustment of camera lenses to improve recognition accuracy.
- ✓ **Record Access:** Retrieving specific historical video clips from storage.

Postconditions:

Camera feed is optimized or evidence is retrieved.

UC-08: Complain (Grievance Handling)

Actors:

Complain Officer

Description:

The workflow for handling disputes raised by citizens regarding penalties.

Internal Scenarios (Included Behaviors):

- ✓ **Request:** Logging a new complaint into the system.
- ✓ **Verification:** The process of validating the complaint against system evidence.
- ✓ **View** (Officer must view the violation record to verify).

Postconditions:

Complaint status is updated (Approved/Rejected).

UC-09: Coordinate

Actors:

Subcity Officer

Description:

Facilitates inter-departmental cooperation, allowing Subcity officers to manage traffic events that cross administrative boundaries.

Relationships:

Extends: UC-01 Login

Postconditions:

Coordination tasks are logged.

UC-10: Payment

Actors:

Traffic Police (Monitor), Subcity (Monitor), Driver (OTP Actor), Payment Gateway (External)

Description:

Handles the financial settlement of penalties. The system triggers a payment request which is processed via an external gateway.

Internal Scenarios (Included Behaviors):

- ✓ **Request Payment:** System initiates the transaction request.
- ✓ **Payment Integration** (Secure handshake with the banking API); Include **OTP Payment:** The validation step where the Driver receives a One-Time Password to authorize the fund transfer.

The Driver is an external actor whose only interaction with the system is receiving/inputting the OTP.

Postconditions:

Penalty status changes to "Paid" and free the active record.

3.12 Analysis-Level Class Diagram (Comprehensive Description and Rationale)

3.12.1 Purpose of the Analysis-Level Class Diagram

The Analysis-Level Class Diagram defines the **static structural architecture** of the **ALPR-Based Smart Traffic Management System (ALPR-STMS)** for Hawassa City.

It provides a **technology-agnostic, business-focused model** that identifies:

- ✓ Core domain entities
- ✓ Authorized human actors
- ✓ Supporting application services
- ✓ Evidence, intelligence, and enforcement components
- ✓ Relationships, multiplicity, and responsibility boundaries

The diagram serves as a **foundational artifact** for:

- ✓ Detailed design and implementation models
- ✓ Database schema derivation
- ✓ API and service contracts
- ✓ Role-Based Access Control (RBAC) enforcement
- ✓ Legal auditing and compliance verification

3.12.2 High-Level System Grouping and Structural Domains

To ensure clarity, maintainability, and legal defensibility, the system is divided into **well-defined functional domains**. Each domain encapsulates logically related responsibilities.

Primary Structural Domains

1. Security & User Management
2. Vehicle & Registration
3. Violation & Enforcement (Core Domain)
4. Surveillance & Evidence
5. Intelligence & Coordination
6. Complaint & Oversight
7. Payment (External Interaction Only)
8. Audit & Accountability

9. Shared Value Objects

This grouping ensures **clear separation of concerns**, minimizes coupling, and enables independent evolution of subsystems.

3.12.3 Design Principles Applied

3.12.3.1 Single Responsibility Principle (SRP)

Each class has **one clearly defined responsibility**.

For example:

- ✓ Violation manages lifecycle state only
- ✓ Penalty manages punishment details only
- ✓ AuditLog manages traceability only

3.12.3.2 Separation of Concerns

The model separates:

- ✓ **Actors** → User hierarchy
- ✓ **Business entities** → Violation, Vehicle, Penalty
- ✓ **Application services** → ALPRProcessor, RoutePrediction
- ✓ **Infrastructure** → CCTVFeed, AuditLog

This prevents role confusion and legal ambiguity.

3.12.3.3 Role-Based Access Control (RBAC)

RBAC is implemented using:

- ✓ An abstract User superclass
- ✓ Controlled inheritance
- ✓ Explicit role responsibilities

Drivers are **explicitly excluded** from system login.

3.12.3.4 Auditability and Legal Validity

Every legally significant action is traceable via:

- ✓ AuditLog
- ✓ Explicit user attribution
- ✓ Timestamped records

This ensures:

- ✓ Court admissibility
- ✓ Internal investigations

- ✓ Anti-corruption compliance

3.12.3.5 Human-in-the-Loop Enforcement

Automated components **support but never replace** human authority.

No violation becomes enforceable without:

- ✓ Officer validation
- ✓ Evidence attachment
- ✓ Supervisor confirmation (where required)

3.12.4 Security & User Management Domain

3.12.4.1 Abstract Class: User

Purpose:

- ✓ Represents all **authorized system personnel**.
- ✓ It does **not** represent drivers or vehicle owners.
- ✓ **Attributes**
 - `userId : String` – Unique system identifier
 - `fullName : String` – Official identity
 - `phoneNumber : String` – Communication and OTP reference
 - `email : String` – Notifications and audit contact
 - `role : String` – RBAC identifier
- ✓ **Responsibilities**
 - Establish authenticated identity
 - Enable controlled access to system functions
 - Ensure accountability for every action

3.12.4.2 TrafficOfficer

Represents on-duty field traffic police.

Additional Attributes

- ✓ `badgeNumber : String`
- ✓ `rank : String`
- ✓ `currentLocation : GeoCoordinate`

Responsibilities

- ✓ Manually report violations
- ✓ Initiate violation records
- ✓ Draw estimated escape routes
- ✓ Receive and respond to alerts

Associations

- ✓ One TrafficOfficer → many Violations
- ✓ One TrafficOfficer → many Alerts

3.12.4.3 SubcityOfficer

Represents jurisdiction-level coordination authority.

Responsibilities

- ✓ Validate violations
- ✓ Coordinate inter-subcity enforcement
- ✓ Authorize escalation
- ✓ Supervise payment initiation

3.12.4.4 CCTVOperator

Responsible for surveillance validation.

Responsibilities

- ✓ Monitor live CCTV feeds
- ✓ Validate ALPR detections
- ✓ Attach video/image evidence
- ✓ Support complaint investigations

3.12.4.5 ComplaintOfficer

Ensures independent oversight.

Responsibilities

- ✓ Review complaints (office-based)
- ✓ Retrieve violation records
- ✓ Request additional evidence
- ✓ Decide confirmation or dismissal

3.12.4.6 SystemAdmin

System governance authority.

Responsibilities

- ✓ Manage users and roles
- ✓ Configure violation rules
- ✓ Maintain system parameters
- ✓ Oversee audit compliance

3.12.5 Vehicle & Registration Domain

3.12.5.1 Vehicle

Represents a registered vehicle (read-only for enforcement).

Attributes

- ✓ plateNumber (Primary Identifier)
- ✓ vehicleType
- ✓ model
- ✓ color
- ✓ ownerName
- ✓ registrationStatus

Relationship

- ✓ One Vehicle → many Violations

3.12.5.2 LicensePlate (Value Object)

Generated by ALPR processing.

Attributes

- ✓ plateNumber
- ✓ plateImage
- ✓ confidenceScore

Used to ensure AI results are **reviewable and auditable**.

3.12.6 Violation & Enforcement Domain (Core)

3.12.6.1 Violation

Central entity of the entire system.

Attributes

- ✓ violationId
- ✓ violationType
- ✓ description
- ✓ timestamp
- ✓ location
- ✓ status

Lifecycle States

- ✓ Pending
- ✓ Confirmed
- ✓ UnderComplaint
- ✓ Dismissed

- ✓ Paid

Responsibilities

- ✓ Maintain complete enforcement lifecycle
- ✓ Link vehicle, evidence, penalties, alerts, and complaints
- ✓ Serve as single source of truth

3.12.6.2 ViolationRule

Represents codified traffic laws.

Attributes

- ✓ ruleCode
- ✓ ruleName
- ✓ penaltyAmount
- ✓ penaltyPoints
- ✓ isOptional

Purpose

- ✓ Enables legal compliance
- ✓ Eliminates hard-coded penalties
- ✓ Supports policy updates

3.12.6.3 Penalty

Represents validated punishment.

Attributes

- ✓ penaltyId
- ✓ amount
- ✓ points
- ✓ status

Constraint

- ✓ Exists **only if violation is confirmed**
- ✓ One-to-one with Violation

3.12.7 Surveillance & Evidence Domain

3.12.7.1 Camera

Represents physical CCTV infrastructure.

Attributes

- ✓ cameraId
- ✓ cameraType
- ✓ location
- ✓ status

3.12.7.2 CCTVFeed

Represents captured footage.

Attributes

- ✓ feedId
- ✓ timestamp
- ✓ videoStream

Associations

- ✓ One Camera → many CCTVFeeds
- ✓ One Violation → zero or many CCTVFeeds

3.12.7.3 ALPRProcessor (Application Service)

Automated detection assistant.

Responsibilities

- ✓ Detect license plates
- ✓ Extract characters
- ✓ Assign confidence scores

3.12.8 Intelligence & Coordination Domain

3.12.8.1 Alert

Represents notifications to enforcement units.

Attributes

- ✓ alertId
- ✓ priority
- ✓ message
- ✓ timestamp

Filtered by

- ✓ Location
- ✓ Jurisdiction
- ✓ Predicted escape routes

3.12.8.2 RoutePrediction (Application Service)

Provides predictive intelligence.

Attributes

- ✓ predictedPaths
- ✓ confidenceLevel

Supports proactive interception and coordination.

3.12.9 Complaint & Oversight Domain

3.12.9.1 Complaint

Represents formal citizen challenges.

Attributes

- ✓ complaintId
- ✓ reason
- ✓ status

Ensures due process and trust.

3.12.9.2 AuditLog

Ensures transparency and legal traceability.

Attributes

- ✓ logId
- ✓ action
- ✓ timestamp
- ✓ performedBy

Coverage

- ✓ Violation creation
- ✓ Validation
- ✓ Complaint decisions
- ✓ Payment confirmation
- ✓ Rule configuration

3.12.10 Payment Domain (External Interaction Only)

3.12.10.1 Payment

Represents OTP-based financial settlement.

Attributes

- ✓ paymentId
- ✓ paymentMethod : OTP
- ✓ amount
- ✓ paymentDate

Critical Design Rules

- ✓ Driver never logs into system
- ✓ Payment initiated only after confirmation
- ✓ OTP handled externally

3.12.11 Shared Value Objects

3.12.11.1 GeoCoordinate / GeoLocation

Reusable location object.

Attributes

- ✓ latitude
- ✓ longitude

Used consistently across:

- ✓ Violations
- ✓ Cameras
- ✓ Officers
- ✓ Route prediction

3.12.12 Conclusion

The merged analysis-level class diagram and documentation provide a **legally defensible, scalable, and human-centered structural foundation** for the ALPR-Based Smart Traffic Management System.

The design:

- ✓ Preserves citizen rights
- ✓ Prevents autonomous enforcement abuse
- ✓ Enables smart-city intelligence
- ✓ Aligns with Ethiopian urban governance and public accountability standards

Notation	Meaning
alt	Legal decision branches
opt	Optional but permitted flows
return	Explicit system response
ref	Logical delegation
note	Legal/administrative clarification
Solid arrows	Mandatory operations
Dashed arrows	Responses / acknowledgements

Table 3.3 Legend and Modeling Conventions

3.13.2 Detailed Sequence Flow Descriptions

3.13.2.1 Security Login Sequence (All Officials – Excluding Driver)

Purpose

To ensure **authenticated, role-based, and auditable access** for all human system users, except drivers, who are explicitly excluded from system access.

Actors Involved

- ✓ Traffic Officer
- ✓ Subcity Officer
- ✓ CCTV Operator
- ✓ Complaint Officer

Detailed Sequence Flow

1, User Access Initiation

- ✓ An authorized official accesses the system through the secured Login UI.
- ✓ Access is only permitted from registered devices or secure networks.

2, Credential Submission

- ✓ Unique personnel ID
- ✓ Username
- ✓ Password
- ✓ Optional second factor (OTP / biometric) depending on role sensitivity

3, Authentication Request

- ✓ Login UI transmits encrypted credentials to the Authentication Service.
- ✓ No credentials are stored or cached at the UI layer.

4, Identity Verification

- ✓ Credential correctness
- ✓ Account status (active, suspended, revoked)
- ✓ Role authorization level

5, Audit Logging

- ✓ User ID
- ✓ Timestamp
- ✓ IP/device fingerprint
- ✓ Outcome

6, Access Grant / Denial

- ✓ If successful:
 - Secure session token is issued
 - Role-based access rights are loaded
- ✓ If failed:
 - Access is denied
 - System enforces retry limits and lockout policies

7, Legal & Security Notes

- ✓ Drivers **never** authenticate into the system.
- ✓ Login records are **legally admissible audit evidence**.

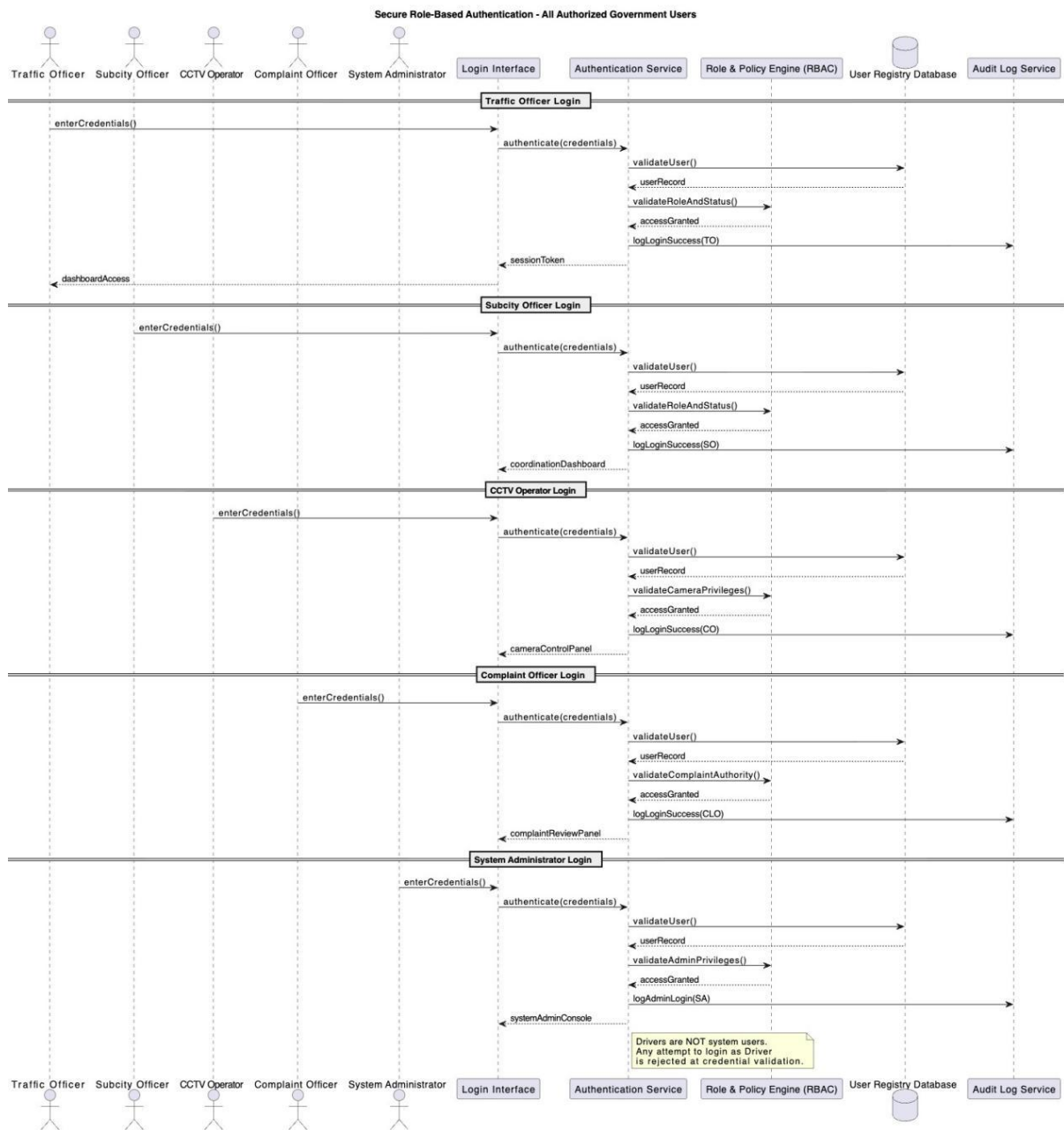


Fig 3.3 Security Login Sequence UML Diagram ALPR STMS, Hawassa

3.13.2.2 Manual Violation Reporting (Traffic Officer)

Purpose

To allow on-site, legally accountable violation creation by traffic officers.

Detailed Sequence Flow

1, Violation Observation

- ✓ Traffic Officer identifies a violation physically on the road.

2, Violation Reporting Initiation

- ✓ Officer selects “Create Violation” from the system dashboard.

3, Automatic Context Capture [Automatic Location Update Service captures]

- ✓ GPS coordinates
- ✓ Time and date
- ✓ Officer identity

4, Violation Data Entry

- ✓ Violation type
- ✓ Vehicle description
- ✓ Plate number (manual or assisted)

5, Violation Record Creation

- ✓ PENDING_CONFIRMATION

6, Penalty Drafting

- ✓ Applicable fine
- ✓ Legal references

7, Audit Trail Creation

- ✓ Full violation creation process is logged.

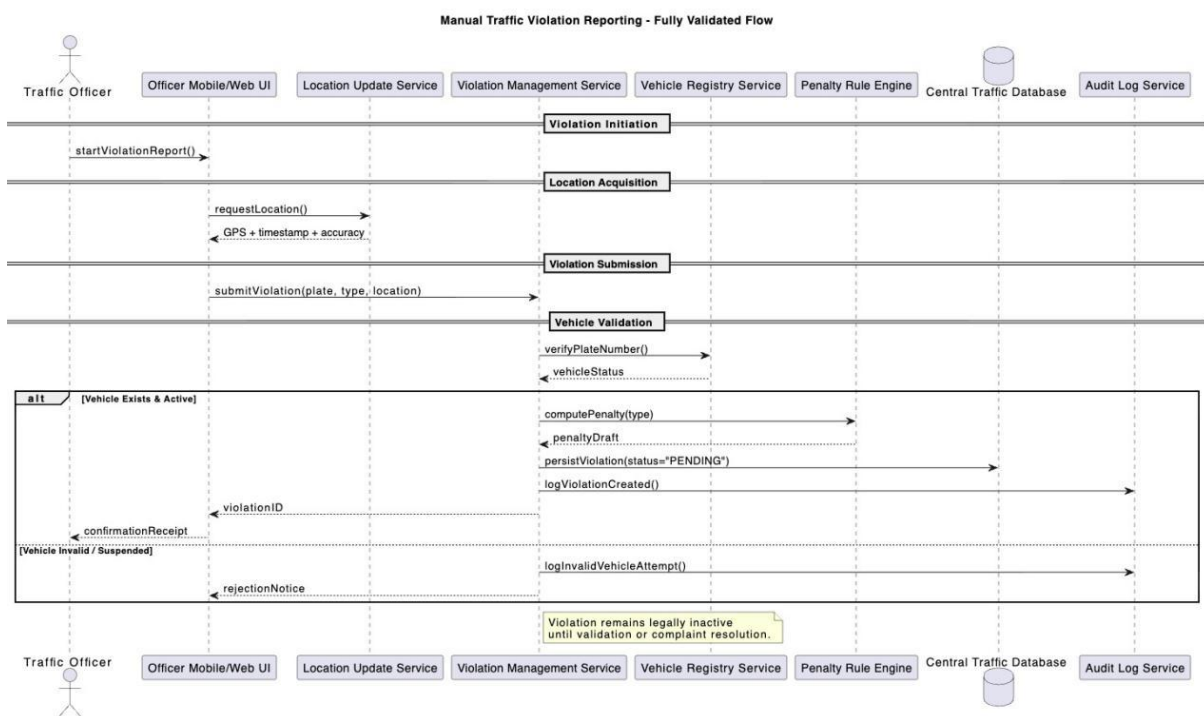


Fig 3.4 Manual Violation Reporting Sequence UML Diagram ALPR STMS, Hawassa

3.13.2.3 CCTV-Assisted Automated Violation Detection (Advanced System-Initiated Detection)

Purpose

To detect violations **without human presence**, while ensuring **human validation before enforcement**.

Detailed Sequence Flow

1, Continuous Video Streaming

- ✓ CCTV cameras stream live footage to the CCTV Feed Manager.

2, Frame Analysis

- ✓ License plates
- ✓ Traffic violations

3, Confidence Scoring

- ✓ Each detection is assigned a confidence score.

4, Decision Gate

- ✓ If confidence < legal threshold:
 - Detection is discarded
 - Logged for transparency
- ✓ If confidence $\geq$ threshold:
 - Proceed to candidate violation creation

5, Location Binding

- ✓ Camera ID
- ✓ Geo-coordinates
- ✓ Timestamp

6, Candidate Violation Creation

- ✓ Stored as UNCONFIRMED
- ✓ No penalty or notification yet

7, Mandatory Human Review

- ✓ CCTV Operator reviews footage and evidence.

8, Validation Outcome

- ✓ Confirm → Violation becomes active
- ✓ Reject → Violation permanently dismissed

3.13.2.4 Educational Route Prediction & Alerting

Purpose

To **predict vehicle movement** and assist officers in **preventive and educational enforcement**, not pursuit alone.

Detailed Sequence Flow

1, Violation Confirmation Trigger

- ✓ Confirmed violation triggers route prediction.

2, Path Analysis

- ✓ Road network
- ✓ Traffic patterns
- ✓ Historical movement data

3, Educational Alerts

- ✓ Subcity officers
- ✓ Nearby traffic officers

Focused on awareness, interception, or guidance

4, Audit Logging

- ✓ Predictions and alerts are logged for accountability.

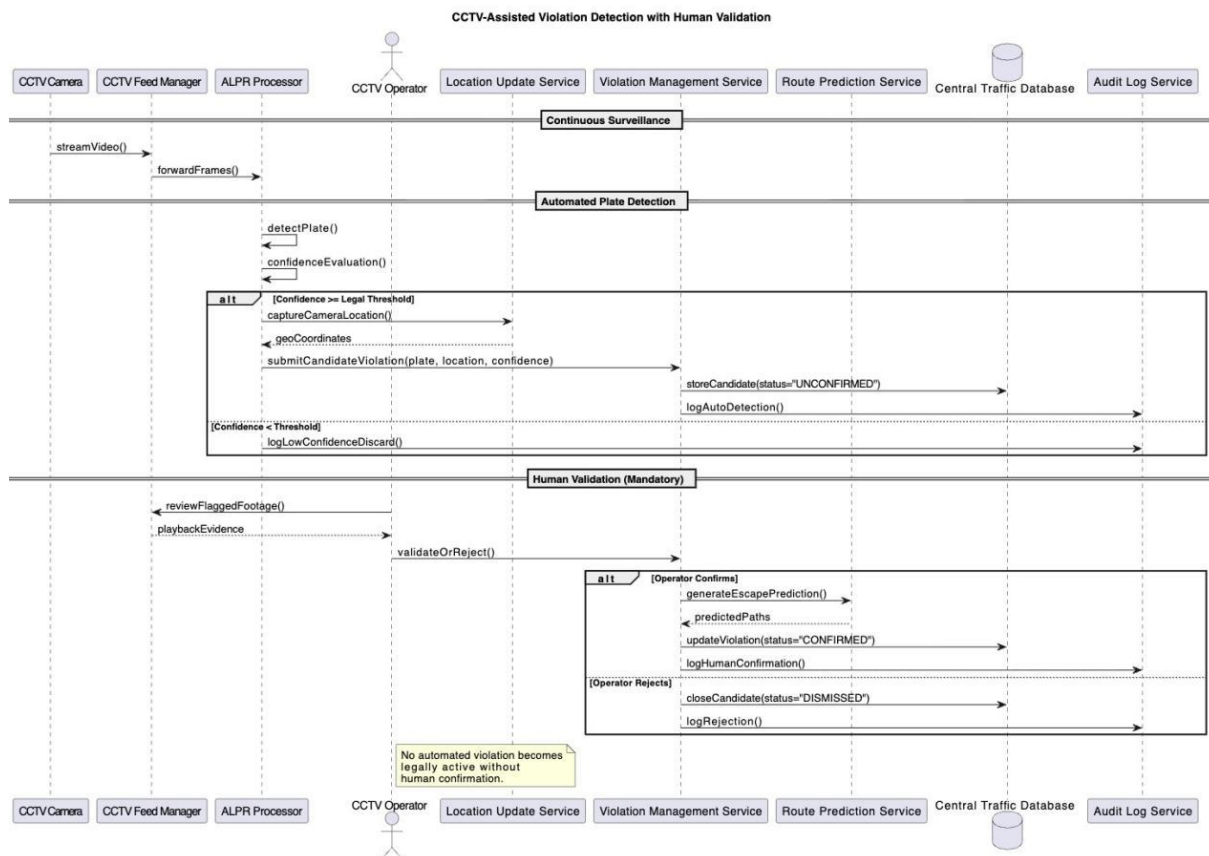


Fig 3.5 CCTV-Assisted Violation Sequence UML Diagram ALPR STMS, Hawassa

3.13.2.5 Inter-Subcity Coordination

Purpose

To ensure **cross-jurisdiction enforcement continuity**.

Detailed Sequence Flow

1, Subcity Escalation

- ✓ Subcity Officer escalates violation when vehicle exits jurisdiction.

2, Central Coordination

- ✓ Alert Service routes information to neighboring subcities.

3, Shared Visibility

- ✓ All involved subcities can view violation status.

4, Audit Logging

- ✓ Escalation paths recorded.

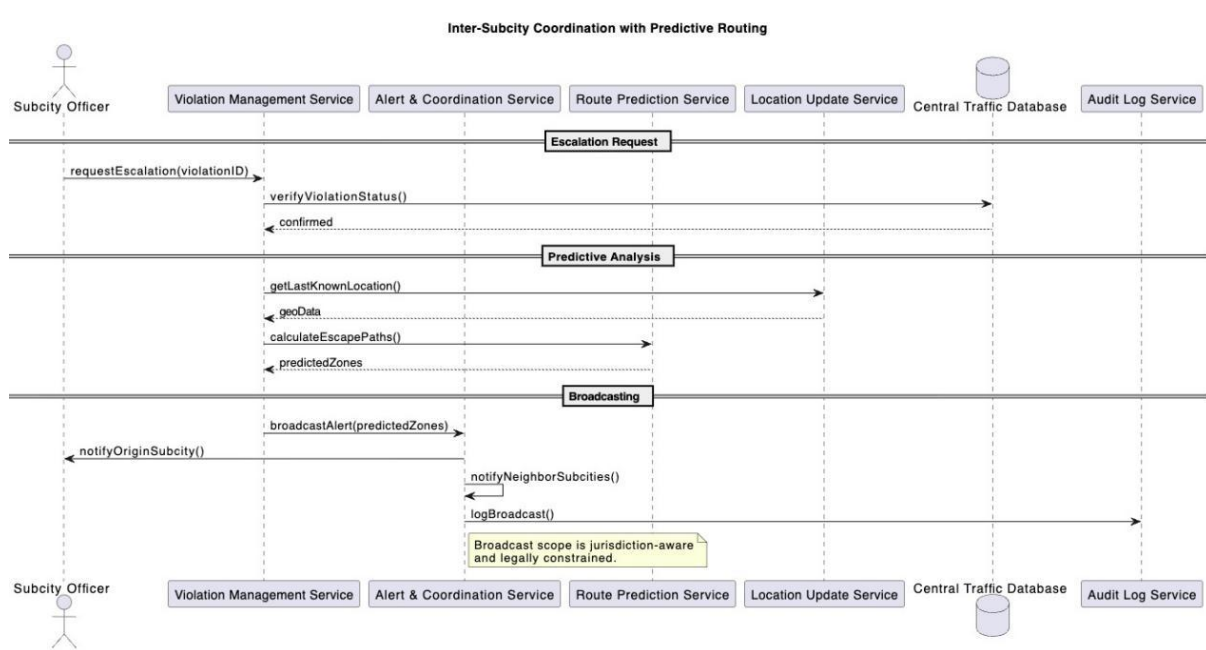


Fig 3.6 Inter-Subcity Coordination Sequence UML Diagram ALPR STMS, Hawassa

3.13.2.6 Complaint Handling (Office-Only, No Driver System Access)

Purpose

To ensure **fair review and legal correction** of violations.

Detailed Sequence Flow

1, Driver Complaint Submission

- ✓ Driver receives violation ID from officer.
- ✓ Driver visits subcity office (no system access).

2, Complaint Retrieval

- ✓ Complaint Officer retrieves violation by ID.

3, Evidence Review

- ✓ CCTV footage requested if applicable.
- ✓ Reporting officer may be contacted.

4, Decision Making

- ✓ Confirm violation → proceed to payment
- ✓ Revoke violation → close permanently

5, Audit Recording

- ✓ Decision rationale recorded.

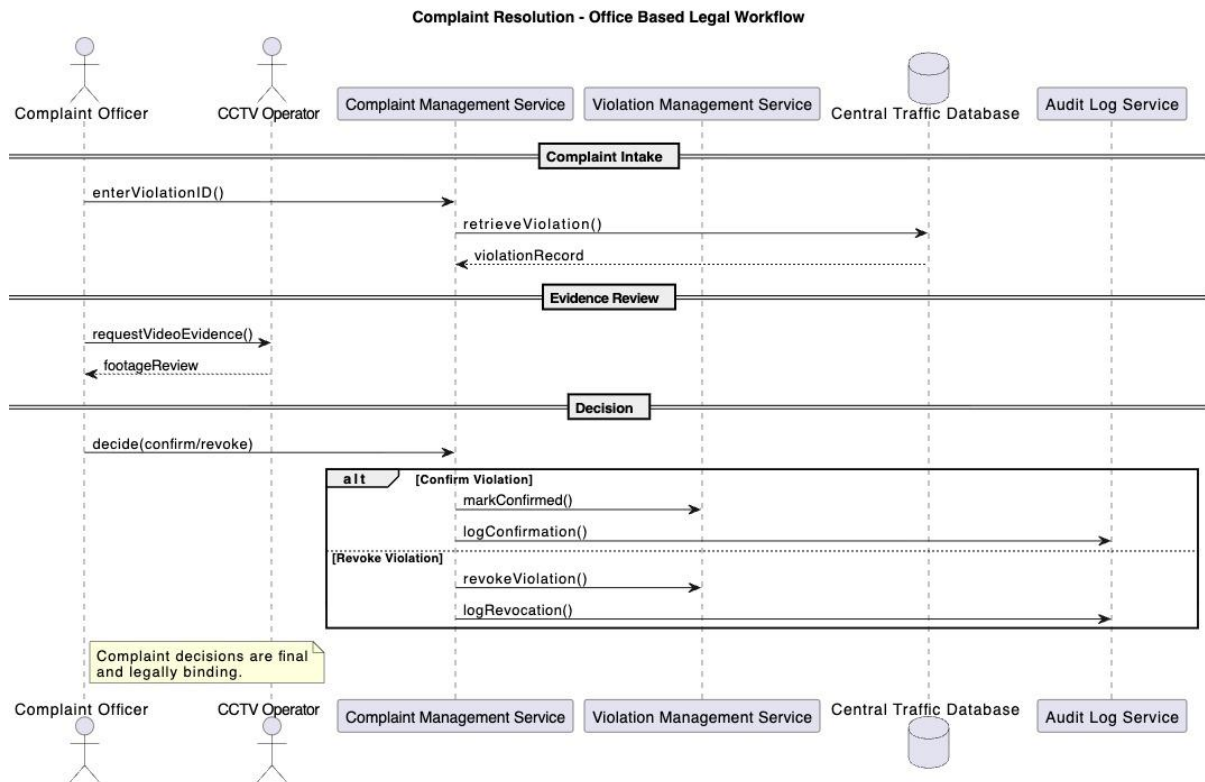


Fig 3.7 Complain Resolution Sequence UML Diagram ALPR STMS, Hawassa

3.13.2.7 Payment Processing (OTP-Only Driver Interaction)

Purpose

To allow **secure, limited, and controlled payment** without system exposure.

Detailed Sequence Flow

1, Payment Initiation

- ✓ Traffic Officer OR
- ✓ Subcity Officer (post-complaint)

2, OTP Generation

- ✓ Payment Service requests OTP from Payment Gateway.

3, Driver Interaction

- ✓ Driver receives OTP.
- ✓ Submits OTP only.

4, Payment Confirmation

- ✓ Gateway confirms payment.

5, System Update

- ✓ Payment stored.
- ✓ Violation marked as PAID.

6, Audit Logging

- ✓ Financial trail fully logged.

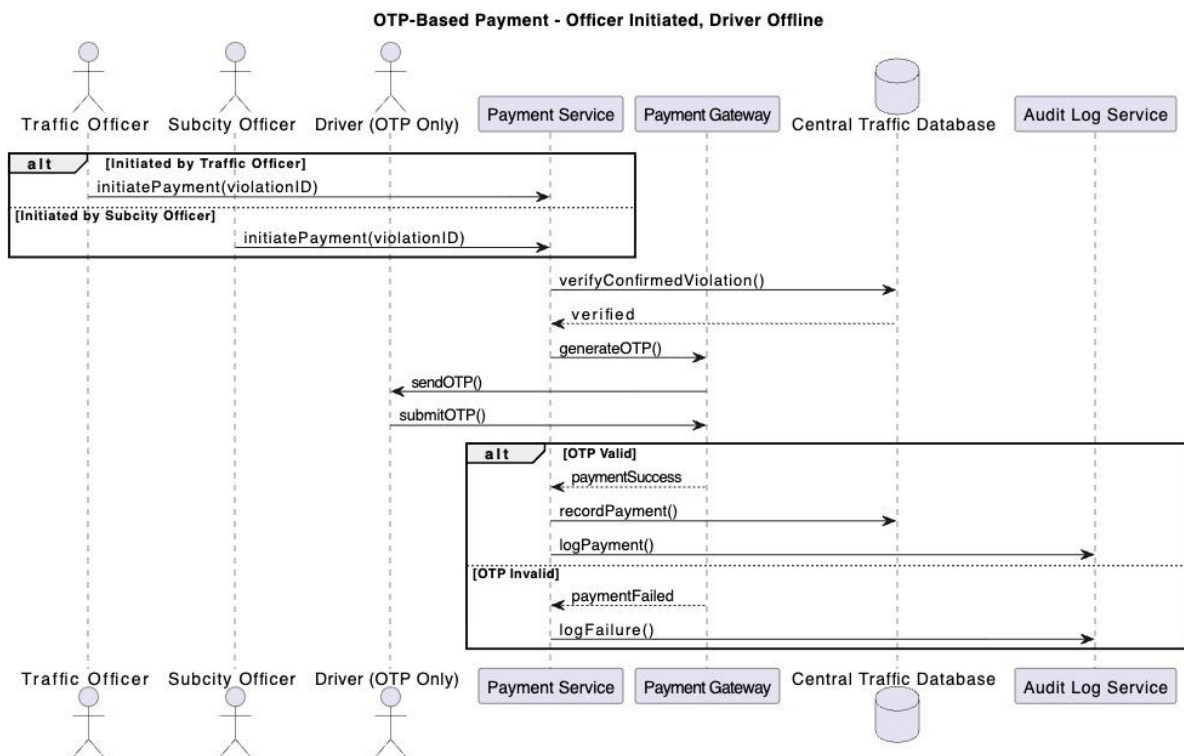


Fig 3.8 OTP based Payment Sequence UML Diagram ALPR STMS, Hawassa

3.13.2.8 System Closure & Reporting

Purpose

To formally conclude enforcement and maintain institutional records.

Detailed Sequence Flow

1, Final Status Review

- ✓ Subcity Officer reviews violation lifecycle.

2, Reporting

- ✓ Legal review
- ✓ Statistical analysis
- ✓ Policy planning

3, Audit Preservation

- ✓ Records retained per government retention laws.

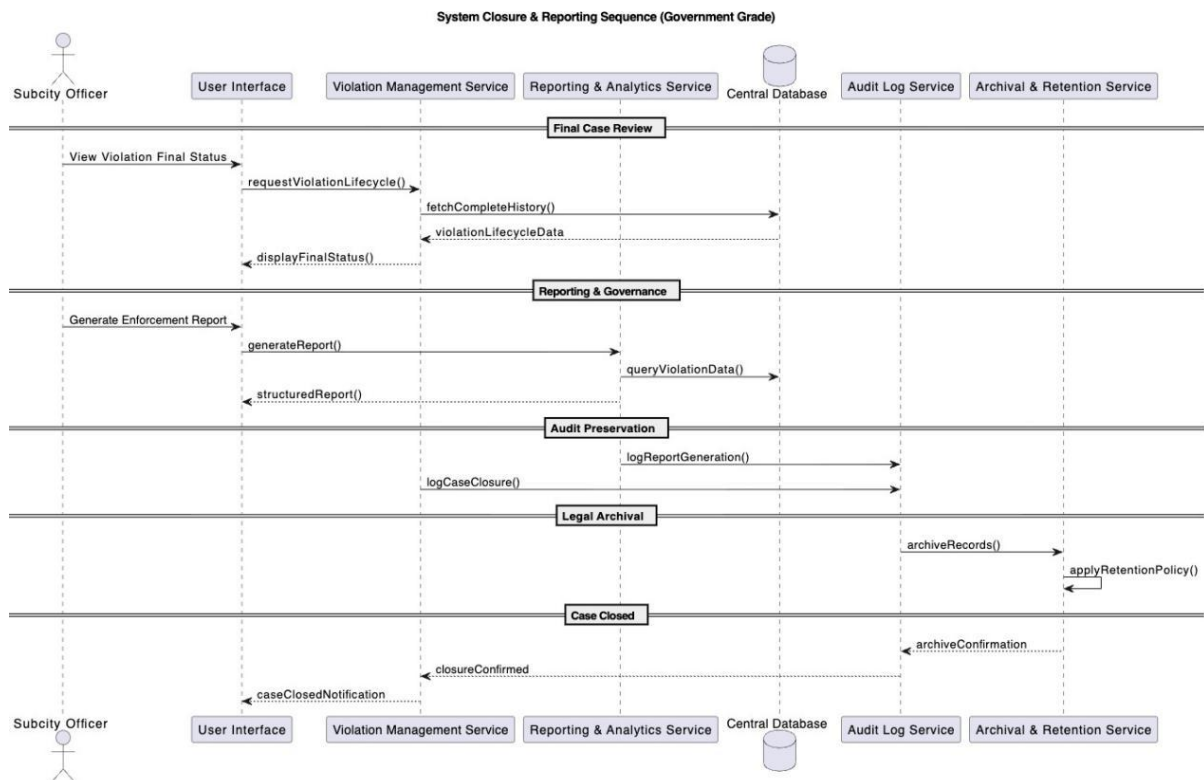


Fig 3.9 System Closure Sequence UML Diagram ALPR STMS, Hawassa

Final Compliance Statement

- ✓ All officials authenticated
- ✓ Driver excluded from system
- ✓ Automation never enforces alone
- ✓ Human accountability preserved
- ✓ Court-defensible audit trails
- ✓ Government-grade operational rigor

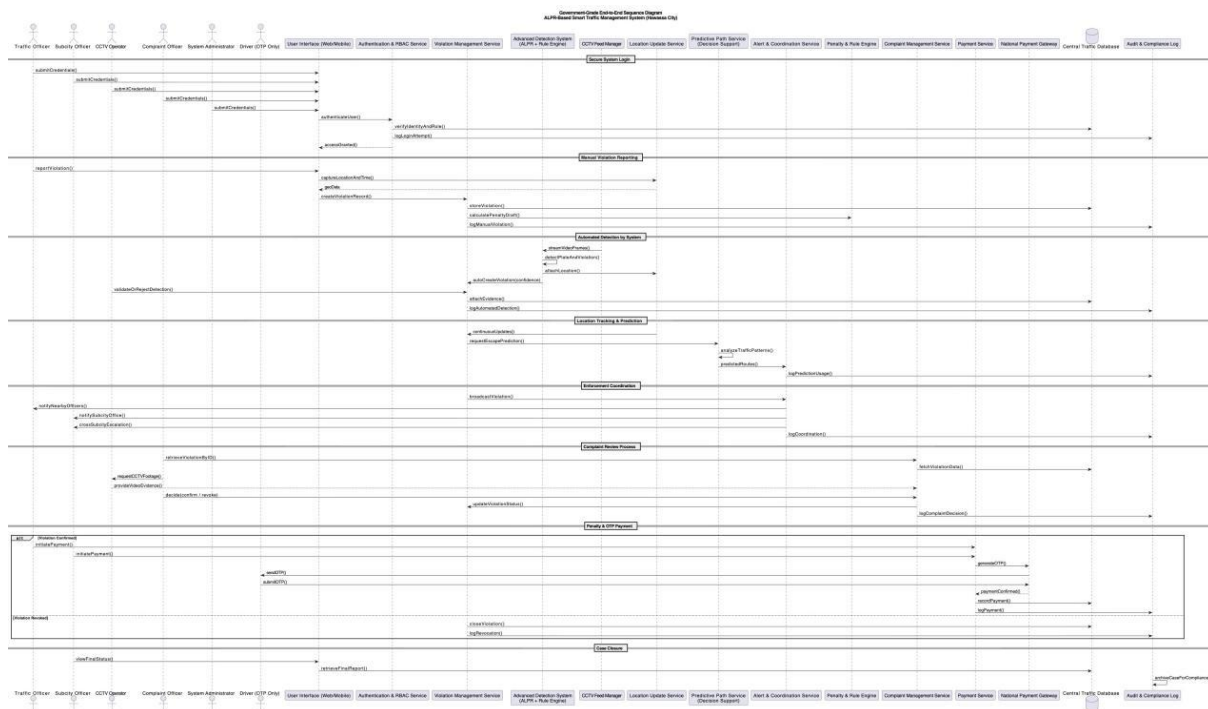


Fig 3.10 Full-System Sequence UML Diagram ALPR STMS, Hawassa

3.14 Activity Diagram – Detailed Functional Documentation

3.14.1 Introduction

The Activity Diagram represents the **end-to-end operational workflow** of the **Automatic License Plate Detection and Recognition–Based Smart Traffic Management System (ALPR-STMS)** deployed for Hawassa City Administration.

This diagram models the **dynamic behavior** of the system by illustrating:

- ✓ The **sequence of actions**
- ✓ The **decision points**
- ✓ The **parallel activities**
- ✓ The **responsibilities of each actor**
- ✓ The **system’s internal processing**

The objective is to ensure that **every operational step is legally valid, traceable, role-controlled, and auditable**, in compliance with government traffic enforcement standards.

3.14.2 Scope and Coverage

The Activity Diagram covers the **complete traffic enforcement lifecycle**, including:

- ✓ Secure system access
- ✓ Manual and automated violation detection
- ✓ Inter-officer coordination
- ✓ Vehicle interception

- ✓ Complaint handling
- ✓ OTP-based payment
- ✓ Case closure and reporting

The diagram explicitly **excludes direct system interaction by drivers**, except during OTP-based payment confirmation, in line with the system's governance model.

3.14.3 Authentication and Role Assignment Phase

Purpose

To ensure that **only authorized government personnel** can access and operate the system.

Process Description

- 1, All system users (Traffic Officers, Subcity Officers, CCTV Operators, Complaint Officers, System Administrators) must **enter valid credentials**.
- 2, The Authentication Service validates credentials against the central authority database.
- 3, Upon successful authentication:
 - A secure session is established.
 - The user is assigned **role-based permissions**.
- 4, If authentication fails, access is denied and the process terminates.

Key Controls

- ✓ Mandatory authentication
- ✓ Role-based access control (RBAC)
- ✓ Audit logging of login events

Legal Significance

Ensures accountability and non-repudiation for every action taken within the system.

3.14.4 Violation Detection Phase (Parallel Activities)

Purpose

To detect traffic violations through **both human observation and automated surveillance**.

Parallel Execution Logic

This phase executes in **parallel**, ensuring maximum coverage and responsiveness.

3.14.4.1 Manual Detection by Traffic Officer

- 1, The Traffic Officer observes a traffic violation.

2, The officer records:

- Vehicle plate number
- Violation type

3, The violation is submitted to the system for processing.

This preserves the **human authority and discretion** required by law.

3.14.4.2 Automated Detection by ALPR & CCTV System

1, CCTV cameras continuously monitor traffic flow.

2, The ALPR subsystem:

- Detects plate numbers
- Analyzes behavior against violation rules

3, If a violation is detected:

- An automated violation report is generated.

4, If no violation is detected, the automated process ends silently.

Governance Principle

Automated detection **supports** enforcement but does not replace human validation.

3.14.5 Violation Registration and Evidence Capture

Purpose

To formally register violations with complete contextual data.

System Actions

- ✓ Capture GPS location and timestamp
- ✓ Store violation with status **Pending**
- ✓ Generate initial audit log entry

Outcome

A legally traceable violation record exists, ready for coordination and enforcement.

3.14.6 Alert Broadcasting and Coordination Phase

Purpose

To enable rapid, structured inter-unit coordination.

Process

- ✓ The Alert & Coordination Service broadcasts the violation.
- ✓ Notifications are sent to:
 - Nearby Traffic Officers
 - Relevant Subcity Officer

Benefits

- ✓ Faster interception
- ✓ Jurisdictional awareness
- ✓ Reduced escape probability

3.14.7 Vehicle Interception and Driver Interaction

Purpose

To safely stop the vehicle and determine the next enforcement path.

Traffic Officer Responsibilities

- ✓ Intercept the vehicle
- ✓ Stop the vehicle using lawful procedures
- ✓ Inform the driver of the violation

Decision Point

The process now diverges based on the driver's response.

3.14.8 Immediate Admission Path (No Complaint)

Condition

Driver admits the violation.

Process

- ✓ Traffic Officer confirms the violation on-site.
- ✓ System finalizes the penalty.
- ✓ Payment authorization is issued.

Key Rule

This path allows **rapid enforcement without administrative delay**.

3.14.9 Complaint Handling Path (Office-Based Only)

Condition

Driver disputes the violation.

Process

- ✓ Traffic Officer issues a **Violation Record ID**.
- ✓ Driver is directed to the Subcity Traffic Office.

3.14.9.1 Complaint Officer Review

- ✓ Complaint Officer logs into the system.
- ✓ Retrieves the violation using the provided ID.
- ✓ Reviews:
 - Officer report
 - Violation details

3.14.9.2 CCTV Evidence Support

- ✓ CCTV Operator provides video evidence **upon request**.
- ✓ Evidence is reviewed for confirmation.

3.14.9.3 Decision Outcomes

Violation Confirmed

- ✓ Penalty is finalized.
- ✓ Payment authorization is issued.

Violation Revoked

- ✓ Violation status updated to **Revoked**.
- ✓ Decision is logged.
- ✓ Process terminates with no payment.

3.14.10 Payment Processing Phase (OTP-Based)

Purpose

To securely collect penalties without exposing drivers to the internal system.

Process

- ✓ Payment Service initiates OTP request.
- ✓ Payment Gateway sends OTP to the driver.
- ✓ Driver submits OTP.
- ✓ Payment is confirmed or marked pending.

Security Controls

- ✓ OTP validation

- ✓ Gateway integration
- ✓ Audit logging

3.14.11 Case Closure and Archival

Purpose

To formally conclude enforcement and preserve records.

Final Actions

- ✓ Subcity Officer reviews final case status.
- ✓ System:
 - Closes the case
 - Archives all records
 - Preserves audit logs
 - Generates analytical and legal reports

Compliance

Data retention follows **government record-keeping laws**.

3.14.12 Summary of Activity Diagram Strengths

The activity diagram ensures:

- ✓ Legal compliance
- ✓ Clear authority boundaries
- ✓ Human-in-the-loop enforcement
- ✓ Secure payment handling
- ✓ Full traceability
- ✓ Audit readiness

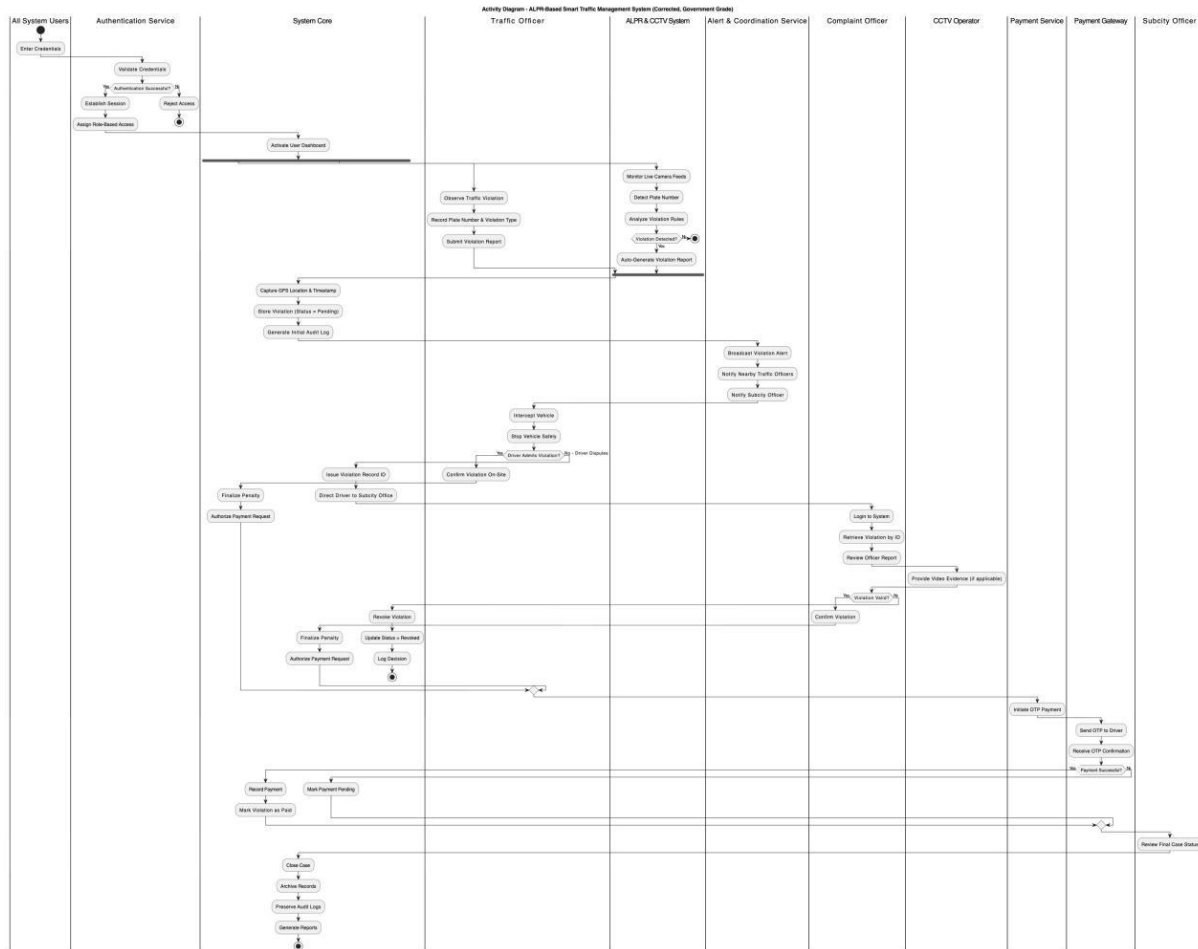


Fig 3.11 Full-System Activity UML Diagram ALPR STMS, Hawassa

3.15 User Interface Prototyping

3.15.1 Introduction

User Interface (UI) Prototyping is a critical activity in system analysis that translates functional and non-functional requirements into **visual, role-specific interaction models**. The objective of UI prototyping in this project is not aesthetic design, but **operational clarity, legal correctness, and usability under real traffic enforcement conditions**.

For the ALPR-Based Smart Traffic Management System (ALPR-STMS), the user interfaces are designed to:

- ✓ Support **fast, error-free enforcement actions**
- ✓ Reflect **institutional roles and authority**
- ✓ Minimize cognitive load on officers in the field
- ✓ Ensure **auditability and traceability**
- ✓ Operate under **low bandwidth and high-stress environments**

All interfaces follow:

- ✓ Role-Based Access Control (RBAC)
- ✓ Government usability standards

- ✓ Minimal input, maximum automation
- ✓ Clear status visibility

3.15.2 User Interface Design Principles

The following principles govern all UI prototypes:

a) Role Separation

Each user role accesses **only the screens relevant to their legal authority**:

- ✓ Traffic Officer
- ✓ Subcity Officer
- ✓ CCTV Operator
- ✓ Complaint Officer
- ✓ System Administrator

Drivers are NOT system users and have no UI access.

b) Field-First Design

Interfaces for traffic officers are:

- ✓ Mobile-first
- ✓ Large buttons
- ✓ Minimal typing
- ✓ GPS-assisted
- ✓ Offline-tolerant

c) Evidence-Centric Enforcement

Every violation screen prioritizes:

- ✓ Plate number
- ✓ Time & location
- ✓ Evidence (image/video)
- ✓ Status lifecycle

d) Audit & Accountability

All screens visibly show:

- ✓ Logged-in officer identity
- ✓ Timestamp
- ✓ Action confirmation
- ✓ Immutable records

3.15.3 UI Prototypes by User Role

3.15.3.1 Login Interface (All Human Operators)

Applicable Users

- ✓ Traffic Officer
- ✓ Subcity Officer
- ✓ CCTV Operator
- ✓ Complaint Officer
- ✓ System Administrator

Purpose

- ✓ To authenticate human users and assign them to their authorized operational environment.

UI Elements

- ✓ Username / Badge Number
- ✓ Password
- ✓ Role auto-resolution (no manual role selection)
- ✓ Login button
- ✓ Error / lockout message

Behavior

- ✓ Credentials validated against central database
- ✓ Role determined automatically
- ✓ Session established with RBAC
- ✓ Login event logged in Audit Log

Security Notes

- ✓ No driver login
- ✓ No OTP here
- ✓ Multi-factor optional (future phase)

3.15.3.2 Traffic Officer Dashboard (Core Operational UI)

Purpose

- ✓ Primary enforcement workspace for field officers.

Main Components

- ✓ Current location (map view)
- ✓ “Report Violation” button
- ✓ Active alerts list
- ✓ Wanted vehicle notifications
- ✓ Shift status

Key Actions

- ✓ Create violation
- ✓ Draw escape path
- ✓ Receive broadcast alerts
- ✓ Initiate penalty request (when driver admits)

3.15.3.3 Violation Reporting Interface

Triggered From

- ✓ Traffic Officer Dashboard

UI Sections

- ✓ **Vehicle Information**
 - Plate number (manual or ALPR)
 - Vehicle type (optional)
- ✓ **Violation Details**
 - Violation type (dropdown)
 - Description (optional)
 - Auto-captured timestamp
- ✓ **Location**
 - Auto GPS
 - Manual adjustment if needed
- ✓ **Evidence**
 - Photo upload
 - CCTV reference (if available)
- ✓ **Prediction**
 - Map canvas to draw escape path
 - Radius suggestion auto-generated
- ✓ **Submit**
 - Confirmation dialog
 - Violation ID generated

3.15.3.4 Alert & Coordination Interface

Used By

- ✓ Traffic Officers
- ✓ Subcity Officers

Features

- ✓ Real-time alert feed
- ✓ Priority color coding
- ✓ Predicted route overlay
- ✓ Inter-subcity escalation controls

Behavior

- ✓ Alerts filtered by proximity
- ✓ Officers acknowledge receipt
- ✓ All actions logged

3.15.3.5 CCTV Operator Interface

Purpose

- ✓ Support ALPR-assisted detection and validation.

UI Components

- ✓ Live camera list
- ✓ Video feed viewer
- ✓ Detected plate panel
- ✓ Confidence score display
- ✓ Approve / Reject detection buttons

Workflow

- ✓ System proposes detection
- ✓ Operator validates
- ✓ Evidence linked to violation
- ✓ Location update broadcasted

3.15.3.6 Complaint Handling Interface (Subcity Office Only)

Used By

- ✓ Complaint Officer

Purpose

- ✓ To ensure legal fairness and due process.

UI Components

- ✓ Violation ID search
- ✓ Violation timeline
- ✓ Evidence viewer
- ✓ CCTV request button

- ✓ Decision panel

Decisions

- ✓ Confirm violation → Payment authorized
- ✓ Revoke violation → Case closed

Driver presence is **physical**, not digital.

3.15.3.7 Payment Interface (Officer-Initiated Only)

Who Can Initiate

- ✓ Traffic Officer (if driver admits)
- ✓ Subcity Officer (after complaint resolution)

UI Elements

- ✓ Violation ID
- ✓ Amount & points
- ✓ “Request OTP Payment” button
- ✓ Payment status indicator

Driver Interaction

- ✓ Receives OTP on phone
- ✓ Enters OTP to payment gateway
- ✓ No system login

3.15.3.8 System Administration Interface

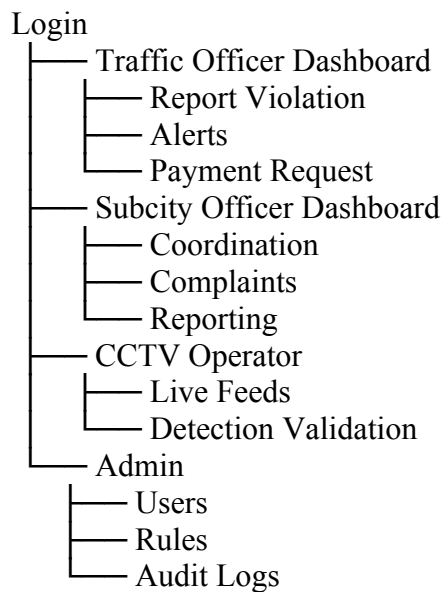
Used By

- ✓ System Administrator

Functions

- ✓ User management
- ✓ Role assignment
- ✓ Violation rule configuration
- ✓ Camera registry
- ✓ Audit review

3.15.4 UI Navigation Structure



3.15.5 Non-Functional UI Requirements

- ✓ Response time < 3 seconds
- ✓ Mobile-friendly (officers)
- ✓ Low bandwidth tolerant
- ✓ Amharic & English ready
- ✓ WCAG-aligned accessibility

3.15.6 User Experience (UX) Justification for Examiners

Purpose of UX Justification

The User Experience (UX) design of the ALPR-Based Smart Traffic Management System is intentionally engineered to support **law enforcement accuracy, legal accountability, operational efficiency, and institutional adoption** rather than consumer convenience.

This section justifies UX decisions based on **government system standards, human factors engineering, and public-sector operational realities**, ensuring the system is **fit for production deployment** within Hawassa City Administration.

3.15.6.1 UX Design Philosophy

The UX design follows a “**Command-and-Control Public System Model**”, not a commercial or citizen-centric application model.

Core Principles Applied

- ✓ Authority-Driven Interaction
- ✓ Error Prevention over Error Recovery
- ✓ Minimal Cognitive Load

- ✓ Legally Traceable Actions
- ✓ Role-Bound Visibility

These principles ensure that **every action is intentional, auditable, and compliant with traffic law enforcement procedures.**

3.15.6.2 Role-Based UX Segregation (Critical Design Decision)

Justification

Each user category operates under **different legal mandates and responsibilities**. Therefore, the UX strictly enforces **role isolation**.

Role	UX Design Rationale
Traffic Officer	Fast, field-ready interfaces prioritizing reporting & alerts
CCTV Operator	High-visibility monitoring, validation-focused screens
Subcity Officer	Coordination dashboards, escalation views
Complaint Officer	Evidence-centric, decision-support interfaces
System Admin	Configuration, governance, audit access
Driver	No system UI access (OTP only)

Table 3.4 Role and UX Design Rationale

3.15.6.3 Login UX Justification (All Human Operators)

Why All Human Operators Must Log In

- ✓ Establishes **individual accountability**
- ✓ Enables **audit trail generation**
- ✓ Supports **role-based access control**
- ✓ Prevents anonymous system actions

Why Drivers Do NOT Log In

- ✓ Drivers are **external subjects**, not system operators
- ✓ Legal procedures require **office-based complaint handling**
- ✓ Prevents digital disputes and identity fraud
- ✓ Aligns with **current Ethiopian traffic enforcement practices**

UX Outcome: Reduced system complexity, higher legal defensibility.

3.15.6.4 Minimalist Interface Design Justification

Why the UI Avoids Visual Complexity

- ✓ Users operate under **stressful, time-critical conditions**
- ✓ Many officers have **varying levels of digital literacy**
- ✓ Government systems prioritize **clarity over aesthetics**

UX Decisions

- ✓ Neutral colors
- ✓ Large actionable elements
- ✓ Explicit labels (no icons-only actions)
- ✓ No hidden or gesture-based controls

UX Outcome: Faster training, fewer operational errors, higher adoption.

3.15.6.5 Evidence-Centric UX Design

Justification

Traffic enforcement decisions must withstand:

- ✓ Legal scrutiny
- ✓ Complaints and appeals
- ✓ Administrative review

UX Implementation

- ✓ Evidence always visible, never hidden
- ✓ No deletion rights at operational levels
- ✓ Clear linkage between:
 - Violation
 - Evidence
 - Decision
 - Payment

UX Outcome: Strong chain of custody and institutional trust.

3.15.6.6 Alert & Broadcast UX Justification

Why Alerts Are Centralized and Prioritized

- ✓ Traffic enforcement is **time-sensitive**
- ✓ Missed alerts can result in:
 - Escaped vehicles
 - Public safety risks
 - Administrative failure

UX Design Choices

- ✓ Priority-coded alerts
- ✓ Geographic relevance filtering
- ✓ Read-only alert messages (no alteration)

UX Outcome: Faster interception and coordinated response.

3.15.6.7 Complaint Handling UX Justification

Why Complaints Are Office-Based Only

- ✓ Ensures **human judgment**
- ✓ Prevents emotional or impulsive digital disputes
- ✓ Allows evidence verification with officers and CCTV operators

UX Characteristics

- ✓ Step-by-step review screens
- ✓ Mandatory decision logging
- ✓ Evidence request workflow embedded

UX Outcome: Fairness, transparency, and legal compliance.

3.15.6.8 Payment UX Justification (OTP-Only for Drivers)

Why OTP Payment Is the Only Driver Interaction

- ✓ Reduces system exposure to the public
- ✓ Eliminates need for driver accounts
- ✓ Aligns with national digital payment practices

Why Payment Can Be Initiated by:

- ✓ Traffic Officer (admitted violation)
- ✓ Subcity Officer (post-complaint)

UX Outcome: Secure, simple, and fraud-resistant payment flow.

3.15.6.9 Predictive Path & Location UX Justification

Why Map-Based Interaction Is Used

- ✓ Matches officers' real-world mental models
- ✓ Improves situational awareness
- ✓ Supports coordination across subcities

UX Safeguards

- ✓ Prediction shown as **assistive**, not authoritative
- ✓ Human override always allowed

UX Outcome: Decision support without removing human authority.

3.15.6.10 Audit & Transparency UX Justification

Why Users See Audit Indicators

- ✓ Reinforces professional responsibility
- ✓ Discourages misuse
- ✓ Aligns with public-sector accountability culture

UX Implementation

- ✓ “All actions are logged” notices
- ✓ No silent system actions

UX Outcome: Ethical use and institutional trust.

3.16 Supplementary Specifications

3.16.1 Scope and Purpose

This section specifies **supplementary system requirements** that apply across the entire ALPR-Based Smart Traffic Management System (ALPR-STMS) and are not fully expressed through functional requirements or UML models.

These requirements address:

- ✓ Quality attributes
- ✓ Legal and regulatory constraints
- ✓ Operational constraints
- ✓ Security, reliability, and usability
- ✓ External system dependencies

3.16.2 Security Requirements

Authentication

The system **shall require authentication** for all human users, including:

- ✓ Traffic Officers
- ✓ Subcity Officers
- ✓ CCTV Operators
- ✓ Complaint Officers
- ✓ System Administrators

Drivers **shall not authenticate** into the system.

Authorization

- ✓ The system **shall enforce role-based access control (RBAC)**.
- ✓ Users **shall only access functions explicitly permitted** by their assigned role.

Auditability

The system **shall log all security-critical actions**, including:

- ✓ Login and logout
- ✓ Violation creation and updates
- ✓ Payment initiation and confirmation
- ✓ Complaint decisions

Audit logs **shall be tamper-resistant and retained** according to government regulations.

3.16.3 Legal and Regulatory Constraints

- ✓ The system **shall comply with**:
- ✓ *Road Transport Traffic Control Council of Ministers Regulation No. 557/2024*
- ✓ National surveillance and data protection laws

CCTV footage **shall only be accessed by authorized personnel** and used solely for traffic enforcement.

Evidence records **shall be retained** according to legally mandated retention periods.

3.16.4 Performance Requirements

- ✓ The system **shall process manual violation reports in near real-time**.
- ✓ Alert broadcasts **shall occur within operationally acceptable latency** after violation confirmation.
- ✓ ALPR processing **shall not degrade core enforcement operations** during peak traffic hours.

3.16.5 Reliability and Availability

- ✓ The system **shall support fault tolerance**, ensuring manual enforcement remains operational if:
 - CCTV feeds fail
 - Network connectivity is disrupted
- ✓ Automated backups **shall be performed regularly** to prevent data loss.
- ✓ The system **shall support graceful recovery** after partial failures.

3.16.6 Data Integrity and Consistency

- ✓ All violations **shall maintain a complete lifecycle history**.
- ✓ Violation records **shall not be physically deleted**.
- ✓ Data consistency **shall be enforced** across:
 - Violation records
 - Penalties
 - Payments
 - Complaints
- ✓ State transitions **shall be logged and auditable**.

3.16.7 External Interface and Dependency Constraints

The system **shall integrate with**:

- ✓ Existing CCTV infrastructure
- ✓ Approved payment gateways (OTP-based)
- ✓ Government vehicle registration databases (read-only)

The system **shall remain operational** if any external interface is temporarily unavailable.

3.16.8 Usability and Human Factors

- ✓ User interfaces **shall be simple, consistent, and task-focused**.
- ✓ Mobile interfaces **shall be optimized** for field officers.
- ✓ System workflows **shall align with existing enforcement procedures** to minimize training requirements.

3.16.9 Maintainability and Scalability

- ✓ The system **shall support incremental deployment** across subcities.
- ✓ New CCTV cameras and users **shall be added without architectural redesign**.
- ✓ Business rules **shall be configurable without source code modification**.

3.16.10 Assumptions and Dependencies

Assumptions

- ✓ Traffic enforcement personnel will adopt digital reporting.
- ✓ Vehicle registration data is available digitally.

Dependencies

- ✓ Network connectivity
- ✓ CCTV feed availability
- ✓ Approved payment service providers

3.17 Change Management Process

This project follows a structured and controlled change management approach to ensure system stability, traceability, and compliance throughout the development lifecycle. Given the evolving operational and regulatory environment of traffic management, changes to system requirements or design may be necessary during analysis, design, or implementation phases.

Any proposed change to the **requirements, system design, workflows, or technical architecture** shall be formally documented and justified. Changes may arise due to operational feedback, regulatory updates (including traffic control regulations), or technical constraints identified during implementation.

All change requests shall undergo the following process:

- ✓ The change is documented, clearly stating the nature of the change and the justification.
- ✓ An impact analysis is conducted to assess effects on system functionality, design, schedule, and compliance.
- ✓ The proposed change is reviewed and approved by the designated supervisory or academic authority before implementation.
- ✓ Approved changes are incorporated into the relevant project artifacts, including the SRS, UML diagrams, and design documentation.

Unapproved changes shall not be implemented. This controlled approach ensures that the system remains consistent with its original objectives while allowing necessary and justified evolution during development.

CHAPTER FOUR

SYSTEM DESIGN

4.1 Introduction

This chapter presents the **System Design** of the ALPR-Based Smart Traffic Management System for Hawassa City. While Chapter Three focused on requirements analysis and behavioral modeling, this chapter translates those validated requirements into a **structured, implementable, and maintainable system design** suitable for **government deployment and long-term operational use**.

System design in this context serves as the **technical blueprint** that bridges stakeholder requirements with actual software construction. It defines how system components are organized, how responsibilities are distributed across architectural layers, how data flows through the system, and how the system ensures **scalability, security, reliability, and legal compliance** in a real-world public-sector environment.

4.1.1 Design Objectives

The primary objectives guiding the system design are:

Modularity and Maintainability

- ✓ To ensure that system components can be independently developed, tested, upgraded, and maintained without disrupting overall operations.

Government-Grade Reliability and Accountability

- ✓ To support auditable enforcement workflows, evidence preservation, and traceability of all actions performed by human operators and automated subsystems.

Security and Role-Based Control

- ✓ To enforce strict separation of responsibilities among traffic officers, subcity officers, CCTV operators, complaint officers, and system administrators, while explicitly excluding drivers from direct system access.

Incremental Automation Support

- ✓ To allow the system to evolve from manual violation reporting to advanced AI-assisted detection, prediction, and coordination without architectural redesign.

Integration with Existing Infrastructure

- ✓ To seamlessly integrate with existing CCTV networks, payment gateways, and government data repositories while respecting institutional boundaries and data governance rules.

4.1.2 Design Scope

The System Design phase defines **how** the ALPR-STMS will be constructed, structured, and deployed, without altering the validated requirements.

The design scope includes:

- ✓ High-level system architecture
- ✓ Subsystem decomposition
- ✓ Component responsibilities and interfaces
- ✓ Data flow and control flow structures
- ✓ Security and access control architecture
- ✓ Integration with external systems (CCTV, payment gateways)
- ✓ Deployment and scalability considerations

Important Design Constraint

The design strictly preserves:

- ✓ Human authority over enforcement decisions
- ✓ Subcity-based administrative hierarchy
- ✓ Auditability and traceability of all actions

4.1.3 Architectural Philosophy

The system design follows **public-sector software engineering principles**, including:

a) Authority-Centric Design

Human officers remain the final decision-makers at every enforcement stage.

b) Layered Responsibility

Clear separation between:

- ✓ Presentation layer (UI)
- ✓ Application services
- ✓ Domain logic
- ✓ Infrastructure services

c) Phased Capability Enablement

Design supports gradual activation of:

- ✓ Manual reporting
- ✓ CCTV-assisted detection
- ✓ Predictive path analysis
- ✓ Automated detection (future phase)

d) Audit-First Architecture

Every critical operation generates immutable audit records.

4.1.4 Compliance and Standards Alignment

The system design aligns with:

- ✓ **ISO/IEC 25010** – Software quality models (security, reliability, maintainability)
- ✓ **ISO 9241** – Human-centered system interaction (UI considerations)
- ✓ **Government IT governance principles** – Auditability, traceability, and data protection
- ✓ **Public safety system best practices** – Human-in-the-loop decision enforcement

These standards guide both architectural decisions and detailed design choices presented in the subsequent sections.

Analysis Artifact	Design Outcome
Use Case Diagrams	Service boundaries and APIs
Sequence Diagrams	Component interaction design
Activity Diagrams	Control flow and state transitions
Class Diagram	Domain model and persistence design
UX Prototypes	Presentation layer structure
Non-Functional Requirements	Architecture constraints

Table 4.1 Relationship Between Analysis Artifacts and Design Artifacts

4.1.5 Design Readiness Statement

Based on the completed analysis and validation activities:

- ✓ Functional requirements are stable
- ✓ Non-functional requirements are measurable
- ✓ Legal and institutional constraints are respected
- ✓ UX is compliant with ISO 9241
- ✓ Operational workflows are verified

The project is therefore **fully prepared to proceed into the System Design phase**.

4.1.6 Chapter Organization

This chapter is structured as follows:

- ✓ **Section 4.2** introduces the **Class-Type Architectural Layers**, defining responsibilities across the user interface, controller, business, persistence, and system layers.
- ✓ **Section 4.3** presents detailed **Class Modeling**, translating domain concepts into structured software entities.
- ✓ **Section 4.4 – 4.9** elaborate on behavioral, collaborative, component, deployment, persistence, and user interface designs, forming a complete production-ready system design specification.

4.1.7 System Architecture Design

4.1.7.1 Introduction to System Architecture

System Architecture defines the **structural blueprint** of the Automatic License Plate Detection and Recognition–Based Smart Traffic Management System (ALPR-STMS). It specifies how system components are organized, how they interact, and how responsibilities are distributed across layers to ensure **scalability, security, reliability, and legal compliance**.

This architecture is designed specifically for **municipal government deployment**, taking into account:

- ✓ Existing CCTV infrastructure
- ✓ Subcity-based administrative governance
- ✓ Human-centered enforcement authority
- ✓ Incremental technological adoption

The architecture translates previously defined **requirements, workflows, and UML models** into a **deployable technical structure**.

4.1.7.2 Architectural Style Selection

Selected Architecture Style: Layered Service-Oriented Architecture (LSOA)

The ALPR-STMS adopts a **Layered Service-Oriented Architecture (LSOA)** combining:

- ✓ **Layered Architecture** for clarity and control
- ✓ **Service-Oriented Principles** for modularity and scalability

This hybrid approach is preferred for government systems because it:

- ✓ Enforces separation of concerns
- ✓ Simplifies auditing and maintenance
- ✓ Supports phased expansion without redesign

4.2 Class Type Architecture

4.2.1 Overview

The ALPR-Based Smart Traffic Management System adopts a **layered class-type architecture** to ensure **separation of concerns, security enforcement, scalability, and maintainability**. This architecture is intentionally aligned with **enterprise and government IT system design standards**, where each layer has **clearly defined responsibilities**, controlled interaction paths, and auditable boundaries.

The architecture is structured into the following logical layers:

- ✓ **User Interface Layer**
- ✓ **Controller / Process Layer**
- ✓ **Business (Domain) Layer**

- ✓ **Persistence Layer**
- ✓ **System / Integration Layer**

Each layer is implemented using **well-defined classes**, and no layer bypasses another. This design prevents unauthorized access, reduces coupling, and enables controlled system evolution.

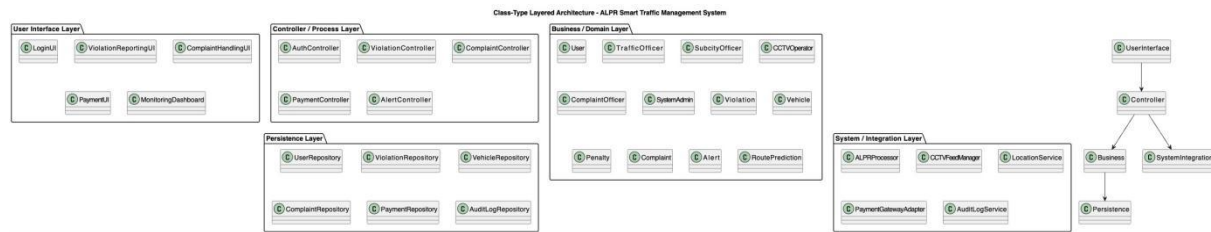


Fig 4.1: Class-type layered architectural Diagram UML Diagram ALPR STMS, Hawassa

4.2.2 Presentation Layer (User Interaction Layer)

Purpose:

The User Interface (UI) Layer provides **role-specific interaction points** for all authorized human operators. It **does not contain business logic** and **never communicates directly with the database**.

Users Served:

- ✓ Traffic Officer
- ✓ Subcity Officer
- ✓ CCTV Operator
- ✓ Complaint Officer
- ✓ System Administrator

Driver is intentionally excluded from this layer as it's only integrated only on OTP payment.

Components:

- ✓ Web-based dashboard (administrative users)
- ✓ Mobile interface (traffic officers)
- ✓ Secure login and session management
- ✓ Role-based UI rendering

Representative Classes

- ✓ LoginUI
- ✓ ViolationReportingUI
- ✓ ComplaintHandlingUI
- ✓ MonitoringDashboard
- ✓ PaymentUI (used by officers during OTP-based payment facilitation)

Key Characteristics:

- ✓ Role-aware interfaces (Traffic Officer, Subcity Officer, CCTV Operator, Complaint Officer, Admin)
- ✓ Secure login for **all human actors except drivers**
- ✓ Read-only or restricted interfaces where enforcement authority is not required
- ✓ Supports desktop and secured workstation environments
- ✓ ISO 9241 compliant
- ✓ Government accessibility standards
- ✓ Minimal cognitive load

4.2.3 Controller / Process Layer

Purpose:

The Controller Layer acts as the orchestration and coordination layer, handling all requests from the UI and enforcing workflow sequencing. Implements business processes and enforces workflow rules.

Core Services:

- ✓ Violation Management Service
- ✓ Alert & Coordination Service
- ✓ Complaint Management Service
- ✓ Payment Service
- ✓ User & Role Management Service
- ✓ Audit Logging Service

Responsibilities:

- ✓ Enforce process order
- ✓ Validate actions against rules
- ✓ Coordinate between subsystems
- ✓ Prevent unauthorized operations
- ✓ Role authorization enforcement
- ✓ Routing requests to appropriate business services
- ✓ Handling success/failure flows

Representative Classes

- ✓ AuthController
- ✓ ViolationController
- ✓ ComplaintController
- ✓ PaymentController
- ✓ AlertController

Key Design Rule

- ✓ **No UI component is allowed to access business logic directly.**
- ✓ **All requests must pass through a controller.**

This ensures **auditability, traceability, and security enforcement.**

4.2.4 Domain & Intelligence (Business/Domain) Layer

Purpose:

This layer represents the **core intelligence and rules of the traffic enforcement domain**. It models real-world concepts and enforces legal and procedural constraints, independent of UI or infrastructure.

Core Domain Entities

- ✓ User (abstract)
 - TrafficOfficer
 - SubcityOfficer
 - CCTVOperator
 - ComplaintOfficer
 - SystemAdmin
- ✓ Violation
- ✓ Vehicle
- ✓ Penalty
- ✓ Complaint
- ✓ Alert
- ✓ RoutePrediction

Responsibilities

- ✓ Violation lifecycle management
- ✓ Penalty calculation rules
- ✓ Complaint decision logic (confirm / revoke)
- ✓ Enforcement status transitions
- ✓ Route prediction logic (decision support only)

Key Modules:

- ✓ Violation Rule Engine
- ✓ Penalty Calculation Engine
- ✓ ALPR Processing Engine
- ✓ Route Prediction & Path Estimation Module
- ✓ Evidence Validation Logic

Special Note:

Even when automated detection is enabled:

- ✓ **Final violation confirmation remains human-controlled**
- ✓ AI outputs are advisory, not authoritative

4.2.5 Data Management (Persistence) Layer

Purpose:

The Persistence Layer abstracts all data storage operations and enforces **data integrity, consistency, auditability, and security**.

Responsibilities

- ✓ CRUD operations
- ✓ Transaction management
- ✓ Historical data retention
- ✓ Legal evidence preservation

Primary Data Stores:

- ✓ Central Violation Database
- ✓ Vehicle & Plate Registry
- ✓ CCTV Evidence Repository
- ✓ Payment Records
- ✓ Complaint Records
- ✓ Audit Logs

Data Governance Rules:

- ✓ Immutable audit records
- ✓ Role-restricted access
- ✓ Retention per government policy
- ✓ Legal traceability of every action

Representative Classes

- ✓ UserRepository
- ✓ ViolationRepository
- ✓ VehicleRepository
- ✓ ComplaintRepository
- ✓ PaymentRepository
- ✓ AuditLogRepository

4.2.6 Infrastructure & Integration (System) Layer

Purpose:

This layer handles **external systems and advanced service** integrations, isolating them from core business logic.

Integrated Services

- ✓ ALPRProcessor – AI-based plate recognition
- ✓ CCTVFeedManager – video stream handling
- ✓ LocationService – GPS and geo-context services
- ✓ PaymentGatewayAdapter – OTP payment integration

- ✓ AuditLogService – centralized audit recording

Design Justification

- ✓ Prevents vendor lock-in
- ✓ Enables replacement or upgrade of AI models
- ✓ Protects core logic from external system failures

Deployment Model:

- ✓ On-premise government servers
- ✓ Secure VPN access
- ✓ Failover and redundancy support

Actor	Presentation	Services	Domain	Data
Traffic Officer	✓	✓	✓	Limited
Subcity Officer	✓	✓	✓	Limited
CCTV Operator	✓	✓	✓	Evidence Only
Complaint Officer	✓	✓	✓	Complaint Scope
System Admin	✓	✓	✓	Full
Driver	X	OTP Only	X	Payment Only

Table 4.2 Actor to Layer Mapping

4.2.7 Architectural Security Principles

The architecture embeds security by design:

- ✓ Role-Based Access Control (RBAC)
- ✓ Encrypted communication (TLS)
- ✓ Multi-factor authentication (officials)
- ✓ OTP-based payment authentication (driver)
- ✓ Full auditability of all actions

Phase	Capability
Phase 1	Manual reporting + alerts
Phase 2	CCTV-assisted detection
Phase 3	Predictive route analysis
Phase 4	Automated detection (human-validated)

Table 4.3 Architectural Support for Phased Deployment

The architecture allows **progressive activation** without structural changes.

4.2.8 Architectural Quality Attributes

The system architecture explicitly satisfies:

- ✓ **Scalability:** Add cameras, users, subcities
- ✓ **Reliability:** Redundant services and backups
- ✓ **Maintainability:** Modular service design
- ✓ **Auditability:** Immutable logs
- ✓ **Legal Compliance:** Human authority preserved

4.2.9 Architecture Validation Statement

- ✓ Fully aligns with system requirements
- ✓ Supports government operational models
- ✓ Preserves legal enforcement authority
- ✓ Enables smart city expansion
- ✓ Is suitable for real-world deployment

Aspect	Benefit
Security	Strict role boundaries and access control
Maintainability	Layer isolation reduces change impact
Scalability	Services can scale independently
Auditability	Centralized logging and traceability
Legal Compliance	Human-in-the-loop enforcement preserved

Table 4.4 Architectural Benefits Summary

4.3 Class Modeling

4.3.1 Purpose of Class Modeling

Class modeling defines the **static structural blueprint** of the system by identifying the key classes, their attributes, operations, and relationships. In the ALPR-Based Smart Traffic Management System, class modeling ensures that:

- ✓ Real-world traffic enforcement entities are accurately represented
- ✓ Responsibilities are clearly assigned
- ✓ Dependencies are controlled and traceable
- ✓ The design supports extensibility, auditability, and legal compliance

This model serves as the **foundation for implementation, testing, and future system evolution**.

4.3.2 Class Modeling Scope

The class model focuses on:

- ✓ **Core domain entities** (e.g., Violation, Vehicle, Penalty)

- ✓ **User hierarchy** representing authorized human actors
- ✓ **Supporting service classes** required for enforcement, prediction, and auditing
- ✓ **Clear separation between data entities and service/process classes**

Drivers are **explicitly excluded** as system users, consistent with legal and operational requirements.

4.3.3 User Class Hierarchy

User (Abstract Class)

The User class defines common identity and accountability attributes shared by all authorized personnel.

Key Attributes

- ✓ User identifier
- ✓ Personal identification details
- ✓ Assigned role

Design Rationale

- ✓ Enforces role-based access control
- ✓ Avoids redundancy
- ✓ Supports centralized authentication and auditing

Class	Responsibility
TrafficOfficer	Report violations, initiate roadside penalties
SubcityOfficer	Coordinate enforcement and review cases
CCTVOperator	Validate ALPR detections and evidence
ComplaintOfficer	Adjudicate disputes and complaints
SystemAdmin	Manage system configuration and users

Table 4.5 Specialized User Classes

4.3.4 Core Domain Classes

Vehicle

Represents a registered vehicle identified by its license plate.

- ✓ Serves as a reference entity
- ✓ Can be linked to multiple violations over time
- ✓ Immutable historical linkage ensures legal traceability

Violation (Central Domain Class)

The Violation class is the **core entity** around which the entire system operates.

Responsibilities

- ✓ Track the lifecycle of a traffic offense
- ✓ Link evidence, penalties, alerts, and complaints
- ✓ Maintain legally auditable status transitions

Typical States

- ✓ Reported
- ✓ Under Review
- ✓ Confirmed
- ✓ Complained
- ✓ Revoked
- ✓ Paid

Penalty

Represents the enforcement outcome when a violation is confirmed.

- ✓ Created only after validation
- ✓ May result in OTP-based payment
- ✓ Status-controlled to prevent duplicate or unauthorized payments

Complaint

Encapsulates a formal dispute raised by a driver through an office process.

- ✓ Linked to exactly one violation
- ✓ Reviewed exclusively by Complaint Officers
- ✓ Decision outcomes directly affect violation and penalty status

Alert

Represents system-generated or officer-triggered notifications.

- ✓ Distributed based on geography and priority

Supports inter-subcity coordination

- ✓ Enhances response efficiency

Route Prediction

A decision-support class that provides **predicted movement paths** for a suspected vehicle.

- ✓ Uses historical and real-time data
- ✓ Assists officers but does not enforce actions
- ✓ Designed as advisory intelligence only

4.3.5 Relationship Design Principles

- ✓ **Aggregation over inheritance** is used where legal independence is required (e.g., Violation–Penalty)
- ✓ **One-to-many relationships** preserve historical enforcement data
- ✓ **Optional associations** reflect real-world uncertainty (e.g., not all violations lead to complaints)

These principles ensure **accuracy, flexibility, and legal defensibility**.

4.3.6 Compliance and Design Quality Considerations

- ✓ Supports forensic auditing and evidence preservation
- ✓ Prevents unauthorized state transitions
- ✓ Aligns with public-sector software accountability requirements
- ✓ Enables future expansion (e.g., additional sensors or analytics)

4.4 State Chart Modeling

4.4.1 Introduction

State chart modeling describes the **dynamic behavior** of critical system entities by illustrating how they transition from one state to another in response to events, conditions, and decisions.

For the ALPR-Based Smart Traffic Management System, state charts are essential to ensure:

- ✓ Legal correctness of enforcement workflows
- ✓ Controlled decision-making authority
- ✓ Auditability and traceability
- ✓ Prevention of invalid or unauthorized transitions

This section focuses on **state-critical domain entities**, primarily the **Violation**, which governs the lifecycle of traffic enforcement actions.

4.4.2 Scope of State Modeling

State chart modeling is applied to entities that:

- ✓ Have a **clear lifecycle**
- ✓ Undergo **formal decision points**
- ✓ Require **legal and administrative control**

Modeled Entities

- ✓ **Violation** (primary and mandatory)
- ✓ **Complaint** (secondary, embedded within violation lifecycle)
- ✓ **Penalty** (state-dependent outcome)

Among these, the **Violation State Chart** is the authoritative control model for the entire system.

4.4.3 Violation State Chart – Conceptual Overview

A **Violation** progresses through a strictly controlled lifecycle beginning at detection/reporting and ending with either **payment completion** or **revocation**.

Key design principles:

- ✓ No state skipping allowed
- ✓ All transitions are event-driven
- ✓ Decisions are role-restricted
- ✓ Every transition is audit-logged

4.4.3.1 Detailed State Descriptions

1. Reported

Entry Condition

- ✓ Violation detected manually by Traffic Officer
- ✓ OR automatically detected by ALPR system

Characteristics

- ✓ Initial immutable record created
- ✓ Contains timestamp and provisional location

2. UnderReview

Trigger

- ✓ Evidence attachment (image/video/location)

Actors

- ✓ Traffic Officer
- ✓ CCTV Operator (if ALPR-based)

Purpose

- ✓ Validate legitimacy before city-wide escalation

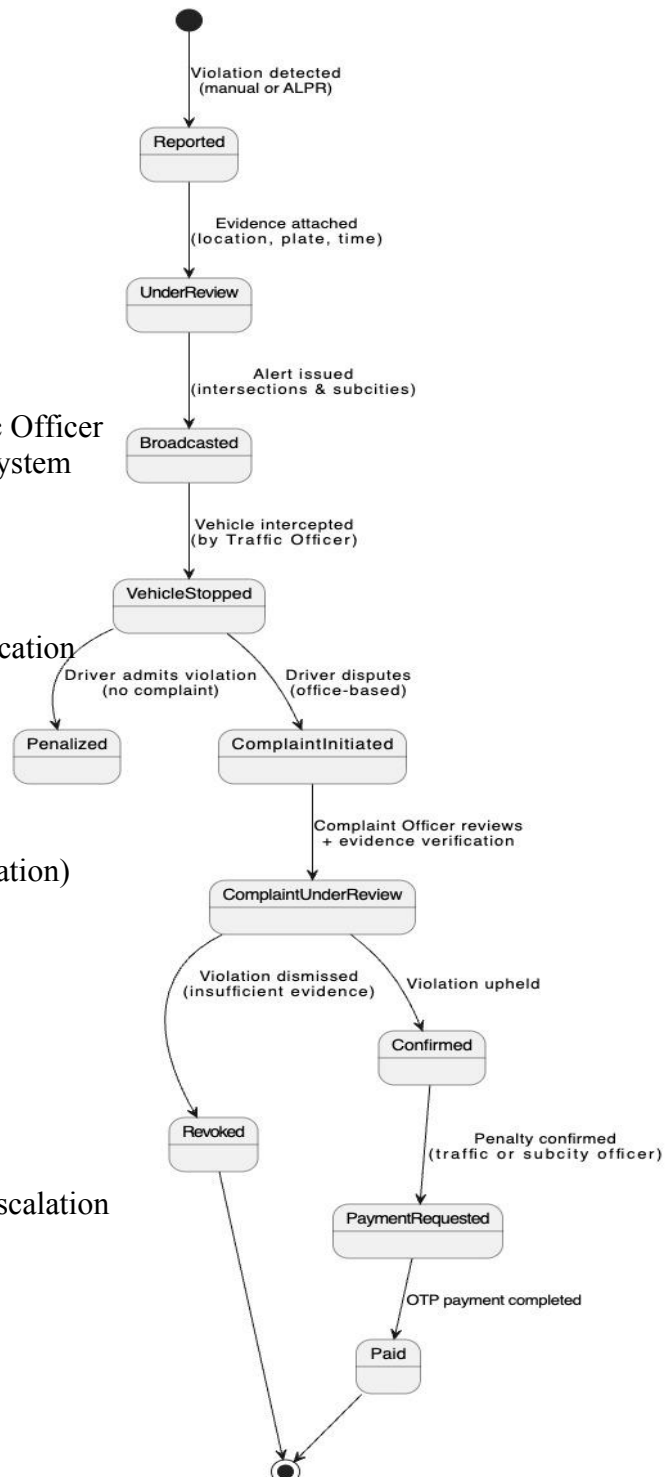
3. Broadcasted

Trigger

- ✓ Confirmation of violation validity

Fig 4.2: Violation State Chart Diagram

Violation State Chart - ALPR Smart Traffic Management System



System Actions

- ✓ Alert dissemination to:
 - Nearby officers
 - Subcity control rooms
 - Border checkpoints (if applicable)

4. VehicleStopped

Trigger

- ✓ Physical interception of the vehicle

Importance

- ✓ No penalty or complaint can proceed without interception
- ✓ Ensures procedural fairness

5. Penalized

Condition

- ✓ Driver admits violation
- ✓ No complaint requested

Action

- ✓ Penalty prepared immediately
- ✓ Payment request initiated by Traffic Officer or Subcity Officer

6. ComplaintInitiated

Condition

- ✓ Driver disputes violation
- ✓ Complaint registered **only at office level**

Important Rule

- ✓ Driver never interacts digitally with enforcement system

7. ComplaintUnderReview

Actors

- ✓ Complaint Officer
- ✓ CCTV Operator (if evidence requested)

Decision Inputs

- ✓ Stored violation data
- ✓ CCTV footage
- ✓ Officer report

8. Revoked

Outcome

- ✓ Violation dismissed
- ✓ Penalty canceled
- ✓ Case closed permanently

Legal Significance

- ✓ Prevents further enforcement or payment attempts

9. Confirmed

Outcome

- ✓ Complaint rejected
- ✓ Violation upheld

Next Step

- ✓ Mandatory payment initiation

10. PaymentRequested

Actors

- ✓ Traffic Officer (roadside)
- ✓ OR Subcity Officer (post-complaint)

System Behavior

- ✓ OTP payment generated
- ✓ Payment gateway engaged

11. Paid

Final State

- ✓ Payment successfully completed
- ✓ Violation closed
- ✓ Records archived per government retention policy

4.4.4 Complaint State Behavior (Embedded)

Although not modeled separately, complaints operate as a **controlled sub-state** within the violation lifecycle.

Key guarantees:

- ✓ One complaint per violation
- ✓ No payment allowed during complaint review
- ✓ Decision authority limited to Complaint Officer

4.4.5 Design & Compliance Considerations

- ✓ Prevents illegal enforcement
- ✓ Supports judicial review
- ✓ Ensures evidence-based decisions
- ✓ Fully auditable for government oversight

The state chart ensures that **technology enforces policy**, not the other way around.

4.5 Collaboration Modeling

4.5.1 Introduction

Collaboration Modeling describes **how system actors and software components work together** to accomplish business objectives.

Unlike class diagrams (static) or state charts (behavioral per entity), collaboration modeling focuses on:

- ✓ Responsibility distribution
- ✓ Interaction correctness
- ✓ Operational authority
- ✓ Information flow integrity

In the ALPR-Based Smart Traffic Management System, collaboration modeling is critical because the system operates in a **multi-agency, legally sensitive environment**, involving:

- ✓ Traffic enforcement officers
- ✓ Subcity administrative units
- ✓ CCTV operations
- ✓ Complaint handling authorities
- ✓ Payment infrastructure

This section ensures that **no actor exceeds authority, no process bypasses legal control, and all interactions are traceable and auditable**.

4.5.2 Collaboration Modeling Scope

Collaboration is modeled across **five primary operational domains**:

- ✓ Violation Detection & Reporting
- ✓ Alert Broadcasting & Coordination
- ✓ CCTV-Assisted Evidence Collaboration
- ✓ Complaint Handling & Decision Making
- ✓ Penalty Payment & Closure

Each collaboration group is described using:

- ✓ Participating actors
- ✓ Collaborating system components
- ✓ Responsibility boundaries
- ✓ Interaction rules

4.5.3 Key Collaboration Participants

Human Actors

- ✓ Traffic Officer
- ✓ Subcity Officer
- ✓ CCTV Operator
- ✓ Complaint Officer
- ✓ Driver (OTP payment only; no system access)

System Components

- ✓ Authentication Service
- ✓ Violation Management Service
- ✓ ALPR Processor
- ✓ CCTV Feed Manager
- ✓ Alert & Coordination Service
- ✓ Penalty Engine
- ✓ Complaint Management Service
- ✓ Payment Service
- ✓ Central Database
- ✓ Audit Log Service

4.5.4 Collaboration Scenario 1: Violation Detection & Reporting

Objective

To collaboratively detect, validate, and formally register a traffic violation.

Participants

- ✓ Traffic Officer
- ✓ ALPR Processor
- ✓ Violation Management Service
- ✓ Location Service
- ✓ Central Database
- ✓ Audit Log Service

Collaboration Flow

- ✓ Traffic Officer reports a violation manually **OR**
- ✓ ALPR Processor detects a violation automatically
- ✓ Location Service provides GPS and timestamp
- ✓ Violation Management Service consolidates all inputs
- ✓ Central Database persists the violation
- ✓ Audit Log records the action

Key Collaboration Rule

- ✓ Detection authority is shared, but registration authority is centralized.

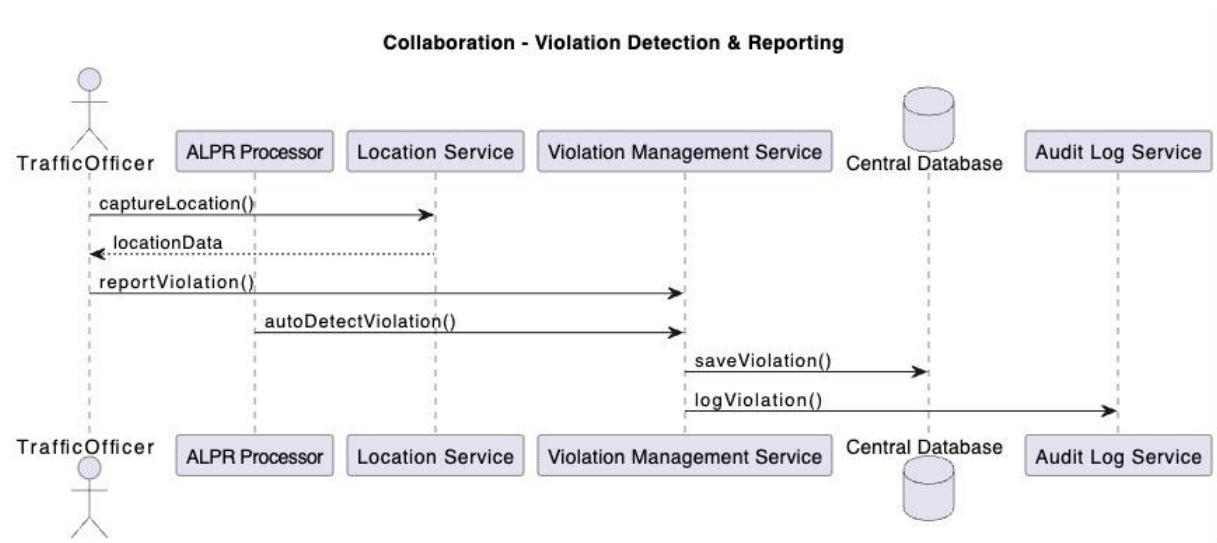


Fig 4.3: Collaboration Diagram – Violation Detection and Reporting

4.5.5 Collaboration Scenario 2: Alert Broadcasting & Inter-Subcity Coordination

Objective

To ensure rapid and geographically targeted enforcement response.

Participants

- ✓ Violation Management Service
- ✓ Alert & Coordination Service
- ✓ Traffic Officers
- ✓ Subcity Officers
- ✓ Route Prediction Service

Collaboration Flow

- ✓ Violation Management Service triggers alert
- ✓ Alert Service requests predicted escape routes
- ✓ Alerts broadcast based on proximity and jurisdiction
- ✓ Subcity Officers coordinate escalation

Key Collaboration Rule

- ✓ Only Subcity Officers may authorize inter-subcity escalation.

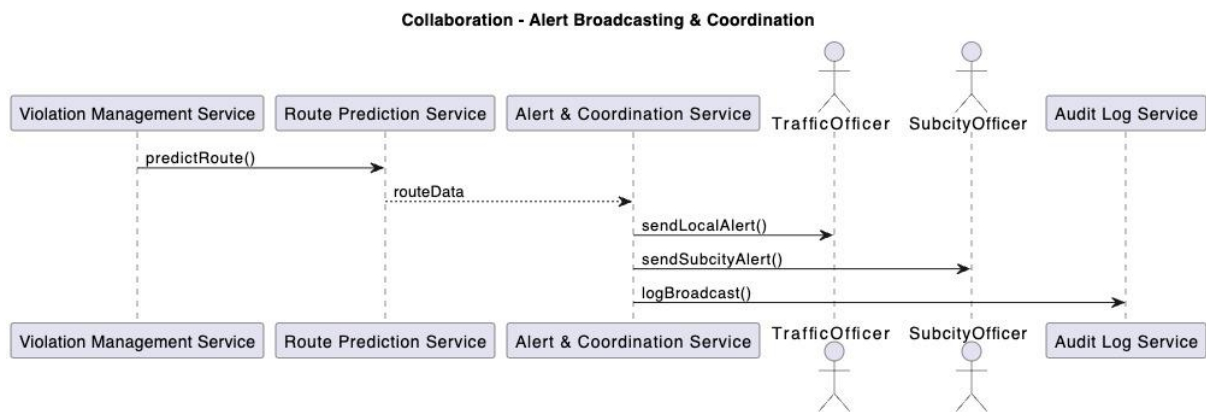


Fig 4.4: Collaboration Diagram Alert & Coordination

4.5.6 Collaboration Scenario 3: CCTV-Assisted Evidence Validation

Objective

To validate violations using video evidence without automated enforcement authority.

Participants

- ✓ CCTV Operator
- ✓ CCTV Feed Manager
- ✓ ALPR Processor
- ✓ Violation Management Service

Collaboration Flow

- ✓ CCTV feeds are monitored continuously
- ✓ ALPR detects candidate plates
- ✓ CCTV Operator validates or rejects detection
- ✓ Evidence attached to violation record

Key Collaboration Rule

- ✓ Human validation is mandatory before legal enforcement.

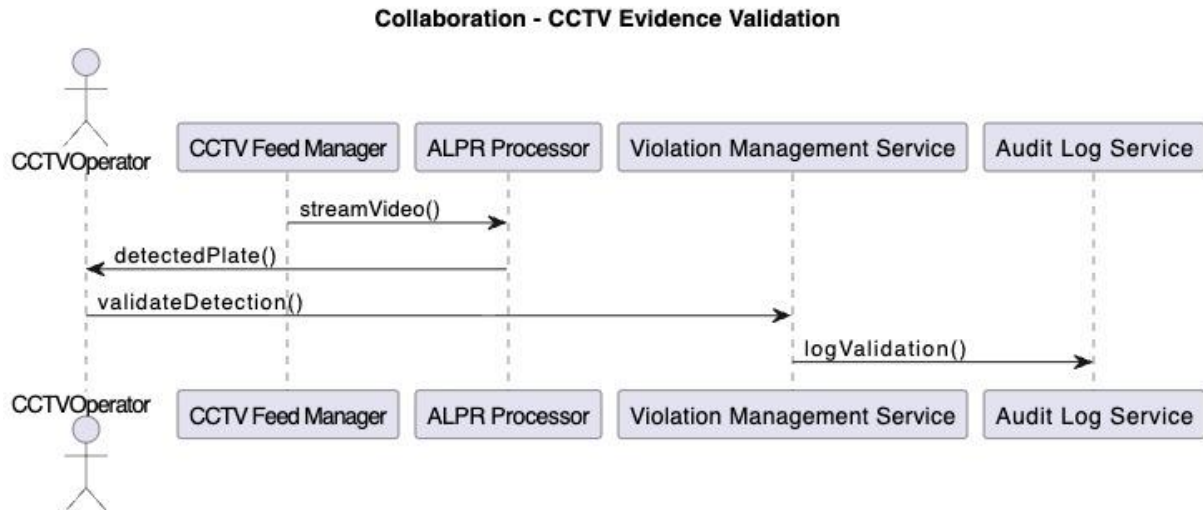


Fig 4.5: Collaboration Diagram – CCTV Validation

4.5.7 Collaboration Scenario 4: Complaint Handling & Legal Decision

Objective

To ensure fair adjudication of disputed violations.

Participants

- ✓ Complaint Officer
- ✓ Complaint Management Service
- ✓ Violation Management Service
- ✓ CCTV Operator
- ✓ Central Database

Collaboration Flow

- ✓ Driver submits violation ID at office
- ✓ Complaint Officer retrieves case
- ✓ Evidence reviewed and optionally requested
- ✓ Decision recorded (Confirm / Revoke)

Key Collaboration Rule

- ✓ Complaint decisions override all automated outcomes.

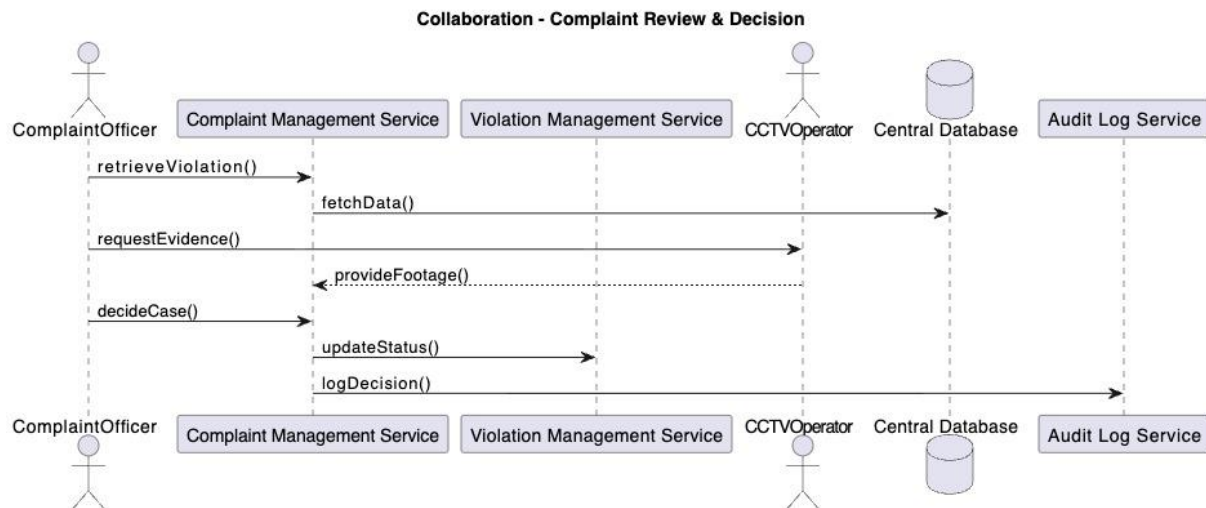


Fig 4.6: Collaboration Diagram – Complaint Handling

4.5.8 Collaboration Scenario 5: Penalty Payment & Case Closure

Objective

To complete enforcement legally and financially.

Participants

- ✓ Traffic Officer / Subcity Officer
- ✓ Payment Service
- ✓ Payment Gateway
- ✓ Driver (OTP only)
- ✓ Audit Log Service

Collaboration Flow

- ✓ Authorized officer initiates payment
- ✓ Payment Service requests OTP
- ✓ Driver receives and submits OTP
- ✓ Payment confirmed and logged
- ✓ Violation closed

Key Collaboration Rule

- ✓ Drivers interact only with payment gateway, never with enforcement system.

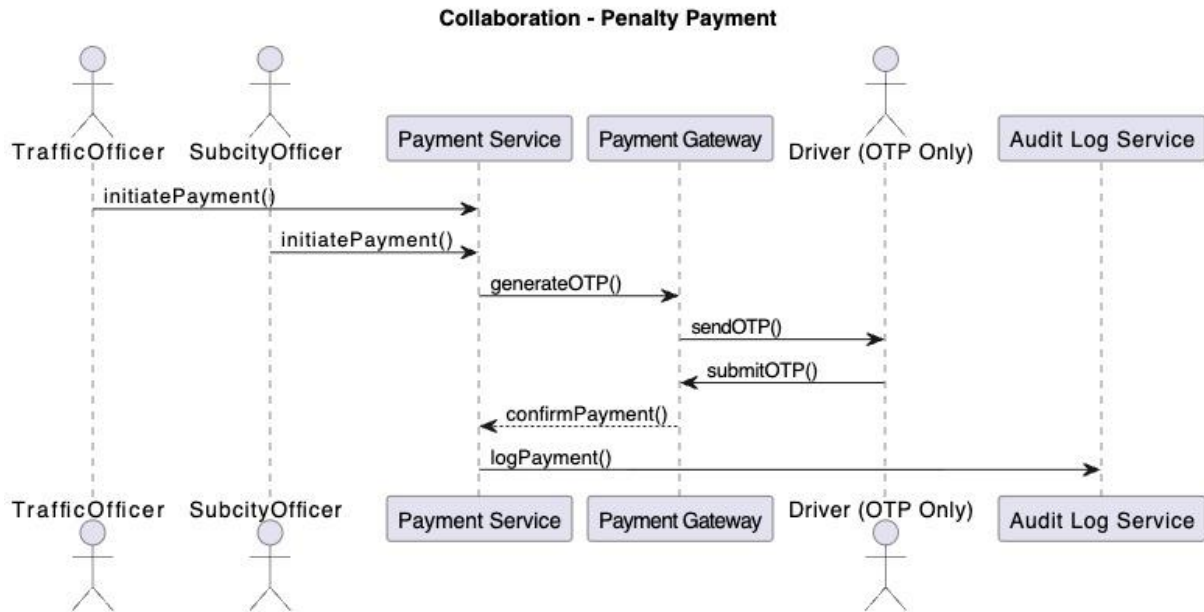


Fig 4.7: Collaboration Diagram – Payment

4.5.9 Collaboration Integrity & Governance Controls

- ✓ All collaborations are **role-restricted**
- ✓ No direct driver-system interaction
- ✓ All decisions are evidence-based
- ✓ Every collaboration produces audit logs
- ✓ Inter-subcity actions require authorization

4.6 Component Modeling

4.6.1 Introduction

Component Modeling defines the **logical and physical decomposition of the system into deployable, replaceable, and maintainable components**.

While collaboration modeling focused on who interacts with whom, component modeling focuses on:

- ✓ **What software units implement those interactions**
- ✓ **How responsibilities are isolated**
- ✓ **How government-grade reliability, security, and auditability are achieved**

For the ALPR-Based Smart Traffic Management System (ALPR-STMS), component modeling is especially critical because the system must operate across **law enforcement, administrative governance, surveillance infrastructure, and financial systems**, each with different legal and operational constraints.

4.6.2 Component Modeling Objectives

The component model is designed to achieve the following objectives:

Separation of Concerns

- ✓ Each component has a single, well-defined responsibility.

Role-Based Enforcement

- ✓ Components enforce authority boundaries defined in the SRS.

Auditability and Legal Traceability

- ✓ Every critical action passes through auditable components.

Scalability and Future Expansion

- ✓ AI-based automation and smart city integration can be added without redesign.

Fault Isolation

- ✓ Failures in one component (e.g., payment gateway) do not compromise core enforcement operations.

4.6.3 High-Level Component Architecture

The system is composed of **five major component groups**:

- ✓ Presentation Components
- ✓ Control & Orchestration Components
- ✓ Core Domain Components
- ✓ Infrastructure & Integration Components
- ✓ Governance & Audit Components

Each group is described in detail below.

4.6.4 Presentation Components

Purpose

To provide secure, role-based access to system functionalities without exposing internal business logic.

Components

- ✓ Web User Interface
- ✓ Mobile Officer Interface
- ✓ CCTV Control Dashboard

Responsibilities

- ✓ Credential input and session initiation
- ✓ Data visualization (violations, alerts, status)
- ✓ Controlled action triggering (no direct data manipulation)

Governance Rule

- ✓ **Presentation components never perform enforcement logic.**

4.6.5 Control & Orchestration Components

Purpose

To coordinate workflows, validate permissions, and route requests to appropriate domain components.

Components

- ✓ Authentication Service
- ✓ Authorization & Role Manager
- ✓ Workflow Controller

Responsibilities

- ✓ Login and session validation (all human actors except drivers)
- ✓ Role-based access enforcement
- ✓ Workflow sequencing (e.g., violation → alert → penalty)

Governance Rule

- ✓ **All user actions must pass through orchestration components.**

4.6.6 Core Domain Components

Purpose

To implement the **legal and operational logic** of traffic enforcement.

Components

- ✓ Violation Management Service
- ✓ Penalty Engine
- ✓ Complaint Management Service
- ✓ Route Prediction Service

Responsibilities

- ✓ Violation lifecycle management
- ✓ Penalty calculation based on legal rules

- ✓ Complaint adjudication support
- ✓ Predictive path analysis (decision support only)

Legal Constraint

- ✓ No component may autonomously enforce penalties without human authorization.

4.6.7 Surveillance & Intelligence Components

Purpose

To support evidence-based enforcement using CCTV and AI-assisted detection.

Components

- ✓ CCTV Feed Manager
- ✓ ALPR Processor
- ✓ Evidence Repository

Responsibilities

- ✓ Video ingestion and stream management
- ✓ Plate detection and confidence scoring
- ✓ Evidence attachment to violations

Governance Rule

- ✓ AI detection assists decision-making but does not replace human judgment.

4.6.8 Financial & External Integration Components

Purpose

To manage legally compliant payment processing and external system communication.

Components

- ✓ Payment Service
- ✓ Payment Gateway Adapter
- ✓ Notification Service (OTP, SMS)

Responsibilities

- ✓ Payment initiation by authorized officers
- ✓ OTP-based driver interaction
- ✓ Secure confirmation handling

Critical Constraint

- ✓ Drivers have no system access beyond OTP interaction.

4.6.9 Governance & Audit Components

Purpose

To ensure accountability, compliance, and traceability.

Components

- ✓ Audit Log Service
- ✓ Reporting & Analytics Service

Responsibilities

- ✓ Immutable logging of actions
- ✓ Government reporting support
- ✓ Legal evidence preservation

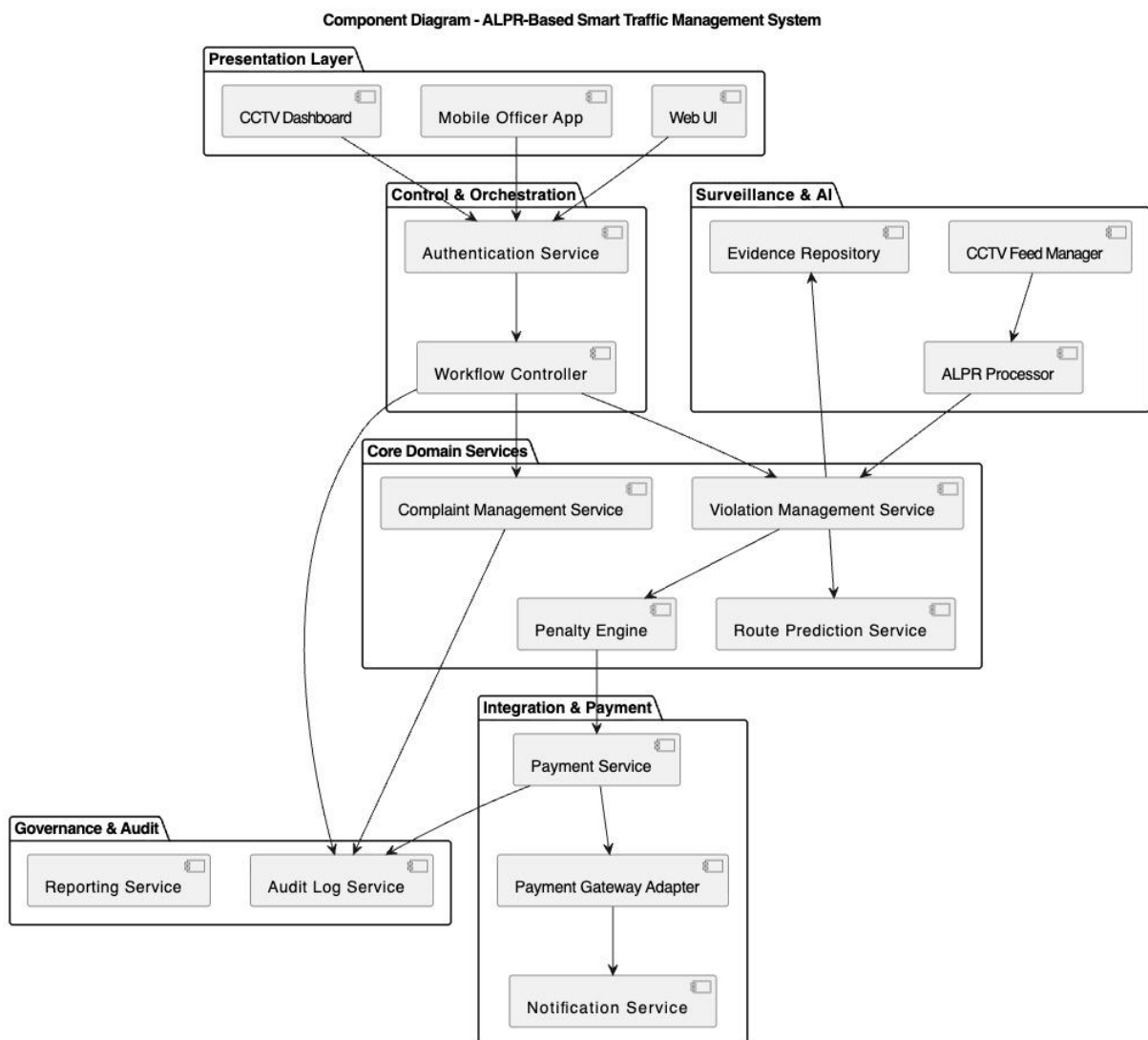


Fig 4.8: Component Interaction Diagram

4.6.10 Component Dependency Rules

- ✓ Presentation components depend only on orchestration services
- ✓ Domain components never depend on UI components
- ✓ Surveillance components never trigger penalties directly
- ✓ Payment components operate only after legal confirmation
- ✓ Audit logging is mandatory for all state-changing operations

4.6.11 Government-Grade Design Assurance

This component architecture ensures:

- ✓ Legal accountability
- ✓ Operational transparency
- ✓ Controlled automation
- ✓ Future AI extensibility
- ✓ Compliance with public-sector IT governance standards

4.6.12 Transition to Next Section

With system components clearly defined, the next section:

4.7 Deployment Modeling

4.7.1 Introduction

Deployment modeling describes the **physical realization of the system architecture**, specifying:

- ✓ Hardware nodes
- ✓ Network boundaries
- ✓ Execution environments
- ✓ Component-to-node mappings

For the **ALPR-Based Smart Traffic Management System (ALPR-STMS)**, deployment modeling is critical because the system operates in a **government environment** involving:

- ✓ Law enforcement field operations
- ✓ Central administrative control
- ✓ Surveillance infrastructure
- ✓ Financial systems
- ✓ Legal audit requirements

The deployment architecture is designed to ensure:

- ✓ High availability
- ✓ Security isolation
- ✓ Legal compliance
- ✓ Future scalability

4.7.2 Deployment Design Principles

The deployment architecture follows these government IT principles:

Centralized Governance, Distributed Operations

- ✓ Core logic centralized
- ✓ Enforcement actions distributed

Network Segmentation

- ✓ Public access strictly limited
- ✓ Surveillance and enforcement isolated

Fail-Safe Operation

- ✓ Manual enforcement continues during outages

Audit-First Architecture

- ✓ Logs stored centrally and immutably

Phased Infrastructure Adoption

- ✓ Supports gradual expansion of CCTV and AI

4.7.3 Deployment Environments

The system is deployed across **four primary environments**:

1. Field Environment

- ✓ Traffic officers
- ✓ Subcity officers
- ✓ Mobile enforcement units

2. Surveillance Environment

- ✓ CCTV cameras
- ✓ CCTV operator workstations

3. Central Data Center

- ✓ Core system services
- ✓ Databases
- ✓ Audit and reporting

4. External Integration Environment

- ✓ Payment gateway

- ✓ SMS/OTP services

4.7.4 Physical Deployment Nodes

4.7.4.1 Officer Mobile Devices

Description

- ✓ Smartphones or tablets used by traffic police officers

Deployed Components

- ✓ Mobile Officer Application (UI only)

Responsibilities

- ✓ Violation reporting
- ✓ Alert reception
- ✓ Payment initiation (on-site admission cases)

Constraints

- ✓ No local data persistence
- ✓ Requires authenticated session
- ✓ Works in online/offline hybrid mode

4.7.4.2 Subcity Traffic Office Workstations

Description

- ✓ Desktop systems located at subcity traffic offices

Deployed Components

- ✓ Web UI
- ✓ Complaint handling interface
- ✓ Reporting dashboard

Responsibilities

- ✓ Complaint adjudication
- ✓ Payment authorization after complaint resolution
- ✓ Inter-subcity coordination

4.7.4.3 CCTV Control Center

Description

- ✓ Secured facility managed by city police surveillance unit

Deployed Components

- ✓ CCTV Dashboard
- ✓ CCTV Feed Manager (local)
- ✓ ALPR preprocessing module (optional edge processing)

Responsibilities

- ✓ Video monitoring
- ✓ Manual validation of ALPR detections
- ✓ Evidence provision upon request

Security Note

- ✓ CCTV operators cannot initiate penalties or payments.

4.7.4.4 Central Application Servers

Description

- ✓ Located at Hawassa City Administration Data Center

Deployed Components

- ✓ Authentication Service
- ✓ Workflow Controller
- ✓ Violation Management Service
- ✓ Penalty Engine
- ✓ Complaint Management Service
- ✓ Route Prediction Service
- ✓ Audit Log Service
- ✓ Reporting Service

Responsibilities

- ✓ Core system logic
- ✓ Legal enforcement processing
- ✓ Central coordination

4.7.4.5 Central Database Servers

Description

- ✓ Secure government-hosted database servers

Stored Data

- ✓ User accounts (excluding drivers)
- ✓ Violations

- ✓ Evidence metadata
- ✓ Complaints
- ✓ Payments
- ✓ Audit logs

Security Measures

- ✓ Encryption at rest
- ✓ Role-based access
- ✓ Backup and disaster recovery

4.7.4.6 External Payment & Notification Services

Description

- ✓ Third-party or government-approved services

Components

- ✓ Payment Gateway
- ✓ SMS/OTP Provider

Interaction Rule

- ✓ Drivers interact only with OTP messages — never directly with system interfaces.

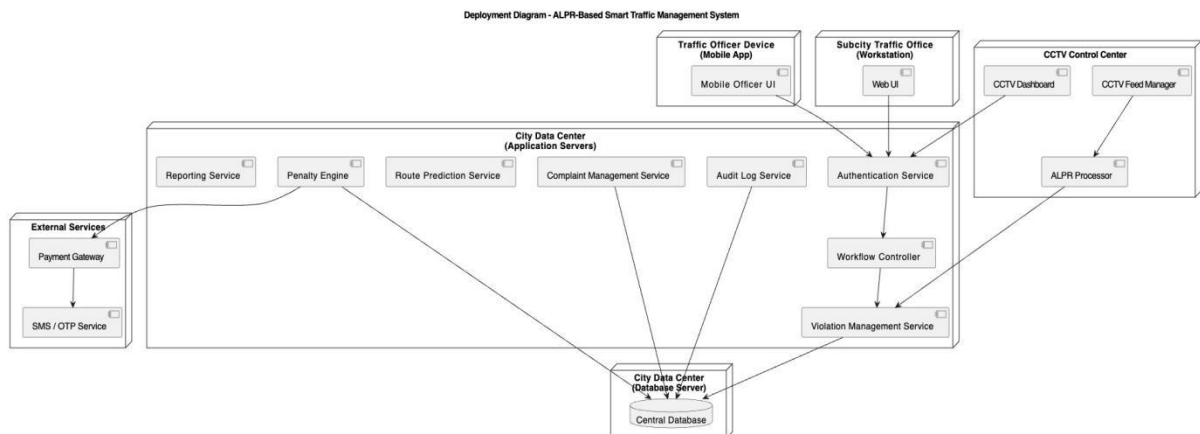


Fig 4.9: Deployment Diagram

Zone	Access Level
Officer Devices	Restricted, authenticated
Subcity Offices	Government intranet
CCTV Center	Highly restricted
Data Center	Fully secured
External Services	One-way integration

Table 4.6 Network & Security Boundaries

4.7.5 Scalability & Future Deployment Considerations

The deployment architecture supports:

- ✓ Adding more CCTV cameras without downtime
- ✓ Migrating ALPR processing to edge devices
- ✓ Integrating national traffic databases
- ✓ Expanding to other cities
- ✓ Cloud or hybrid deployment (future phase)

4.7.6 Government Compliance Assurance

This deployment model ensures:

- ✓ Data sovereignty
- ✓ Legal traceability
- ✓ Operational continuity
- ✓ Security-by-design
- ✓ Compliance with public-sector IT standards

4.8 Persistence Modeling

4.8.1 Introduction

Persistence modeling defines **how system data is stored, managed, protected, retained, and retrieved** throughout the operational lifecycle of the ALPR-Based Smart Traffic Management System (ALPR-STMS).

Given that the system supports:

- ✓ Legal enforcement
- ✓ Financial penalties
- ✓ Surveillance evidence
- ✓ Audit and accountability

the persistence layer is treated as a **mission-critical government asset**, not merely a technical storage mechanism.

This section describes:

- ✓ Data categories
- ✓ Storage architecture
- ✓ Data lifecycle
- ✓ Security controls
- ✓ Legal retention policies

4.8.2 Persistence Design Objectives

The persistence model is designed to achieve the following objectives:

Legal Admissibility

- ✓ Evidence and records must be legally defensible

Data Integrity

- ✓ No unauthorized modification
- ✓ Full traceability of changes

Controlled Access

- ✓ Role-based data visibility

Operational Performance

- ✓ Fast retrieval for enforcement and coordination

Long-Term Retention

- ✓ Compliance with government record-keeping laws

4.8.3 Data Categories and Classification

All system data is classified into the following categories:

4.8.3.1 Master Data

Relatively stable reference data.

Examples:

- ✓ User accounts (officers, operators, admins)
- ✓ Vehicle registry records
- ✓ Violation rules and penalty definitions
- ✓ Camera metadata
- ✓ Subcity configurations

Persistence Characteristics:

- ✓ Updated infrequently
- ✓ Strict administrative control
- ✓ Version-controlled

4.8.3.2 Transactional Data

Generated during daily operations.

Examples:

- ✓ Violations

- ✓ Alerts
- ✓ Penalties
- ✓ Payments
- ✓ Complaints
- ✓ Route predictions

Persistence Characteristics:

- ✓ High volume
- ✓ Time-sensitive
- ✓ Strong consistency required

4.8.3.3 Evidence Data

Legally sensitive multimedia and metadata.

Examples:

- ✓ CCTV snapshots
- ✓ Video frame references
- ✓ ALPR detection results
- ✓ Confidence scores

Persistence Characteristics:

- ✓ Immutable once attached
- ✓ Stored with cryptographic hashes
- ✓ Referenced, not duplicated

4.8.3.4 Audit and Log Data

System accountability records.

Examples:

- ✓ Login/logout events
- ✓ Data access logs
- ✓ Decision logs
- ✓ Payment confirmations
- ✓ Complaint outcomes

Persistence Characteristics:

- ✓ Append-only
- ✓ Tamper-resistant
- ✓ Long-term retention

4.8.4 Database Architecture

4.8.4.1 Centralized Relational Database

The core persistence layer uses a **centralized relational database** hosted in the city data center.

Primary Characteristics:

- ✓ ACID compliance
- ✓ Strong referential integrity
- ✓ Structured schema enforcement

Used For:

- ✓ Violations
- ✓ Users
- ✓ Complaints
- ✓ Penalties
- ✓ Payments
- ✓ Audit logs (metadata)

4.8.4.2 Evidence Storage (Hybrid Model)

- ✓ Evidence data is handled using a **hybrid persistence approach**:
- ✓ Large video files stored in Secure file storage or object storage
- ✓ Metadata stored in Central database

This avoids database overload while maintaining traceability.

4.8.5 Core Persistent Entities

4.8.5.1 User Persistence

Stores all system users except drivers.

Key Fields:

- ✓ User ID
- ✓ Role
- ✓ Subcity assignment
- ✓ Status (active/suspended)
- ✓ Authentication credentials (hashed)

Drivers are **not persisted as system users**.

4.8.5.2 Violation Persistence

The Violation entity is the **central persistence anchor**.

Stored Attributes:

- ✓ Violation ID
- ✓ Plate number

- ✓ Violation type
- ✓ Timestamp
- ✓ Location
- ✓ Reporting source (manual / CCTV)
- ✓ Current status

Each violation links to:

- ✓ Evidence
- ✓ Penalty
- ✓ Alerts
- ✓ Complaints
- ✓ Payments

4.8.5.3 Evidence Persistence

Evidence is persisted as:

- ✓ Reference ID
- ✓ Source camera
- ✓ Capture timestamp
- ✓ Confidence score
- ✓ Hash value (integrity verification)

Evidence cannot be deleted manually.

4.8.5.4 Penalty and Payment Persistence

Penalty Data:

- ✓ Amount
- ✓ Points (if applicable)
- ✓ Status

Payment Data:

- ✓ OTP reference
- ✓ Payment gateway transaction ID
- ✓ Timestamp
- ✓ Confirmation status

Payments are immutable once confirmed.

4.8.5.5 Complaint Persistence

Complaint records include:

- ✓ Complaint ID
- ✓ Violation reference
- ✓ Decision officer

- ✓ Decision outcome
- ✓ Decision justification

Each decision is logged and irreversible.

4.8.6 Data Lifecycle Management

Violation Lifecycle

- ✓ Created
- ✓ Pending confirmation
- ✓ Confirmed or revoked
- ✓ Paid or closed

Evidence Lifecycle

- ✓ Captured → Linked → Archived → Retained

Audit Lifecycle

- ✓ Generated → Stored → Archived → Retained indefinitely

4.8.7 Data Security Controls

Security measures applied at persistence level:

- ✓ Encryption at rest
- ✓ Role-based access enforcement
- ✓ Read/write separation
- ✓ Immutable audit tables
- ✓ Hash-based evidence verification

4.8.8 Backup and Disaster Recovery

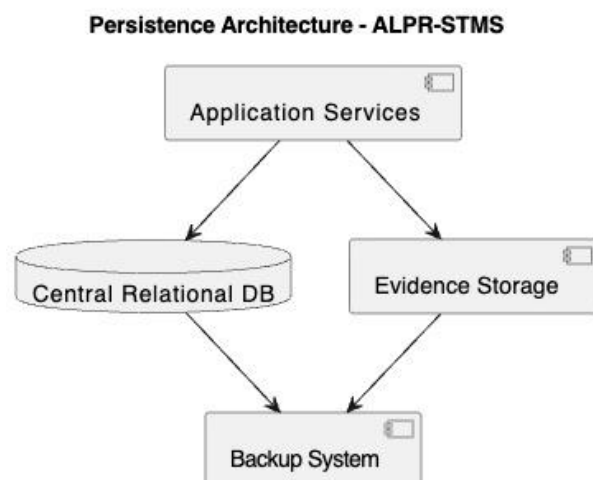
Fig 4.10: Persistence Architecture Diagram

Backup Strategy:

- ✓ Daily incremental backups
- ✓ Weekly full backups
- ✓ Off-site secure storage

Recovery Objectives:

- ✓ RPO: ≤ 24 hours
- ✓ RTO: ≤ 4 hours



Data Type	Retention
Violations	≥ 5 years
Payments	≥ 7 years
Complaints	≥ 7 years
Audit Logs	Permanent
Evidence	As per legal requirement

Table 4.7 Retention and Archival Policy

4.8.9 Compliance and Governance Assurance

The persistence model ensures:

- ✓ Legal defensibility
- ✓ Operational continuity
- ✓ Government audit readiness
- ✓ Privacy protection
- ✓ Long-term institutional memory

4.9 User Interface design

4.9.1 Introduction

The User Interface (UI) of the ALPR-Based Smart Traffic Management System (ALPR-STMS) is designed as a **mission-critical government interface**, prioritizing:

- ✓ Accuracy over aesthetics
- ✓ Clarity over complexity
- ✓ Accountability over convenience

The UI serves as the **primary interaction point** between authorized government personnel and the system, directly supporting enforcement, coordination, complaint resolution, and administrative governance.

The design strictly adheres to:

- ✓ Government IT usability standards
- ✓ ISO 9241 human-centered design principles
- ✓ Role-based operational separation
- ✓ Legal evidentiary requirements

4.9.2 UI Design Objectives

The UI design aims to achieve the following objectives:

Operational Efficiency

- ✓ Minimize steps for critical actions
- ✓ Reduce cognitive load during field operations

Role-Based Clarity

- ✓ Each user sees only relevant functions
- ✓ No feature overlap between unrelated roles

Legal Defensibility

- ✓ Clear timestamps
- ✓ Explicit confirmations
- ✓ Immutable records

Error Prevention

- ✓ Validation at every critical input
- ✓ Mandatory confirmation dialogs

Scalability

- ✓ Supports future automation phases
- ✓ Easily extendable interfaces

Role	UI Access
Traffic Officer	Violation reporting, alerts, route drawing
Subcity Officer	Coordination dashboard, escalation
CCTV Operator	Camera feeds, ALPR validation
Complaint Officer	Complaint review, decision panels
System Admin	Configuration, user management
Driver	No system UI access (OTP payment only)

Table 4.8 User Roles and UI Access Scope

4.9.3 UI Architectural Pattern

The UI follows a **Modular Role-Based Dashboard Architecture**:

- ✓ Common authentication interface
- ✓ Separate dashboards per role
- ✓ Shared visual language
- ✓ Centralized notification framework

This prevents unauthorized access and supports auditability.

4.9.4 Common UI Components

4.9.4.1 Authentication Interface

- ✓ Username and password login
- ✓ Role verification
- ✓ Session timeout enforcement

- ✓ Multi-factor support (future-ready)

Drivers are explicitly excluded from login.

4.9.4.2 Notification & Alert Panel

- ✓ High-visibility alerts
- ✓ Priority indicators (color-coded)
- ✓ Timestamped messages
- ✓ Geographic relevance indicators

Alerts cannot be dismissed without acknowledgment.

4.9.4.3 Map & Location Interface

- ✓ Real-time GIS integration
- ✓ Officer location tracking
- ✓ Route drawing canvas
- ✓ CCTV camera overlay

Used for:

- ✓ Escape path prediction
- ✓ Alert prioritization
- ✓ Interception planning

4.9.5 Role-Specific UI Designs

4.9.5.1 Traffic Officer Interface

Purpose:

- ✓ Field enforcement and reporting

Key Screens:

- ✓ Violation reporting form
- ✓ Plate number input
- ✓ Automatic GPS capture
- ✓ Escape route drawing map
- ✓ Alert inbox

Design Characteristics:

- ✓ Large buttons
- ✓ Minimal text
- ✓ Mobile-first layout
- ✓ Offline fallback indicators

4.9.5.2 Subcity Officer Dashboard

Purpose:

- ✓ Coordination and escalation

Key Screens:

- ✓ Active violations overview
- ✓ Inter-subcity alerts
- ✓ Escalation controls
- ✓ Status tracking

Design Characteristics:

- ✓ Tabular summaries
- ✓ Filters by priority
- ✓ Region-based visualization

4.9.5.3 CCTV Operator Interface

Purpose:

- ✓ Surveillance and validation

Key Screens:

- ✓ Live camera feeds
- ✓ ALPR detection review panel
- ✓ Confidence score display
- ✓ Manual confirmation controls

Design Characteristics:

- ✓ Multi-feed layouts
- ✓ Zoom and playback tools
- ✓ Evidence tagging controls

4.9.5.4 Complaint Officer Interface

Purpose:

- ✓ Legal review and decision-making

Key Screens:

- ✓ Violation search by ID
- ✓ Evidence review
- ✓ Officer statements
- ✓ Decision submission form

Design Characteristics:

- ✓ Read-only evidence
- ✓ Mandatory justification fields
- ✓ Decision locking after submission

4.9.5.5 System Administrator Interface

Purpose:

- ✓ Governance and configuration

Key Screens:

- ✓ User management
- ✓ Role assignment
- ✓ Rule configuration
- ✓ Audit log viewer

Design Characteristics:

- ✓ Restricted access
- ✓ Confirmation dialogs
- ✓ Change history panels

4.9.6 UI Validation and Error Handling

- ✓ Mandatory fields enforced
- ✓ Inline validation messages
- ✓ Explicit error explanations
- ✓ Prevention of duplicate submissions

All critical actions require confirmation.

4.9.7 Accessibility and Government Standards Compliance

The UI complies with:

- ✓ ISO 9241 (ergonomics of human-system interaction)
- ✓ Government accessibility requirements
- ✓ Color contrast standards
- ✓ Keyboard navigation support
- ✓ Clear iconography with text labels

4.9.8 UI Security Considerations

- ✓ Session timeout enforcement
- ✓ Role-based UI rendering
- ✓ No client-side role logic
- ✓ Sensitive data masking

- ✓ Screenshot and export restrictions (where applicable)

4.9.9 UI Evolution and Future Readiness

The UI is designed to support:

- ✓ Full ALPR automation
- ✓ AI-assisted decision hints
- ✓ National system integration
- ✓ Smart city dashboards

4.9.10 UI Design Summary

The ALPR-STMS UI:

- ✓ Supports real-world enforcement workflows
- ✓ Ensures legal and administrative compliance
- ✓ Minimizes operational risk
- ✓ Enhances coordination and accountability
- ✓ Reflects government-grade professionalism

4.10 System Transition and Deployment Strategy

4.10.1 Introduction

System transition is the structured process through which the proposed Automatic License Plate Detection and Recognition–Based Smart Traffic Management System (ALPR-STMS) moves from a designed solution into an operational, institutionalized, and legally compliant production system within Hawassa City Administration.

This section defines the **technical, operational, organizational, and procedural activities** required to introduce the system into real traffic enforcement operations while ensuring continuity of service, legal validity, data integrity, and institutional acceptance. The transition strategy is intentionally incremental to accommodate existing manual practices, infrastructure constraints, and human resource readiness.

4.10.2 Transition Objectives

The primary objectives of the system transition are to:

- ✓ Ensure **seamless migration** from manual and semi-manual traffic enforcement to a digital system without service disruption.
- ✓ Preserve **legal authority and human decision-making** throughout all phases.
- ✓ Enable **progressive adoption** of ALPR and predictive intelligence based on infrastructure readiness.
- ✓ Maintain **operational continuity** during system rollout.
- ✓ Guarantee **data security, auditability, and regulatory compliance**.
- ✓ Achieve **organizational acceptance** across traffic police, subcity offices, and administrative bodies.

4.10.3 Transition Scope

The transition covers the following domains:

Technical Scope

- ✓ System deployment and configuration
- ✓ Integration with existing CCTV infrastructure
- ✓ Database initialization and data validation
- ✓ Network and security setup

Operational Scope

- ✓ Traffic violation reporting workflows
- ✓ Inter-subcity coordination
- ✓ Complaint handling and resolution
- ✓ OTP-based payment processing

Organizational Scope

- ✓ Role assignment and access control
- ✓ Training and capacity building
- ✓ Change management and policy alignment

4.10.4 Phased Transition Strategy

To reduce risk and ensure adoption, the ALPR-STMS follows a **four-phase transition model**.

Phase 1: Manual Digital Enforcement (Baseline Deployment)

Description

This phase introduces digital reporting while preserving all existing enforcement practices.

Key Characteristics

- ✓ Traffic officers log in and manually report violations
- ✓ Location and timestamp captured automatically
- ✓ Alerts broadcast digitally instead of via walkie-talkie
- ✓ No automated detection

Outcome

- ✓ Immediate improvement in documentation, coordination, and traceability
- ✓ Zero disruption to legal and operational workflows

The implementation timeline for Phase One is aligned with the detailed project schedule presented in **Appendix D**.

Phase 2: CCTV-Assisted Detection (Human-Validated)

Description

CCTV feeds are integrated, but all decisions remain human-validated.

Key Characteristics

- ✓ ALPR suggests candidate plate numbers
- ✓ CCTV operators validate or dismiss detections
- ✓ Evidence stored digitally
- ✓ No automated penalty issuance

Outcome

- ✓ Improved detection accuracy
- ✓ Strengthened evidentiary support
- ✓ Increased enforcement efficiency

Phase 3: Predictive Tracking and Smart Alerts

Description

The system introduces predictive intelligence while maintaining officer authority.

Key Characteristics

- ✓ Route prediction based on last known locations
- ✓ Priority alerts for officers along predicted paths
- ✓ Enhanced inter-subcity coordination

Outcome

- ✓ Higher interception success rate
- ✓ Reduced escape incidents
- ✓ Smarter resource allocation

Phase 4: Semi-Automated Enforcement (Future-Ready)

Description

Advanced automation is introduced under strict governance controls.

Key Characteristics

- ✓ Automated violation detection (subject to validation)
- ✓ Automated penalty calculation
- ✓ Full digital lifecycle management

Outcome

- ✓ Maximum efficiency

- ✓ Policy-driven automation
- ✓ Scalable smart-city readiness

4.10.5 User Transition and Role Enablement

Authorized System Users

Only government-authorized personnel interact with the system:

- ✓ Traffic Officers
- ✓ Subcity Officers
- ✓ CCTV Operators
- ✓ Complaint Officers
- ✓ System Administrators

Drivers are NOT system users.

Their interaction is limited strictly to:

- ✓ OTP-based payment via external payment gateway
- ✓ Physical complaint submission at subcity offices

This ensures:

- ✓ Legal clarity
- ✓ Reduced system abuse
- ✓ Strong institutional control

4.10.6 Training and Capacity Building

A structured training program is mandatory prior to each phase transition.

Training Coverage

- ✓ Role-specific system usage
- ✓ Legal implications of digital evidence
- ✓ Complaint handling procedures
- ✓ Data privacy and security responsibilities

Training Approach

- ✓ Classroom sessions
- ✓ Practical simulations
- ✓ Supervised live operations
- ✓ Continuous refresher programs

4.10.7 Data Migration and Initialization

Data Sources

- ✓ Vehicle registration records

- ✓ Traffic rule and penalty definitions
- ✓ User accounts and roles

Data Controls

- ✓ Validation against authoritative government sources
- ✓ Controlled access during migration
- ✓ Audit logging of all data initialization activities

No historical paper records are altered or deleted; digital records supplement existing archives.

Risk	Mitigation Strategy
Network failure	Offline capture & delayed synchronization
CCTV malfunction	Manual reporting fallback
User resistance	Incremental rollout & training
Legal disputes	Human-validated decisions & audit trails
Data breach	Encryption, RBAC, audit logging

Table 4.9 Risk Management and Mitigation

4.10.8 Compliance, Governance, and Audit Readiness

The system transition ensures:

- ✓ Full compliance with **Road Transport Traffic Control COM Regulation 557-2024**
- ✓ Role-based access control (RBAC)
- ✓ Immutable audit logs for all critical actions
- ✓ Traceability for every violation, decision, and payment

All records are retained in accordance with government data retention and evidence laws.

4.10.9 Transition Completion Criteria

The transition is considered successful when:

- ✓ All authorized users operate independently.
- ✓ Violations are processed end-to-end digitally.
- ✓ Alerts and coordination occur in real time.
- ✓ Complaints are resolved using digital evidence.
- ✓ Payment processing functions reliably.
- ✓ Audit logs demonstrate accountability and traceability.

Conclusion

This project has presented a comprehensive analysis, specification, and system design of an **Automatic License Plate Recognition (ALPR) Based Smart Traffic Management System** tailored for urban environments such as **Hawassa City**. Beginning from Unit 1, the documentation has systematically progressed from problem identification and requirement analysis through architectural design, modeling, and interface considerations, culminating in a fully integrated, government-grade system specification.

The study initially identified critical challenges in existing traffic management practices, including heavy reliance on manual enforcement, limited inter-subcity coordination, absence of real-time intelligence, and lack of reliable digital records. These limitations were shown to negatively impact enforcement efficiency, legal accountability, and strategic traffic planning. In response, this project proposed a structured, scalable, and legally defensible system that incrementally transitions from manual reporting to advanced automated enforcement supported by computer vision and predictive analytics.

Through **Unit 2 (System Requirement Specifications)**, the system's functional and non-functional requirements were defined with precision, ensuring alignment with real-world operational constraints, government procedures, and institutional responsibilities. Clear use case definitions, sequence diagrams, activity diagrams, and conceptual class models were developed to reflect authentic workflows involving traffic officers, subcity officers, CCTV operators, complaint officers, and system administrators—while deliberately excluding drivers from system access except through OTP-based payment mechanisms. This strict role separation enhances security, accountability, and compliance with public-sector IT governance principles.

In **Unit 3 (Analysis and Modeling)**, the system behavior was explored in depth using production-ready UML artifacts. Detailed sequence diagrams illustrated end-to-end workflows, including login authentication for all human operators, manual and automated violation detection, CCTV-assisted validation, predictive path analysis, complaint resolution, penalty decision-making, and secure payment processing. Activity diagrams further clarified the logical flow of enforcement actions, decision points, and exception handling, ensuring that every possible operational scenario is addressed with clarity and precision.

Unit 4 (System Design) translated these analytical models into a robust architectural framework. The class-type architecture defined clear separation between presentation, controller, business, persistence, and system layers, ensuring maintainability and scalability. Additional modeling—covering state charts, collaboration, components, deployment, persistence, and user interface design—demonstrated how the system can be implemented in a real government IT environment while maintaining performance, security, and legal traceability. Special emphasis was placed on audit logging, data retention, and compliance with standards such as ISO 9241 and public-sector usability and accessibility requirements.

The proposed system is not merely a theoretical construct but a **deployment-ready blueprint** that respects existing institutional structures, integrates with current CCTV infrastructure, and supports phased adoption. Its modular design allows gradual enhancement—from manual reporting and coordination to fully automated ALPR detection and predictive traffic intelligence—without disrupting ongoing operations. Furthermore, the system provides long-

term value by generating structured data that can support legal review, policy formulation, urban planning, and smart city initiatives.

In conclusion, this project successfully delivers a **holistic, accurate, and government-aligned smart traffic management solution**. It demonstrates how modern software engineering practices, when applied with contextual awareness and institutional discipline, can significantly improve traffic enforcement efficiency, transparency, and accountability. The documentation produced serves both as an academic achievement and as a practical foundation for real-world implementation, positioning the system as a viable contribution to the modernization of urban traffic governance.

RECOMMENDATIONS

Based on the analysis, design, and proposed implementation of the **ALPR-Based Smart Traffic Management System for Hawassa City**, the following recommendations are made to ensure effective deployment, sustainability, and institutional acceptance:

Phased Implementation Strategy

It is recommended that the system be deployed incrementally, beginning with:

- ✓ Manual digital violation reporting
- ✓ Centralized violation and vehicle database
- ✓ OTP-based payment processing
- ✓ Advanced features such as automated CCTV-based detection and predictive route tracking should be introduced after sufficient infrastructure readiness and staff training.

Legal and Policy Alignment

Continuous alignment with national and regional traffic regulations, particularly **Road Transport Traffic Control Council of Ministers Regulation No. 557/2024**, is essential. Any updates to traffic laws should be reflected promptly in the system's violation rules and penalty engine.

Capacity Building and Training

Comprehensive training programs should be provided for:

- ✓ Traffic police officers
- ✓ Subcity officers
- ✓ CCTV operators
- ✓ Complaint and administrative officers

This will ensure correct system usage, reduce resistance to change, and improve enforcement accuracy.

Infrastructure Enhancement

Gradual expansion of CCTV coverage, secure network connectivity, and reliable data center infrastructure is recommended to support advanced ALPR processing and real-time coordination.

Data Governance and Security

Strong data protection policies, role-based access control, audit logging, and compliance with government data retention laws should be enforced to maintain public trust and legal defensibility.

Future Integration

The system should be prepared for future integration with:

- ✓ National vehicle registration databases
- ✓ Smart city platforms
- ✓ Inter-city and federal traffic enforcement systems

APPENDIX

Appendix A: Actors Summary

- ✓ Traffic Officer
- ✓ Subcity Officer
- ✓ CCTV Operator
- ✓ Complaint Officer
- ✓ System Administrator
- ✓ Driver (OTP payment participant only)

Appendix B: Legal Basis

The system is legally grounded in Ethiopian traffic enforcement frameworks, with specific reference to:

- ✓ Federal and regional traffic laws
- ✓ Official authority of traffic police
- ✓ Recognized penalty and violation classifications
- ✓ Driver complaint and appeal rights

Appendix C: Assumptions and Limitations

- ✓ Initial CCTV coverage is limited
- ✓ Network availability may vary across subcities
- ✓ Drivers do not directly interact with the system except for OTP payment
- ✓ Complaint handling is strictly office-based and officer-mediated

Appendix D: Detailed Project Task and Schedule

D.1 Purpose of This Appendix

This appendix provides the **detailed, task-level project schedule** for the *ALPR-Based Smart Traffic Management System for Hawassa City*. It expands upon the **high-level schedule overview presented in Chapter 1** and documents the **chronological execution plan**, task dependencies, academic constraints, and milestone alignment for the project.

The schedule presented herein:

- ✓ Covers **end-to-end project activities** (all phases)
- ✓ Emphasizes **Phase One (Manual Digital Traffic Reporting System)** during design, implementation, and testing stages
- ✓ Accounts for **academic holidays, final examinations, and institutional delays**
- ✓ Supports **schedule feasibility analysis** and **project governance review**

D.2 Scheduling Assumptions and Constraints

The detailed schedule is prepared based on the following assumptions:

- ✓ The project follows an **incremental and phased development approach**.
- ✓ Early activities (planning, authorization, documentation) support **all system phases**.
- ✓ Design, implementation, and testing activities after the academic break are **primarily focused on Phase One**.
- ✓ National holidays, academic breaks, and examination periods are treated as **non-development intervals**.
- ✓ Task durations are estimated conservatively to ensure **realistic delivery** under academic and institutional conditions.

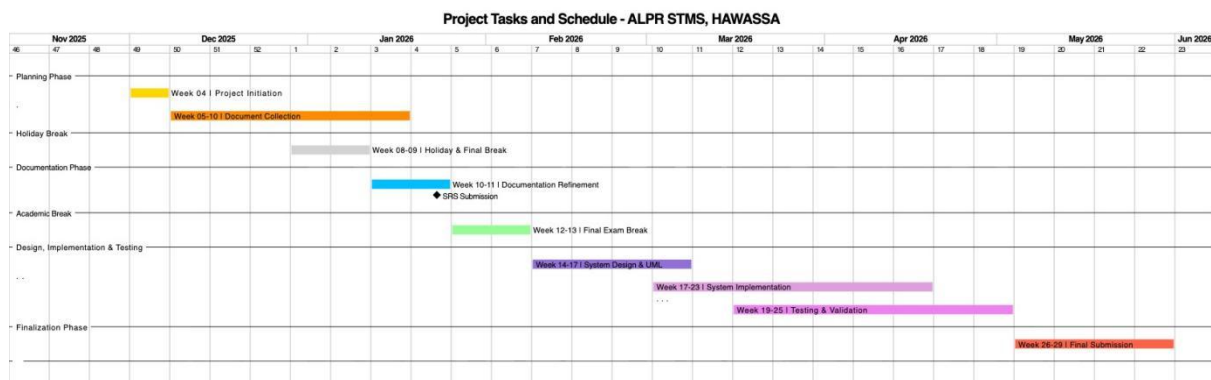


Figure A.1: Detailed Project Tasks and Schedule – ALPR STMS, Hawassa

D.3 Detailed Task Description and Timeline Mapping

D.3.1 Planning Phase

Duration: Week 04 (December 1–7, 2025)

Key Activities:

- ✓ Project initiation
- ✓ Stakeholder coordination
- ✓ Scope confirmation
- ✓ Preliminary feasibility alignment

This phase establishes the **administrative and organizational foundation** for the entire project lifecycle.

D.3.2 Authorization and Document Collection Phase

Duration: Weeks 05–10 (December 8, 2025 – January 18, 2026)

Key Activities:

- ✓ Collection of traffic regulations and legal documents
- ✓ Authorization requests from relevant government bodies
- ✓ Institutional approvals and permissions
- ✓ Data access agreements

This phase is critical for ensuring **legal compliance**, **data legitimacy**, and **governmental alignment**.

D.3.3 Holiday and Final Examination Break

Duration: Weeks 08–09 (December 29, 2025 – January 11, 2026)

This period reflects **non-productive time** due to academic and national holidays. No development or documentation activities are scheduled.

D.3.4 Documentation Refinement Phase

Duration: Weeks 10–11 (January 12–25, 2026)

Key Activities:

- ✓ Finalization of SRS
- ✓ UML refinement
- ✓ Requirements validation
- ✓ Consistency checks and academic formatting

D.3.5 Requirements Specification Milestone

Milestone: *SRS Submission*

Date: January 21, 2026

This milestone represents the **formal completion and submission** of the System Requirements Specification document.

D.3.6 Academic Break

Duration: Weeks 12–13 (January 26 – February 8, 2026)

This scheduled break accommodates **final examinations** and institutional downtime.

D.3.7 System Design Phase (Phase One Focus)

Duration: Weeks 14–17 (February 9 – March 8, 2026)

Key Activities:

- ✓ Architectural design
- ✓ UML modeling (class, sequence, activity, deployment)
- ✓ Design validation against SRS
- ✓ Phase One scope alignment

This phase translates requirements into **formal system design artifacts**.

D.3.8 System Implementation Phase (Phase One)

Duration: Weeks 17–23 (March 2 – April 20, 2026)

Key Activities:

- ✓ Core system development
- ✓ Manual traffic reporting workflows
- ✓ User role management
- ✓ Payment initiation logic (non-driver interaction)

Implementation overlaps partially with testing to support **incremental validation**.

D.3.9 Testing and Validation Phase

Duration: Weeks 19–25 (March 16 – May 4, 2026)

Key Activities:

- ✓ Functional testing
- ✓ Integration testing
- ✓ Validation against traffic enforcement workflows
- ✓ Error correction and performance tuning

D.3.10 Finalization and Submission Phase

Duration: Weeks 26–29 (May 4 – May 31, 2026)

Key Activities:

- ✓ Final documentation consolidation
- ✓ Corrections based on review feedback
- ✓ Formal submission
- ✓ Archival preparation

Phase	Activity	Start Date	End Date	Duration	Phase Coverage
Planning	Project initiation	Dec 1, 2025	Dec 7, 2025	1 week	All phases
Authorization	Document collection	Dec 8, 2025	Jan 18, 2026	6 weeks	All phases
Documentation	Refinement & SRS	Jan 12, 2026	Jan 25, 2026	2 weeks	All phases
Milestone	SRS submission	Jan 23, 2026	Jan 23, 2026	Milestone	All phases
Design	System design	Feb 9, 2026	Mar 8, 2026	4 weeks	Phase One
Implementation	Development	Mar 2, 2026	Apr 20, 2026	7 weeks	Phase One
Testing	Validation	Mar 16, 2026	May 4, 2026	7 weeks	Phase One
Finalization	Submission	May 4, 2026	May 31, 2026	4 weeks	Phase One

Table A.1: Summary of Project Tasks, Duration, and Focus

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